

# NTRU+Sign: Compact NTRU-Based Signatures Using Bimodal Distributions

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## Abstract

We present a new lattice-based signature scheme, called ‘NTRU+Sign’, using the Fiat-Shamir with Aborts framework. The proposed scheme is designed based on a novel NTRU-based key structure that fits well with bimodal distributions, enabling efficiency improvements compared to its predecessor, BLISS. The novel NTRU-based key structure is characterized by: (1) effectively changing a modulus from  $2q$  to  $q$ , which is different from the existing usage of  $2q$  for bimodal distributions, and (2) drastically reducing the magnitude of a secret key, which directly leads to compactness of signature sizes. We provide two concrete parameter sets for NTRU+Sign, supporting 93-bit and 211-bit security levels. Using the technique from GALACTICS (that was suggested as the constant-time implementation of BLISS), our analysis shows that NTRU+Sign achieves a good balance between computational efficiency and signature compactness, with constant-time implementation. For instance, at the NIST-3 security level, NTRU+Sign produces signatures that are significantly smaller than Dilithium and HAETAE, while providing faster verification speeds. These advantages position NTRU+Sign as a competitive and practical solution for real-world deployments.

## 1 Introduction

Lyubashevsky [Lyu09, Lyu12] proposed lattice-based signature schemes based on the Fiat-Shamir with Aborts (FSwA) framework. Their underlying lattice-based identification scheme involves a publicly shared matrix  $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$  and a public key  $\mathbf{T} = \mathbf{AS} \bmod q$ , where the secret key  $\mathbf{S} \in \mathbb{Z}^{m \times k}$  consists of small entries. In response to a challenge  $\mathbf{c} \in \mathbb{Z}^k$ , a response  $\mathbf{z}$  is computed as  $\mathbf{z} = \mathbf{y} + \mathbf{Sc}$ , where  $\mathbf{y} \in \mathbb{Z}^m$  is a small masking vector sampled from a certain distribution. A verifier checks if the relation  $\mathbf{Ay} = \mathbf{Az} - \mathbf{Tc} \bmod q$  holds. Unlike the Schnorr identification scheme, the distribution of  $\mathbf{z}$  remains inherently biased by  $\mathbf{Sc}$ , leading to a potential leakage of information about the secret key  $\mathbf{S}$ . To address this, Lyubashevsky [Lyu09, Lyu12] employed the technique of *rejection sampling* which outputs a candidate value of  $\mathbf{z}$  with probability  $f(\mathbf{z})/(Mg(\mathbf{z}))$ , where  $f$  indicates a target distribution,  $g$  indicates a source distribution, and  $M$  is a constant. Using rejection sampling, the source distribution of  $\mathbf{z}$  is forced to align with the target distribution, effectively removing any dependency on the secret key  $\mathbf{S}$  from  $\mathbf{z}$ .

In 2013, Ducas *et al.* [DDL13] introduced a novel lattice-based signature scheme, BLISS, which leveraged the so-called *bimodal distributions*, achieving significantly more compact signature sizes. In BLISS, the bimodal distributions are characterized by using a random bit  $b$  such that  $\mathbf{z} = \mathbf{y} + (-1)^b \mathbf{S}\mathbf{c}$ , where  $\mathbf{y}$  is sampled from a discrete Gaussian distribution  $D_\sigma^m$  with standard deviation  $\sigma$ . This construction allowed the source distribution of  $\mathbf{z}$  to better align with the target distribution of  $\mathbf{y}$ , improving the efficiency of the rejection sampling process. To ensure correctness regarding the usage of  $(-1)^b$ , BLISS incorporates arithmetic modifications, such as replacing the modulus  $q$  with  $2q$  and changing the key generation algorithm to satisfy the condition  $\mathbf{A}\mathbf{S} = q\mathbf{I} \bmod 2q$ . These modifications guarantee that  $\mathbf{A}\mathbf{z} - q\mathbf{c} = \mathbf{A}\mathbf{y} + (-1)^b \mathbf{A}\mathbf{S}\mathbf{c} - q\mathbf{c} = \mathbf{A}\mathbf{y} \bmod 2q$ , regardless of whether  $b = 0$  or  $b = 1$ . While BLISS was considered one of the most efficient lattice-based signature scheme, its practical implementations revealed significant vulnerabilities to side-channel timing attacks [BDE<sup>+</sup>18, BHLY16a, EFGT17, PBY17]. Such attacks mainly exploited flaws (from a side-channel attack perspective) in the implementation of transcendental functions that are necessary for both discrete Gaussian samplings for  $\mathbf{y}$  and the rejection sampling process involved in computing  $f(\mathbf{z})/(Mg(\mathbf{z}))$ .

To address these vulnerabilities, Barthe *et al.* [BBE<sup>+</sup>19] proposed GALACTICS, a secure implementation of BLISS that mitigates the side-channel vulnerabilities in [BDE<sup>+</sup>18, BHLY16a, EFGT17, PBY17], while maintaining efficiency comparable to the original implementation. In GALACTICS, polynomial approximations of transcendental functions ensure that all operations related to discrete Gaussian sampling and the computation of  $f(\mathbf{z})/(Mg(\mathbf{z}))$  are executed in constant time. Following the basic structure of BLISS, another approach has been made by HAETAE [CCD<sup>+</sup>24], a module lattice-based signature scheme that uses a uniform hyperball sampling [DFPS22]. HAETAE simplifies the rejection sampling process by simply checking if the Euclidean norms of  $\mathbf{z}$  are sufficiently small within a preset bound, rather than computing  $f(\mathbf{z})/(Mg(\mathbf{z}))$  as in BLISS. However, uniform hyperball sampling requires a large number of discrete Gaussian samplings with a large standard deviation and high-precision arithmetic, using the polynomial approximation of the exponential function  $e^{-x}$  essentially similar to GALACTICS, and involves additional computations such as inverse of square roots and the  $\ell_2$  norm of a vector, adding computational complexity.

Ducas *et al.* [DKL<sup>+</sup>18] proposed Dilithium, a lattice-based signature scheme that goes back to the form of  $\mathbf{z} = \mathbf{y} + \mathbf{S}\mathbf{c}$  (that is, avoiding bimodal distributions). Instead of using discrete Gaussian samplings, Dilithium uses uniform sampling in hypercubes to generate  $\mathbf{y}$ , significantly simplifying the sampling part. Additionally, the rejection sampling process is also simplified by checking whether the infinite norm of  $\mathbf{z}$  is sufficiently small within a preset bound. Dilithium was later selected by NIST (FIPS 204) as the module-lattice-based digital signature standard. Although uniform sampling simplifies implementation and makes it easier to protect against side-channel timing attacks, it results in less compact signatures than BLISS. In theory, it has been justified by prior work [DFPS22], which shows that bimodal Gaussian distributions provide better signature compactness than uniform distributions in hypercubes.

## 1.1 Our Contribution

The goal of this paper is to revisit BLISS [DDL13] so as to achieve its full potential. We suggest a novel lattice-based signature scheme, called NTRU+Sign, building upon BLISS by relying on the NTRU problem [HHGP<sup>+</sup>03, HPS98]. At a similar security level, NTRU+Sign achieves more compact signatures and faster verification speeds than Dilithium, HAETAE, and re-parametrized BLISS. To achieve our goal, we made several new changes to BLISS.

First, we introduce a novel NTRU-based key structure suitable for bimodal distributions, which eliminates the need for modulus  $2q$  operations by transitioning to modulus  $q$  operations. Let  $R = \mathbb{Z}[x]/(x^n + 1)$  be a polynomial-based ring for the NTRU-based setting. For a public key  $\mathbf{A} \in R^{1 \times 2}$  and its relevant

signing key  $\mathbf{S} \in R^{2 \times 1}$ , BLISS satisfies the condition of  $\mathbf{A}\mathbf{S} = q \bmod 2q$ , whereas NTRU+Sign satisfies the condition of  $\mathbf{A}\mathbf{S} = \hat{q} \bmod q$ , where  $\hat{q} = 2^{-1} \bmod q$ . In both schemes, signatures are generated as  $(\mathbf{z} = \mathbf{y} + (-1)^b \mathbf{S}\mathbf{c}, \mathbf{c})$ , leveraging bimodal discrete Gaussian distributions. The commitment (corresponding to the identification scheme) uses only the high-order bits of  $\mathbf{A}\mathbf{y}$ . The key distinction lies in the verification process. BLISS checks if the high-order bits of  $\mathbf{A}\mathbf{z} + \mathbf{c}q \bmod 2q$  match the commitment, and NTRU+Sign checks if the high-order bits of  $\mathbf{A}\mathbf{z} + \mathbf{c}\hat{q} \bmod q$  match the commitment. The verification of NTRU+Sign succeeds from the following observation. When  $b = 1$ , the value  $\mathbf{z} = \mathbf{y} - \mathbf{S}\mathbf{c}$  results in  $\mathbf{A}\mathbf{z} + \mathbf{c}\hat{q} = \mathbf{A}\mathbf{y} \bmod q$ , which satisfies the verification equation. Also, when  $b = 0$ , the value  $\mathbf{z} = \mathbf{y} + \mathbf{S}\mathbf{c}$  results in  $\mathbf{A}\mathbf{z} + \mathbf{c}\hat{q} = \mathbf{A}\mathbf{y} + 2\mathbf{c}\hat{q} = \mathbf{A}\mathbf{y} + \mathbf{c} \bmod q$ . Since the coefficients of the challenge  $\mathbf{c} \in R$  are 0 or 1, the high-order bits of  $\mathbf{A}\mathbf{y}$  and  $\mathbf{A}\mathbf{y} + \mathbf{c}$  are equal with high probability. Furthermore, the signing process in NTRU+Sign incorporates an equality check in advance, ensuring this condition holds without requiring modulus  $2q$  operations.

One of the main challenges in achieving small signature sizes within the FSwA framework is to make the upper bound of the Euclidean norm of  $\mathbf{S}\mathbf{c}$  as small as possible. This is crucial because a smaller norm of  $\mathbf{S}\mathbf{c}$  results in a source distribution that more closely resembles the target distribution, allowing for a more compact signature size under a fixed rejection rate. We address this challenge through two key approaches: (1) Compared to BLISS, the NTRU-based key structure in NTRU+Sign significantly reduces the magnitude of  $\mathbf{S}$  itself. For small polynomials  $\mathbf{f}$  and  $\mathbf{g}$  in  $R$ , the public and signing key pairs of BLISS are  $\mathbf{A} = (2((2\mathbf{f}+1)/\mathbf{g} \bmod q), q-2)$  and  $\mathbf{S} = (\mathbf{g}, 2\mathbf{f}+1)^T$ , and this structure has been made in the process of harmonizing bimodal structure and the classical NTRU problem *over*  $q$ . But, NTRU+Sign can avoid such artificial structure by setting the public key as  $\mathbf{A} = ((\mathbf{f} + \hat{q})/\mathbf{g} \bmod q, 1)$ , and so the corresponding signing key  $\mathbf{S} = (\mathbf{g}, -\mathbf{f})^T$  has smaller norm than that of BLISS. For example, for polynomials  $\mathbf{f}$  and  $\mathbf{g}$  with coefficients sampled from  $\{-1, 0, 1\}$ , the coefficients of  $2\mathbf{f} + 1$  in BLISS belong to  $\{-1, 1, 3\}$ , leading to a higher upper bound on  $\|\mathbf{S}\|$  and, consequently,  $\|\mathbf{S}\mathbf{c}\|$ . In contrast, NTRU+Sign ensures that the coefficients of  $\mathbf{S} = (\mathbf{g}, -\mathbf{f})^T$  still belong to  $\{-1, 0, 1\}$ . (2) We further tighten the bound on  $\|\mathbf{S}\mathbf{c}\|$  by leveraging the canonical embedding into  $\mathbb{C}^n$ , as studied in HAETAE [CCD<sup>+</sup>24]. Compared to BLISS, this significantly improves the estimation of  $\|\mathbf{S}\mathbf{c}\|$ , further contributing to the compactness of the signature. Combining these improvements, NTRU+Sign achieves the smallest upper bound on  $\|\mathbf{S}\mathbf{c}\|$  among comparable schemes. At the 180-bit security level, for example, the upper bound is 341 for NTRU+Sign, but 1270 for BLISS and 516 for HAETAE. Separately, compared to HAETAE, NTRU+Sign benefits from a more moderate increase in the Hamming weight  $\tau$  of  $\mathbf{c}$  as the security parameter  $\lambda$  increases. Since  $\tau$  directly affects the upper bound on  $\|\mathbf{S}\mathbf{c}\|$ , this moderate increase is significant. In HAETAE,  $\mathbf{c}$  is a polynomial with the fixed degree of 256, and thus  $\tau$  is required to satisfy  $\binom{256}{\tau} \approx 2^\lambda$ . In NTRU+Sign, however,  $\tau$  is determined by  $\binom{n}{\tau} \approx 2^\lambda$ , where the polynomial degree  $n$  grows with  $\lambda$ . For example, HAETAE requires  $\tau = 80$  for  $\lambda = 180$ , while NTRU + Sign requires only  $\tau = 36$  for  $\lambda = 211$  when  $n = 1024$ .

Lastly, we provide two parameter sets: NTRU+Sign- $\{512, 1024\}$  based on the rings  $\mathbb{Z}_q[x]/(x^n + 1)$ , targeting 93 bits and 211 bits of security levels, respectively. Table 1 presents a comparison between Dilithium, HAETAE, and NTRU+Sign in terms of public key sizes, signature sizes, and performance evaluations. Even at higher security levels, Table 1 demonstrates that NTRU+Sign achieves the shortest signature sizes, compared to Dilithium and HAETAE. For instance, at NIST security level 3 (aiming at 192-bit classical security), the signature size of NTRU+Sign-1024 is approximately 55% smaller than Dilithium and 35% smaller than HAETAE. Furthermore, when considering the combined size of the signature and public key, NTRU+Sign-1024 is about 40% smaller than Dilithium and 15% smaller than HAETAE. All parameter sets for NTRU+Sign are designed to support efficient Number Theoretic Transform (NTT) operations in

Table 1: Comparison between Dilithium, HAETAE and NTRU+Sign

|                | Classical<br>Security | Sig <br>(bytes) | pk <br>(bytes) | Sig + pk <br>(bytes) | KeyGen<br>(K cycle) | Sign<br>(K cycle) | Verify<br>(K cycle) |
|----------------|-----------------------|-----------------|----------------|----------------------|---------------------|-------------------|---------------------|
| Dilithium-2    | 123                   | 2,420           | 1,312          | 3,732                | 342                 | 1,487             | 370                 |
| Dilithium-3    | 182                   | 3,293           | 1,952          | 5,245                | 589                 | 2,780             | 644                 |
| HAETAE-120     | 119                   | 1,474           | 992            | 2,466                | 2,109               | 9,204             | 417                 |
| HAETAE-180     | 180                   | 2,349           | 1,472          | 3,821                | 3,091               | 11,106            | 682                 |
| NTRU+Sign-512  | 93                    | 751             | 768            | 1,519                | 801                 | 1,689             | 73                  |
| NTRU+Sign-1024 | 211                   | 1,551           | 1,664          | 3,215                | 1,716               | 5,377             | 140                 |

their underlying rings. Our reference implementation, based on the open source<sup>1</sup> of GALACTICS, ensures constant-time implementations of Gaussian sampling and rejection sampling, mitigating potential vulnerabilities to side-channel attacks. The implementation demonstrates that NTRU+Sign-1024 outperforms HAETAE-180, achieving 1.7 times faster key generation and signing speeds, and verification speeds approximately 4.9 times faster. In terms of signature compactness, we give another comparison with other lattice-based signature schemes, including re-parametrized BLISS<sup>2</sup>, G + G [DPS23], Patronus [BBRS24] and Falcon [FHK<sup>+</sup>17] in Section 4.3. Our results highlight NTRU+Sign as a highly efficient and compact lattice-based signature scheme, establishing a new benchmark for post-quantum cryptographic applications. Its compactness and computational efficiency make it a strong candidate for real-world implementation in high-security environments.

## 1.2 Related Works

Lyubashevsky introduced the first efficient lattice-based signature scheme, using rejection sampling with uniform distribution in hypercubes [Lyu09]. Since then, extensive research has been conducted on lattice-based signatures under the FSWA framework, focusing on optimizing sampling methods to both achieve signature compactness and simplify rejection processes. Signature schemes with uniform sampling were refined in [GLP12, BG14], achieving reduced signature sizes. Further progress was achieved by Dilithium [DKL<sup>+</sup>18], which optimized public key sizes through a truncation technique. While uniform sampling over hypercubes simplifies rejection sampling by utilizing infinite norm bounds, maintaining low rejection rates often necessitates larger parameters, leading to increased public key and signature sizes.

Gaussian-based signature schemes have also been extensively studied for achieving more compact signatures. Following Lyubashevsky’s FSWA-based digital signature scheme [Lyu12], Ducas *et al.* presented Dilithium-G, a variation described in the first ePrint version of [DKL<sup>+</sup>18]. Dilithium-G employed a unimodal Gaussian distribution. This approach was later optimized by [DFPS22] through the analysis of imperfect rejection sampling using Rényi divergence instead of statistical distance. The adoption of bimodal Gaussian distributions in BLISS [DDLL13] significantly reduced signature sizes. However, the original security analysis of BLISS resulted in parameter sets that did not meet the claimed security levels under more rigorous analyzes.

<sup>1</sup><https://github.com/espitau/GALACTICS>

<sup>2</sup>The original BLISS paper employed a different security analysis, resulting in parameter sets that fail to meet the claimed security levels under the analysis presented in this paper. To achieve at least 128-bit security, we recalibrated the parameters, doubling the dimension to 1024 to ensure alignment with modern security standards. Additionally, we conducted an analysis of the rANS encoding process to estimate the signature size under the revised parameters. More details, including the full parameter table, can be found in Appendix A.

Devevey *et al.* [DFPS22] derived generic optimal bounds for two critical metrics: minimal rejection rate and proof-of-knowledge size compactness. Their results demonstrated that both Gaussian distributions and uniform distributions within hyperballs achieve optimal bounds in the bimodal setting. Using this observation, Cheon *et al.* [CCD<sup>+</sup>24] proposed HAETAE, a signature scheme based on a bimodal hyperball distribution. HAETAE simplifies rejection sampling to a straightforward Euclidean norm check, achieving compact signature sizes. However, its hyperball sampling still depends on discrete Gaussian sampling as a subroutine, which requires high-precision fixed-point arithmetic. This dependency, coupled with additional operations like inverse square root and  $\ell_2$  norm calculations, introduces notable computational overhead.

An alternative approach to prevent secret value leakage without relying on rejection sampling was introduced by G + G [DPS23]. This scheme generates signatures following a centered spherical Gaussian distribution, achieved through the convolution of two distinct Gaussian distributions, each with covariance dependent on the secret key. The primary advantage of G + G is its ability to generate signatures without requiring rejection sampling while ensuring no leakage of secret key information. However, as the scheme requires secret-dependent Gaussian sampling, its secure implementation can be non-trivial and may be vulnerable to side-channel attacks [Gä24]. Raccoon [dPKPR24], submitted to the first round of NIST’s additional post-quantum digital signature standardization process, also explored eliminating the rejection condition. In Raccoon, the parameters are carefully chosen so that rejection sampling can be securely omitted. This design choice, however, results in signatures that are approximately five times larger than those of Dilithium.

Recently, Bambury *et al.* [BBRS24] proposed a new sampler based on uniform sampling within polytopes, leading to a novel signature scheme Patronus. This approach maintains the simplicity of uniform sampling while reducing proof-of-knowledge size, thereby decreasing overall signature sizes. Furthermore, the rejection sampling process is simplified by relying solely on checks involving  $\ell_1$  norm, infinite norm, and  $\ell_2$  norm in optimized versions, eliminating the need for transcendental function computations. However, Patronus suffers from significant computational overhead due to the high cost of uniform sampling within polytopes, and its signature sizes still remain relatively large.

Gärtner [Gär25] proposed a compact lattice-based signature scheme that improves efficiency through finely tuned rejection probabilities combined with an iterative method. While this design achieves reduced signature sizes, the dependence of the rejection probability on secret values complicates constant-time implementation and might introduce potential side-channel leakage. Further research is needed to determine whether secure and practical constant-time implementations can be achieved.

## 2 Preliminaries

Let  $\{0, 1\}^*$  denote the set of all binary strings. Let  $\mathbb{Z}_q = \mathbb{Z}/q\mathbb{Z}$  denote the quotient ring of integers modulo  $q$ . Define  $R$  and  $R_q$  as the rings  $\mathbb{Z}[x]/(x^n + 1)$  and  $\mathbb{Z}_q[x]/(x^n + 1)$ , respectively, where  $q$  is a prime and  $n$  is a power of two. Elements in  $R_q$  are denoted in bold lowercase. Occasionally, elements of  $R_q$  that are either 0 or have all coefficients equal to 0 except for a nonzero constant term, such as 1 or  $\hat{q}$ , are written in lowercase rather than bold lowercase. Let  $R_{n,\tau}$  denote the set of elements in  $R_q$  that have all zero coefficients except for  $\tau$  out of  $n$  coefficients, which are 1. We have  $|R_{n,\tau}| = \binom{n}{\tau}$ . For a positive integer  $k$ , the centered binomial distribution (CBD), denoted by  $\psi_k$ , is a distribution over the integers  $\mathbb{Z}$ , which outputs  $\sum_{i=1}^k (b_i - b'_i)$  using uniformly and independently sampled bits  $b_i, b'_i$ . For  $\mathbf{f} \in R_q$ , we use the notation ‘ $\mathbf{f} \leftarrow \psi_k^n$ ’ to represent that each coefficient of  $\mathbf{f}$  is sampled from  $\psi_k$ . In general, for a probability distribution  $\mathcal{D}$ ,  $a \leftarrow \mathcal{D}$  denotes that  $a$  is sampled according to  $\mathcal{D}$ . For a set  $S$ ,  $a \leftarrow S$  denotes that  $a$  is sampled uniformly at random from  $S$ .

For any integer  $x$  within the range  $[0, q)$  and any positive integer  $d$ ,  $x$  can be uniquely written as  $x = [x]_d \cdot 2^d + [x \bmod 2^d]$ , where  $[x \bmod 2^d] \in [-2^{d-1}, 2^{d-1})$  and  $[x]_d = (x - [x \bmod 2^d])/2^d$  to drop

the  $d$  least significant bits. These functions can be extended to polynomials, i.e.,  $[\mathbf{f}]_d = \sum_{i=0}^{n-1} [f_i]_d x^i$  for  $\mathbf{f} = \sum_{i=0}^{n-1} f_i x^i$ . We define the infinite norm of a polynomial  $\mathbf{f}$  as  $\|\mathbf{f}\|_\infty = \max_{0 \leq i \leq n-1} |f_i|$ . We define  $\hat{q} = 2^{-1} \bmod q$  for an odd prime  $q$ , and  $p = (q-1)/2^d$  for an integer  $d$  satisfying  $q \equiv 1 \bmod 2^d$ . Specifically, given the modulus  $p$ , and an integer  $k \in \mathbb{Z}$ , define  $k' = k \bmod p$  as the unique element  $k'$  such that  $-p/2 \leq k' < p/2$ .

## 2.1 Discrete Gaussian Distribution

The Gaussian distribution with standard deviation  $\sigma \in \mathbb{R}$  and center  $c \in \mathbb{R}$  evaluated at  $x \in \mathbb{R}$  is defined by  $\rho_{c,\sigma}(x) = \exp(-(x-c)^2/(2\sigma^2))$ , and more generally by  $\rho_{\mathbf{c},\sigma}(\mathbf{x}) = \exp(-\|\mathbf{x} - \mathbf{c}\|^2/(2\sigma^2))$  for  $\mathbf{x}, \mathbf{c} \in \mathbb{R}^n$ . The discrete Gaussian distribution over  $\mathbb{Z}^n$  centered at  $\mathbf{c}$  is defined by  $D_{\mathbf{c},\sigma}^n(\mathbf{x}) = \rho_{\mathbf{c},\sigma}(\mathbf{x})/\rho_{\mathbf{c},\sigma}(\mathbb{Z}^n)$ . When the center  $\mathbf{c}$  is  $\mathbf{0}$  or  $n = 1$ , we generally omit it from the notation and simply write  $\rho_\sigma(\mathbf{x})$ ,  $D_\sigma^n(\mathbf{x})$ ,  $D_\sigma(x)$ . Identifying  $R$  with  $\mathbb{Z}^n$  via coefficient embedding,  $D_{\mathbf{c},\sigma}^n$  also denotes a probability distribution on  $R$ .

## 2.2 Cryptographic Definitions

We recall the definitions of an identification scheme and a signature scheme.

**Definition 2.1.** Let  $\mathcal{X}$  and  $\mathcal{Y}$  be two finite sets. A canonical identification scheme ID for an NP relation  $R \subseteq \mathcal{X} \times \mathcal{Y}$  is a 3-round interactive proof system between a prover P and a verifier V, with a commitment set  $\mathcal{W}$ , challenge set  $\mathcal{C}$ , and response set  $\mathcal{Z}$ . The prover holds a pair  $(x, y) \in R$ , while the verifier only has  $x$ , where the pair  $(x, y)$  is generated by a PPT algorithm Gen, called the instance generator. The event " $z = \perp$ " is called an abort, and its probability  $\beta$  is called the probability of aborting. The prover is written as  $P = (P_1, P_2)$ , and the verifier as  $V = (V_1, V_2)$ , with the following specifications:

- $P_1: (x, y) \rightarrow (w, st)$  is a PPT algorithm that takes as input a pair of strings in  $\mathcal{X} \times \mathcal{Y}$  and outputs a commitment  $w \in \mathcal{W}$  and a state  $st \in \{0, 1\}^*$ .
- $V_1: (x, w) \rightarrow c$  is a PPT algorithm that takes as input a string  $x$  and a commitment  $w$ , and outputs a challenge  $c \in \mathcal{C}$ .
- $P_2: (x, y, w, c, st) \rightarrow z$  is a PPT algorithm that takes as input a pair of strings  $(x, y)$ , a commitment  $w$ , a challenge  $c$ , and a state  $st$ , and outputs a response  $z \in \mathcal{Z} \cup \{\perp\}$  (we say that  $P_2$  aborts if it outputs  $\perp$ ).
- $V_2: (x, w, c, z) \rightarrow b \in \{0, 1\}$  is a deterministic polynomial-time algorithm that takes as inputs a string  $x$ , a commitment  $w$ , a challenge  $c$ , and a response  $z$ , and outputs a bit  $b$ , which represents acceptance (1) or rejection (0); in the case that  $z = \perp$ , it returns 0.

We then recall the two definitions of statistical properties of an identification scheme, which are perfect accepting honest-verifier zero-knowledge (paHVZK) and the min-entropy of the commitment.

**Definition 2.2** (Perfect Accepting Honest-verifier Zero-knowledge [BBD<sup>+</sup>23]). An identification scheme is said to be paHVZK if there exists a poly-time algorithm Sim that, when given the public key  $pk$ , outputs  $(w, c, z)$  with a distribution that is identical to the distribution of a transcript  $(w, c, z)$  produced by an honest execution of the protocol conditioned on  $z \neq \perp$ . We refer to this honest execution procedure as Trans.

**Definition 2.3** (Commitment Min-Entropy [KLS18]). For  $\alpha \geq 0$ , we say that an identification scheme  $((P_1, P_2), (V_1, V_2))$  with an instance generator Gen has commitment min-entropy  $\alpha$  if  $H_\infty[w|(w, st) \leftarrow P_1(x, y)] \geq \alpha$ , for all  $(x, y) \leftarrow \text{Gen}(1^\lambda)$ .

**Definition 2.4** (Computational Unique Response (CUR)). The advantage of an adversary  $\mathcal{A}$  against the CUR property of an identification scheme is given by the probability that  $\mathcal{A}$  outputs a commitment  $w$ , a challenge  $c$  and two separate responses  $z$  and  $z'$  such that the verifier accepts both  $(w, c, z)$  and  $(w, c, z')$  as valid transcripts.

**Definition 2.5.** A digital signature (DS) scheme for a message space  $\mathcal{M}$  consists of three algorithms: KeyGen, Sign, and Verify, which are defined as follows:

- $\text{KeyGen}(\lambda)$ : The key generation algorithm takes as input a security parameter  $\lambda$  and outputs a public key and a secret key  $(pk, sk)$ .
- $\text{Sign}(sk, \mu)$ : The signing algorithm takes as input the secret key  $sk$  and a message  $\mu \in \mathcal{M}$ , and then outputs a signature  $\sigma$ .
- $\text{Verify}(pk, \mu, \sigma)$ : The verification algorithm takes as input the public key  $pk$ , a message  $\mu$ , and a signature  $\sigma$ , and then outputs 1 if the signature is valid or 0 otherwise.

We say that a signature scheme is  $(1 - \eta)$ -correct if the following condition holds: for all  $(pk, sk) \in \text{KeyGen}(\lambda)$  and all messages  $\mu \in \mathcal{M}$ ,

$$\Pr [\text{Verify}(pk, \mu, \sigma) = 1 \mid (pk, sk) \leftarrow \text{KeyGen}(\lambda); \sigma \leftarrow \text{Sign}(sk, \mu)] > 1 - \eta(\lambda)$$

where  $\eta$  is a negligible function for the security parameter  $\lambda$ .

**Definition 2.6** (Unforgeability). Let  $\text{DS} = (\text{KeyGen}, \text{Sign}, \text{Verify})$  be a signature scheme. The unforgeability against chosen-message attacks (UF-CMA) is defined via the following experiment  $\text{UF-CMA}_{\text{DS}}^{\mathcal{A}}(\lambda)$  between a challenger  $\mathcal{C}$  and an adversary  $\mathcal{A}$ :

1.  $\mathcal{C}$  runs  $(pk, sk) \leftarrow \text{KeyGen}(\lambda)$  and gives  $pk$  to  $\mathcal{A}$ .
2.  $\mathcal{A}$  queries the signing oracle  $\text{Sign}(sk, \mu)$  with a message  $\mu$ .
3. Finally,  $\mathcal{A}$  outputs a signature  $\sigma^*$  and a message  $\mu^*$  that was not previously queried on the signing oracle.  $\mathcal{C}$  returns 1 if  $\text{Verify}(pk, \mu^*, \sigma^*) = 1$ , and otherwise returns 0 as the output of the game.

The advantage of  $\mathcal{A}$  in breaking the UF-CMA security of DS is defined as  $\text{Adv}_{\text{DS}}^{\text{UF-CMA}}(\mathcal{A}) = \Pr[\text{UF-CMA}_{\text{DS}}^{\mathcal{A}} \Rightarrow 1]$ . We say that a signature scheme is UF-CMA secure if, for any polynomial-time adversary  $\mathcal{A}$ , we have  $\text{Adv}_{\text{DS}}^{\text{UF-CMA}}(\mathcal{A}) \leq \epsilon(\lambda)$ , where  $\epsilon$  is a function of the security parameter  $\lambda$ .

The unforgeability against no-message attack, denoted by UF-NMA is defined similarly except that the adversary is not allowed to query the signing oracle at all. Moreover, Strong existential unforgeability against chosen-message attacks (sUF-CMA) allows the adversary to obtain signatures on its target message, as long as it does not output an identical signature it previously obtained.

**Lemma 2.7** (Rejection Sampling [DDLL13]). Let  $V$  be an arbitrary set, and let  $h : V \rightarrow \mathbb{R}$  and  $f : \mathbb{Z}^m \rightarrow \mathbb{R}$  be probability distributions. If  $g_v : \mathbb{Z}^m \rightarrow \mathbb{R}$  is a family of probability distributions indexed by  $v \in V$  with the property that there exists a constant  $M \in \mathbb{R}$  such that

$$\forall v \in V, \forall z \in \mathbb{Z}^m, \Pr[M \cdot g_v(z) \geq f(z) \mid z \leftarrow f] \geq 1 - \epsilon,$$

then the output distributions of the following two algorithms are within a statistical distance of  $\epsilon/M$ :

1.  $v \leftarrow h, z \leftarrow g_v$ , output  $(z, v)$  with probability  $f(z)/(M \cdot g_v(z))$ .
2.  $v \leftarrow h, z \leftarrow f$ , output  $(z, v)$  with probability  $1/M$ .

## 2.3 Hardness Assumptions

The security of our construction is based on the hardness of two lattice problems, namely the NTRU problem and the BimodalSelfTargetRSIS problem.

**Definition 2.8** (NTRU $_{n,q,\psi}$ ). Let  $\psi$  be a distribution over  $R_q$  and let  $\hat{q} = 2^{-1} \bmod q$ . An adversary  $\mathcal{A}$  solving the NTRU $_{n,q,\psi}$  problem has advantage

$$\text{Adv}_{n,q,\psi}^{\text{NTRU}}(\mathcal{A}) := \left| \frac{\Pr[b = 1 \mid \mathbf{u} \leftarrow R_q; b \leftarrow \mathcal{A}(\mathbf{u})] - \Pr[b = 1 \mid \mathbf{f}, \mathbf{g} \leftarrow \psi; b \leftarrow \mathcal{A}((\mathbf{f} + \hat{q})/\mathbf{g})]}{\Pr[b = 1 \mid \mathbf{f}, \mathbf{g} \leftarrow \psi; b \leftarrow \mathcal{A}((\mathbf{f} + \hat{q})/\mathbf{g})]} \right|.$$

This variant of the NTRU problem is closely related to the more general assumption that  $\mathbf{h} = \mathbf{g}/\mathbf{f}$  (or  $\mathbf{h} = \mathbf{g}/(p\mathbf{f} + 1)$  for a small prime  $p$ ) is indistinguishable from a uniformly random element in  $R_q$  [DHK<sup>+</sup>23]. In our case, setting  $\mathbf{h} = (\mathbf{f} + \hat{q})/\mathbf{g}$  implies that  $2\mathbf{h} = (2\mathbf{f} + 1)/\mathbf{g}$ . Recent results [DHK<sup>+</sup>23, ZFY23] show that the inverse of  $2\mathbf{h}$ , that is,  $\mathbf{g}/(2\mathbf{f} + 1)$ , is also indistinguishable from a random element in  $R_q$ . Thus, it can be inferred that  $2\mathbf{h} = (2\mathbf{f} + 1)/\mathbf{g}$  is also indistinguishable from a random element in  $R_q$ , and so is  $\mathbf{h}$ , because 2 is coprime to  $q$ .

The hardness of this assumption is further supported by concrete security estimates based on known lattice attacks on NTRU, particularly primal and hybrid approaches. For our analysis, we use the NTRU estimator provided by [CDH<sup>+</sup>20], which is based on these attack strategies. A more detailed discussion of the attacks and parameter selection is given in Section 4.2.

Before introducing a new lattice-based problem, we define the ring version of Short Integer Solution problem (RSIS) first.

**Definition 2.9** (RSIS $_{n,q,\beta}$ ). Let  $\beta > 0$  and  $q$  be a positive modulus. An adversary  $\mathcal{A}$  solving the RSIS $_{n,q,\beta}$  problem has advantage

$$\text{Adv}_{n,q,\beta}^{\text{RSIS}}(\mathcal{A}) := \Pr[[1 \mid \mathbf{a}] \cdot \mathbf{y} = 0 \wedge 0 < \|\mathbf{y}\| \leq \beta \mid \mathbf{a} \leftarrow R_q; \mathbf{y} \in R_q^2 \leftarrow \mathcal{A}(\mathbf{a})].$$

Next, we introduce a new lattice-based problem similar to the SelfTargetMSIS problem [KLS18], which allows us to directly prove the UF-NMA security of our signature scheme in the Quantum Random Oracle Model (QROM).

**Definition 2.10** (BimodalSelfTargetRSIS $_{H,n,q,\beta}$ ). Let  $H : \{0, 1\}^* \rightarrow R_{n,\tau}$  be a cryptographic hash function, and let  $q > 0$  be an odd modulus. An adversary  $\mathcal{A}$  solving the BimodalSelfTargetRSIS $_{H,n,q,\beta}$  problem has advantage

$$\text{Adv}_{H,n,q,\beta}^{\text{BimodalSelfTargetRSIS}}(\mathcal{A}) := \Pr \left[ \begin{array}{l} \|\mathbf{Y}\| < \beta \wedge \|\mathbf{Y}\|_\infty < (q-2)/4 \\ \wedge H([1 \mid \mathbf{a}] \cdot \mathbf{Y} + \mathbf{c}\hat{q}, \mu) = \mathbf{c} \end{array} \mid \mathbf{a} \leftarrow R_q; \left( \mathbf{Y} := \begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \end{bmatrix}, \mathbf{c}, \mu \right) \leftarrow \mathcal{A}^{(H)}(\mathbf{a}) \right]$$

where  $\hat{q} = 2^{-1} \bmod q$ ,  $(\mathbf{y}_1, \mathbf{y}_2) \in R_q^2$ ,  $\mathbf{c} \in R_{n,\tau}$ , and  $\mu \in \{0, 1\}^*$ .

Now, we will show that the BimodalSelfTargetRSIS problem is reduced to the RSIS, assuming that  $H$  is a random oracle. If  $\mathcal{A}$  only has classical access to  $H$ , then there is a reduction, using the forking lemma [BN06], to prove that  $\text{Adv}_{H,n,q,\beta}^{\text{BimodalSelfTargetRSIS}}(\mathcal{A}) \approx \sqrt{\text{Adv}_{n,q,4\beta+2\sqrt{\tau}}^{\text{RSIS}}(\mathcal{B})/Q_h}$ , where  $Q_h$  is the number of classical queries to  $H$ . We give its proof sketch below.

Assume that  $\mathcal{A}$  is a solver for the BimodalSelfTargetRSIS $_{H,n,q,\beta}$  problem. Then,  $\mathcal{B}$  passes the polynomial  $\mathbf{a}$  from its RSIS $_{n,q,4\beta+2\sqrt{\tau}}$  instance to  $\mathcal{A}$  and replies to  $\mathcal{A}$ 's queries  $H(\mathbf{w}, \mu)$  with a uniformly random



$\mathbf{c} \in R_{n,\tau}$ . If  $\mathcal{A}$  returns a solution  $(\mathbf{Y} = [\mathbf{y}_1 | \mathbf{y}_2]^T, \mathbf{c}, \mu)$  to  $\text{BiModalSelfTargetRSIS}_{H,n,q,\beta}$ , then  $\mathcal{B}$  reprograms the "winning" query  $H([1|\mathbf{a}] \cdot \mathbf{Y} + \mathbf{c}\hat{q}, \mu)$  to a different random element  $\mathbf{c}'$ . The forking lemma states that  $\mathcal{A}$  outputs another solution  $(\mathbf{Y}' = [\mathbf{y}'_1 | \mathbf{y}'_2]^T, \mathbf{c}', \mu)$  with probability  $1/Q_h$ . Now, we have

$$\begin{cases} [1 | \mathbf{a}] \cdot \mathbf{Y} + \mathbf{c}\hat{q} = \mathbf{w} = [1 | \mathbf{a}] \cdot \mathbf{Y}' + \mathbf{c}'\hat{q} \bmod q \\ \|\mathbf{Y}\| < \beta, \|\mathbf{Y}'\| < \beta, \|\mathbf{Y}\|_\infty < (q-2)/4, \|\mathbf{Y}'\|_\infty < (q-2)/4. \end{cases}$$

Thus, we obtain  $[1 | \mathbf{a}] \cdot (\mathbf{Y} - \mathbf{Y}') + (\mathbf{c} - \mathbf{c}')\hat{q} = 0 \bmod q$ . Since  $\hat{q}$  satisfies  $2\hat{q} = 1 \bmod q$  by definition, we derive  $2[1 | \mathbf{a}] \cdot (\mathbf{Y} - \mathbf{Y}') + \mathbf{c} - \mathbf{c}' = 0 \bmod q$ , following that  $[1 | \mathbf{a}] \cdot \begin{bmatrix} 2\mathbf{y}_1 - 2\mathbf{y}'_1 + \mathbf{c} - \mathbf{c}' \\ 2\mathbf{y}_2 - 2\mathbf{y}'_2 \end{bmatrix} = 0 \bmod q$ .

The condition  $\mathbf{c} \neq \mathbf{c}' \bmod 2$  implies that  $2\mathbf{y}_1 - 2\mathbf{y}'_1 + \mathbf{c} - \mathbf{c}'$  is non-zero over  $R$ . Since  $\|\mathbf{Y}\|_\infty, \|\mathbf{Y}'\|_\infty < (q-2)/4$ , we obtain  $\|2\mathbf{y}_1 - 2\mathbf{y}'_1 + \mathbf{c} - \mathbf{c}'\|_\infty < q$ . Therefore,  $2\mathbf{y}_1 - 2\mathbf{y}'_1 + \mathbf{c} - \mathbf{c}' \neq 0$ . Moreover, we have  $0 < \|(2\mathbf{y}_1 - 2\mathbf{y}'_1 + \mathbf{c} - \mathbf{c}', 2\mathbf{y}_2 - 2\mathbf{y}'_2)\| \leq \|(2\mathbf{y}_1 - 2\mathbf{y}'_1, 2\mathbf{y}_2 - 2\mathbf{y}'_2)\| + \|(\mathbf{c} - \mathbf{c}', 0)\| \leq \|2\mathbf{Y} - 2\mathbf{Y}'\| + 2\sqrt{\tau} < 4\beta + 2\sqrt{\tau}$ .

This provides an  $\text{Adv}_{n,q,4\beta+2\sqrt{\tau}}^{\text{RSIS}}$  solution for  $[1 | \mathbf{a}]$ , which is a polynomial vector  $[2\mathbf{y}_1 - 2\mathbf{y}'_1 + \mathbf{c} - \mathbf{c}' | 2\mathbf{y}_2 - 2\mathbf{y}'_2]^T$ .

### 3 Our NTRU+Sign Signature Scheme

#### 3.1 Construction

We present the NTRU+Sign scheme in Figure 1.

#### 3.2 Correctness

The Verify algorithm will accept the signature  $(\mathbf{z}_1, \mathbf{h}, \mathbf{c})$  for a message  $\mu \in \{0, 1\}^*$  to be signed if the following three conditions hold:

1.  $\|(\mathbf{z}_1, 2^d \mathbf{h})\| \leq B_2$ ,
2.  $\|(\mathbf{z}_1, 2^d \mathbf{h})\|_\infty \leq B_\infty$ ,
3.  $\mathbf{c} = H([\mathbf{a}\mathbf{z}_1 + \mathbf{c}\hat{q}]_d + \mathbf{h} \bmod p, \mu)$ .

The first two conditions are guaranteed, because these conditions are already checked by the Sign algorithm in the same manner. The third one is also guaranteed by the following equations:

$$\begin{aligned} \mathbf{a}\mathbf{z}_1 + \mathbf{c}\hat{q} &= \mathbf{a}(\mathbf{y}_1 + (-1)^b \mathbf{s}_1 \mathbf{c}) + (\mathbf{y}_2 + (-1)^b \mathbf{s}_2 \mathbf{c}) + \mathbf{c}\hat{q} - \mathbf{z}_2 \bmod q \\ &= \mathbf{u} + (-1)^b \mathbf{c}(\mathbf{a}\mathbf{s}_1 + \mathbf{s}_2) + \mathbf{c}\hat{q} - \mathbf{z}_2 \bmod q & (\because \mathbf{u} = \mathbf{a}\mathbf{y}_1 + \mathbf{y}_2 \bmod q) \\ &= \mathbf{u} + (-1)^b \mathbf{c}\hat{q} + \mathbf{c}\hat{q} - \mathbf{z}_2 \bmod q & (\because \mathbf{a}\mathbf{s}_1 + \mathbf{s}_2 = \hat{q} \bmod q) \\ &= \mathbf{u} - \mathbf{z}_2 + (1 - b)\mathbf{c} \bmod q. \end{aligned}$$

The last equality can be verified by the fact that: if  $b = 0$ ,  $(-1)^b \hat{q} + \hat{q}$  becomes  $2\hat{q} = 1$  in  $R_q$ , and if  $b = 1$ ,  $(-1)^b \hat{q} + \hat{q}$  becomes 0 in  $R_q$ , which is equal to  $1 - b$  for the same bit  $b \in \{0, 1\}$ . Now, we can consider the

KeyGen( $1^\lambda$ )

- 1:  $\mathbf{f}, \mathbf{g} \leftarrow \psi_1^n$
- 2: **if**  $\mathbf{g}$  is not invertible in  $R_q$ , **then** restart
- 3:  $\mathbf{S} := (\mathbf{g}, -\mathbf{f})$
- 4: **if**  $\mathcal{N}(\mathbf{S}) > \gamma^2 n$ , **then** restart
- 5:  $\mathbf{a} := (\mathbf{f} + \hat{q})/\mathbf{g} \in R_q$  //  $\hat{q} = 2^{-1} \bmod q$
- 6:  $pk := \mathbf{a}$
- 7:  $sk := \mathbf{S} = (\mathbf{s}_1, \mathbf{s}_2)$
- 8: **return**  $(pk, sk)$

Sign( $sk = \mathbf{S}, \mu$ )

- 1:  $\mathbf{y} = (\mathbf{y}_1, \mathbf{y}_2) \leftarrow \mathcal{D}_\sigma^n \times \mathcal{D}_\sigma^n$
- 2:  $\mathbf{u} := \mathbf{a}\mathbf{y}_1 + \mathbf{y}_2 \in R_q$
- 3:  $\mathbf{c} := H([\mathbf{u}]_d \bmod p, \mu) \in R_{n,\tau}$  //  $p = (q - 1)/2^d$
- 4: Choose a random bit  $b \leftarrow \{0, 1\}$
- 5:  $\mathbf{z} = (\mathbf{z}_1, \mathbf{z}_2) = \mathbf{y} + (-1)^b \mathbf{S}\mathbf{c}$  //  $\mathbf{z}_i = \mathbf{y}_i + (-1)^b \mathbf{s}_i \mathbf{c}$
- 6: **Continue** with probability  $1/(M \exp(-\frac{\|\mathbf{S}\mathbf{c}\|^2}{2\sigma^2}) \cosh(\frac{\langle \mathbf{z}, \mathbf{S}\mathbf{c} \rangle}{\sigma^2}))$  // rejection sampling
- 7: **otherwise** restart
- 8: **if**  $[\mathbf{u}]_d \neq [\mathbf{u} + (-1)^b \mathbf{c}]_d$ , **then** restart // equality check
- 9:  $\mathbf{h} := [\mathbf{u}]_d - [\mathbf{u} - \mathbf{z}_2 + (1 - b)\mathbf{c}]_d \bmod p$
- 10: **if**  $\|(\mathbf{z}_1, 2^d \mathbf{h})\| > B_2$ , **then** restart
- 11: **if**  $\|(\mathbf{z}_1, 2^d \mathbf{h})\|_\infty > B_\infty$ , **then** restart
- 12: **return**  $(\mathbf{z}_1, \mathbf{h}, \mathbf{c})$

Verify( $pk, \mu, (\mathbf{z}_1, \mathbf{h}, \mathbf{c})$ )

- 1: **if**  $\|(\mathbf{z}_1, 2^d \mathbf{h})\| > B_2$  **then** Reject
- 2: **if**  $\|(\mathbf{z}_1, 2^d \mathbf{h})\|_\infty > B_\infty$  **then** Reject
- 3: Accept if  $H([\mathbf{a}\mathbf{z}_1 + \mathbf{c}\hat{q} \bmod q]_d + \mathbf{h} \bmod p, \mu) = \mathbf{c}$

Figure 1: Description of NTRU+Sign

hint  $\mathbf{h}$  as being  $\mathbf{h} = [\mathbf{u}]_d - [\mathbf{u} - \mathbf{z}_2 + (1 - b)\mathbf{c}]_d \bmod p = [\mathbf{u}]_d - [\mathbf{a}\mathbf{z}_1 + \mathbf{c}\hat{q}]_d \bmod p$ , using the above last equality. Then, it holds that

$$\begin{aligned} [\mathbf{a}\mathbf{z}_1 + \mathbf{c}\hat{q}]_d + \mathbf{h} &= [\mathbf{a}\mathbf{z}_1 + \mathbf{c}\hat{q}]_d + [\mathbf{u}]_d - [\mathbf{a}\mathbf{z}_1 + \mathbf{c}\hat{q}]_d \bmod p \\ &= [\mathbf{u}]_d \bmod p, \end{aligned}$$

as required in line 3 of the Verify algorithm.

### 3.3 Equality Check

Unlike BLISS, the Sign algorithm of NTRU+Sign includes an additional equality check in line 8, which checks whether or not  $[\mathbf{u}]_d = [\mathbf{u} + (-1)^b \mathbf{c}]_d$ . The purpose of this equality check is to conceal the information of the bit  $b$  from the hint  $\mathbf{h}$ , ensuring that the distribution of  $\mathbf{h}$  remains the same regardless of the value of  $b$ . More concretely,  $\mathbf{h}$  is calculated as  $\mathbf{h} = [\mathbf{u}]_d - [\mathbf{u} - \mathbf{z}_2 + (1 - b)\mathbf{c}]_d \bmod p$ ; when  $b = 0$ ,  $\mathbf{h} = [\mathbf{u}]_d - [\mathbf{u} - \mathbf{z}_2 + \mathbf{c}]_d \bmod p$ , and when  $b = 1$ ,  $\mathbf{h} = [\mathbf{u}]_d - [\mathbf{u} - \mathbf{z}_2]_d \bmod p$ . This creates a difference in the distributions of  $\mathbf{h}$  based on the choice of  $b$ . However, if the condition  $[\mathbf{u}]_d = [\mathbf{u} + \mathbf{c}]_d$  is met for  $b = 0$ , then it holds that  $\mathbf{h} = [\mathbf{u}]_d - [\mathbf{u} - \mathbf{z}_2 + \mathbf{c}]_d = [\mathbf{u} + \mathbf{c}]_d - [\mathbf{u} + \mathbf{c} - \mathbf{z}_2]_d$ , which has the same distribution as in the case when  $b = 1$ , by setting  $\mathbf{u}' = \mathbf{u} + \mathbf{c}$ .

Technically, the equality check increases the expected number of the repetition rate  $M$  (heuristically) by  $(2^d / (2^d - 1))^\tau$  times, where  $(2^d - 1)/2^d$  is the probability that the equality holds for each coefficient of  $\mathbf{u}$ <sup>3</sup>. Since  $\mathbf{c} \in R_{n,\tau}$ , only  $\tau$  coefficients are considered, resulting in a probability of  $((2^d - 1)/2^d)^\tau$  for passing the equality check. Indeed, this probability does not significantly affect the repetition rate. For example, at the (classical) 211-bit security level (in case of NTRU+Sign-1024), with parameters  $d = 8$  and  $\tau = 36$ , the probability is approximately 0.86, leading to an increase in the repetition rate  $M$  by a factor of about 1.15.

### 3.4 Rejection Sampling

The rejection sampling of NTRU+Sign, detailed in line 6 of the Sign algorithm, follows the approach used in BLISS [DDLL13]. For the discrete Gaussian distribution  $D_\sigma^{2n}$  with standard deviation  $\sigma$ , the source distribution is  $g_{\mathbf{Sc}} = \frac{1}{2}D_{\mathbf{Sc},\sigma}^{2n} + \frac{1}{2}D_{-\mathbf{Sc},\sigma}^{2n}$  where  $\mathbf{Sc}$  is computed in line 5 of the Sign algorithm over the space of all  $(b, \mathbf{y})$  where  $b \leftarrow \{0, 1\}$  and  $\mathbf{y} \leftarrow D_\sigma^{2n}$ , whereas the target distribution is the centered discrete Gaussian distribution  $D_\sigma^{2n}$ , that is,  $f = D_\sigma^{2n}$ . Then, Lemma 2.7 shows that, for a fixed  $\mathbf{Sc}$ , the computed  $\mathbf{z}$  in line 5 of the Sign algorithm should be accepted with probability:

$$P_{\mathbf{z}} = \frac{f(\mathbf{z})}{Mg_{\mathbf{Sc}}(\mathbf{z})} = 1 / \left( M \exp \left( - \frac{\|\mathbf{Sc}\|^2}{2\sigma^2} \right) \cosh \left( \frac{\langle \mathbf{z}, \mathbf{Sc} \rangle}{\sigma^2} \right) \right),$$

where  $M$  is a predetermined positive real constant such that, for all possible values of  $\mathbf{Sc}$ , the inequality  $f \leq Mg_{\mathbf{Sc}}$  holds with probability 1 (i.e.,  $\epsilon = 0$  in Lemma 2.7). Indeed,  $M$  is the number of repetition rate, more specifically, repeating from line 1 to 6 in the Sign algorithm. According to [DDLL13], it suffices to set  $M = \exp(1/2\hat{\alpha}^2)$ , where  $\hat{\alpha} \leq \sigma / \|\mathbf{Sc}\|$ . To set an appropriate value  $M$ , [DDLL13] (and also NTRU+Sign) first selects an appropriate value  $\hat{\alpha}$  that makes  $M$  acceptable, for instance,  $M = 3.25$  or  $M = 4.28$ , and next sets  $\sigma$  to be  $\hat{\alpha} \cdot \|\mathbf{Sc}\|$ , using the equality of  $\sigma = \hat{\alpha} \cdot \|\mathbf{Sc}\|$ . This indicates that, for a fixed  $\hat{\alpha}$  (and thus  $M$ ), reducing the bound on  $\|\mathbf{Sc}\|$  lowers  $\sigma$ , leading to a more compact signature.

<sup>3</sup>It is assumed that the low  $d$  bits of  $\mathbf{u}$  is uniformly distributed.

### 3.5 Secret Key Constraint

To bound on  $\|\mathbf{Sc}\|$  tightly, we follow the method of HAETAE [CCD<sup>+</sup>24] based on the *canonical embedding* into  $\mathbb{C}^n$ , which is a classical concept from algebraic number theory. In the canonical embedding, both addition and multiplication in  $R_q$  correspond to their coordinate-wise counterparts in  $\mathbb{C}^n$ , yielding tight bounds on geometric quantities such as Euclidean norms and inner products [LPR13]. Under the canonical embedding,  $\|\mathbf{Sc}\|^2$  can be represented as  $\|\mathbf{Sc}\|^2 = (\sum_{j=1}^n \|\mathbf{S}(w_j)\|^2 \cdot \|\mathbf{c}(w_j)\|^2)/n$ , where  $w_j$  is the  $2n$ -th primitive root of unity for  $1 \leq j \leq n$ . To achieve a tight bound, [CCD<sup>+</sup>24] uses  $m$ -largest values of  $\|\mathbf{S}(w_j)\|^2$  where  $m = \lfloor n/\tau \rfloor$ , rather than  $\max_j (\|\mathbf{S}(w_j)\|^2)$ . Additionally, to ensure that the bound holds regardless of the specific choice of  $\mathbf{c}$  (i.e., to eliminate the terms related to  $\mathbf{c}$ ), the right-hand side of the equation is bounded from above in a worst-case manner.

**Lemma 3.1** ([CCD<sup>+</sup>24]). For any  $\mathbf{c} \in \{0, 1\}^n$  with Hamming weight  $\tau$  and a secret  $\mathbf{S} \in R$ , the value  $\|\mathbf{Sc}\|^2$  is upper bounded by

$$\frac{\tau}{n} \cdot \mathcal{N}(\mathbf{S}) = \frac{\tau}{n} \left( \sum_{i=0}^m \max_{0 \leq j < n}^{i\text{-th}} \|\mathbf{S}(w_j)\|^2 + r \cdot \max_{0 \leq j < n}^{(m+1)\text{-th}} \|\mathbf{S}(w_j)\|^2 \right)$$

where  $m = \lfloor n/\tau \rfloor$ ,  $r = n \bmod \tau$ , and  $w_j$ 's are the  $2n$ -th primitive roots of unity.

Using this lemma, we obtain a direct upper-bound on  $\|\mathbf{Sc}\|$  by simply fixing  $\mathcal{N}(\mathbf{S}) \leq \gamma^2 n$  in line 4 of the KeyGen algorithm for some positive real number  $\gamma$ .

### 3.6 Security Proof

**Theorem 3.2** (UF-CMA Security [BBD<sup>+</sup>23]). Assume that  $\text{NTRU+ident} = ((P_1, P_2), (V_1, V_2))$  is a paHVZK public-coin identification protocol with probability of aborting  $\beta$  and that the commitment message of the prover has min-entropy  $\alpha$ , described in Figure 2. Let  $\mathcal{A}$  be any arbitrary adversary against UF-CMA security of  $\text{DS} = \text{FS}[\text{NTRU+ident}, H]$  that issues at most  $Q_h$  queries to the random oracle  $H$  and  $Q_s$  classical queries to the signing oracle. Then there exists an adversary  $\mathcal{B}$  against UF-NMA security of  $\text{DS}$  with running time  $\text{Time}(\mathcal{B}) \approx \text{Time}(\mathcal{A}) + Q_s T_{\text{Sim}}$  such that:

$$\begin{aligned} \text{Adv}_{\text{DS}}^{\text{UF-CMA}}(\mathcal{A}) &\leq \text{Adv}_{\text{DS}}^{\text{UF-NMA}}(\mathcal{B}) + \frac{2^{-\alpha/2+1} Q_s}{1-\beta} \sqrt{Q_h + 1 + \frac{Q_s}{1-\beta}} \\ &\quad + 2^{-\alpha/2+1} (Q_h + 1) \sqrt{\frac{Q_s}{1-\beta}} + Q_s \epsilon_{zk}. \end{aligned}$$

Moreover, if  $\mathcal{A}$  is an adversary against the sUF-CMA security of  $\text{DS}$ , then there exist an additional adversaries  $\mathcal{C}$  against the CUR property of identification such that the above bound holds with the following additional term  $\text{Adv}_{\text{ID}}^{\text{CUR}}$ .

We recall that an identification scheme  $\text{ID}$  can be transformed into a digital signature via the Fiat-Shamir transform, and let  $\text{FS}[\text{ID}, H]$  denote the resulting signature scheme (see [DPS23] for more details). Based on the analysis of [DFPS23], which reduces UF-CMA security to UF-NMA security, we need to show that the commitment min-entropy  $\alpha$  is high and the underlying identification scheme is paHVZK. The underlying identification, with repetition parameter  $M \geq 1$ , Euclidean norm bound  $B_2$  and infinite norm bound  $B_\infty$  is given in Figure 2.

|                                                                                 |                                                                                                         |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| $P_1(sk = (s_1, s_2))$                                                          | $P_2(sk = S, w, c, st)$                                                                                 |
| 1: $y = (y_1, y_2) \leftarrow \mathcal{D}_\sigma^n \times \mathcal{D}_\sigma^n$ | 1: $b \leftarrow U(\{0, 1\})$                                                                           |
| 2: $u = ay_1 + y_2$                                                             | 2: $z := (z_1, z_2) = y + (-1)^b Sc$                                                                    |
| 3: $w = [u]_d$                                                                  | 3: With prob. $1/(M \exp(-\frac{\ Sc\ ^2}{2\sigma^2}) \cosh(\frac{\langle z, Sc \rangle}{\sigma^2}))$ : |
| 4: <b>return</b> $(W = w, st = (u, y))$                                         | 4: <b>if</b> $[u]_d = [u + (-1)^b c]_d$                                                                 |
| $V_1(pk = a, w)$                                                                | 5: $h := [u]_d - [u - z_2 + (1 - b)c]_d \bmod p$                                                        |
| 1: $c \leftarrow U(\mathcal{C})$                                                | 6: <b>if</b> $\ (z_1, 2^d h)\  \leq B_2$                                                                |
| 2: <b>return</b> $c$                                                            | 7: <b>if</b> $\ (z_1, 2^d h)\ _\infty \leq B_\infty$                                                    |
|                                                                                 | 8: <b>return</b> $(z_1, h)$                                                                             |
|                                                                                 | 9: <b>return</b> $\perp$                                                                                |
|                                                                                 | $V_2(pk, (w, c, z_1, h))$                                                                               |
|                                                                                 | 1: <b>if</b> $\ (z_1, 2^d h)\  \leq B_2, \ (z_1, 2^d h)\ _\infty \leq B_\infty$                         |
|                                                                                 | 2: <b>if</b> $[az_1 + c\hat{q}]_d + h = w \bmod p$                                                      |
|                                                                                 | 3: <b>return</b> <b>Accept</b>                                                                          |
|                                                                                 | 4: <b>return</b> <b>Reject</b>                                                                          |

Figure 2: NTRU+ident protocol

### 3.6.1 paHVZK

In this section, we show that the underlying NTRU+ident scheme from Figure 2 satisfies the paHVZK property in the non-aborting case. Specifically, we demonstrate that the statistical distance  $\epsilon_{zk}$  between the output  $(w, c, z) = (w, c, (z_1, h))$  distributions of Trans and Sim from Figure 3 is 0, following the approach used in previous works such as [DKL<sup>+</sup>18, CCD<sup>+</sup>24, BBR24].

First, it is straightforward to observe that  $z_1$  in the Trans algorithm and  $z_1$  in the Sim algorithm are statistically identical, based on Lemma 2.7 and the analysis in Chapter 3.4. Next, we show that the distribution of the commitment  $w$  in the Trans algorithm and the distribution of  $w$  in the Sim algorithm are identical. To achieve this, we show that the condition in line 7 in the Trans algorithm corresponds to the condition in line 5 in the Sim algorithm from Figure 3. Recall that

$$az_1 + z_2 + c\hat{q} = \begin{cases} u + c & (\text{if } b = 0) \\ u & (\text{if } b = 1) \end{cases} \quad (1)$$

where  $u = ay_1 + y_2$ . Notably, checking if  $[az_1 + z_2 + c\hat{q} - c]_d = [az_1 + z_2 + c\hat{q}]_d \bmod p$  is equivalent to checking if  $[u]_d = [u + c]_d \bmod p$  in case that  $b = 0$  and  $[u - c]_d = [u]_d \bmod p$  in case that  $b = 1$ . In other words, the equality check simplifies to checking whether  $[u]_d = [u + (-1)^b c]_d \bmod p$ . Therefore, through this equality check, we guarantee that  $w$  in the Trans algorithm and  $w$  in Sim algorithm are identical, as it always holds that  $[u]_d = [az_1 + z_2 + c\hat{q}]_d \bmod p$  for both cases that  $b = 0$  or 1.

Lastly, we need to prove that the distribution of  $h$  in the Trans algorithm is identical to that in the Sim. To do so, we show that the hint  $h$  in line 8 of the Trans algorithm corresponds to the hint  $h$  in line 5 of the Sim algorithm (as shown in Figure 3). For  $b = 1$ , we observe that  $[az_1 + z_2 + c\hat{q}]_d - [az_1 + c\hat{q}]_d = [u]_d - [u - z_2]_d \bmod p$  by equation (1). For  $b = 0$ , it follows that  $[az_1 + z_2 + c\hat{q}]_d - [az_1 + c\hat{q}]_d = [u + c]_d - [u - z_2 + c]_d \bmod p$  again by equation (1). Given the equality  $[u]_d = [u + c]_d \bmod p$  when  $b = 0$ , the latter expression can be rewritten as  $[u]_d - [u - z_2 + c]_d \bmod p$ . In summary, we conclude

| $\text{Trans}(sk = (\mathbf{s}_1, \mathbf{s}_2), \mathbf{c})$                                                                                        | $\text{Sim}(pk, \mathbf{c})$                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 1: $\mathbf{y} = (\mathbf{y}_1, \mathbf{y}_2) \leftarrow \mathcal{D}_\sigma^n \times \mathcal{D}_\sigma^n$                                           | 1: With probability $1/M$ :                                                 |
| 2: $\mathbf{u} = \mathbf{a}\mathbf{y}_1 + \mathbf{y}_2$                                                                                              | 2: $\mathbf{z}_1, \mathbf{z}_2 \leftarrow \mathcal{D}_\sigma^n$             |
| 3: $\mathbf{w} = [\mathbf{u}]_d$                                                                                                                     | 3: $\mathbf{v} = \mathbf{a}\mathbf{z}_1 + \mathbf{z}_2 + \mathbf{c}\hat{q}$ |
| 4: $b \leftarrow \{0, 1\}$                                                                                                                           | 4: <b>if</b> $[\mathbf{v}]_d = [\mathbf{v} - \mathbf{c}]_d$                 |
| 5: $\mathbf{z} := (\mathbf{z}_1, \mathbf{z}_2) = \mathbf{y} + (-1)^b \mathbf{S}\mathbf{c}$                                                           | 5: $\mathbf{h} = [\mathbf{v}]_d - [\mathbf{v} - \mathbf{z}_2]_d \bmod p$    |
| 6: With prob. $1/(M \exp(-\frac{\ \mathbf{S}\mathbf{c}\ ^2}{2\sigma^2}) \cosh(\frac{\langle \mathbf{z}, \mathbf{S}\mathbf{c} \rangle}{\sigma^2}))$ : | 6: <b>if</b> $\ (\mathbf{z}_1, 2^d \mathbf{h})\  \leq B_2$                  |
| 7: <b>if</b> $[\mathbf{u}]_d = [\mathbf{u} + (-1)^b \mathbf{c}]_d$                                                                                   | 7: <b>if</b> $\ (\mathbf{z}_1, 2^d \mathbf{h})\ _\infty \leq B_\infty$      |
| 8: $\mathbf{h} = [\mathbf{u}]_d - [\mathbf{u} - \mathbf{z}_2 + (1 - b)\mathbf{c}]_d \bmod p$                                                         | 8: <b>return</b> $(\mathbf{w} = [\mathbf{v}]_d, \mathbf{z}_1, \mathbf{h})$  |
| 9: <b>if</b> $\ (\mathbf{z}_1, 2^d \mathbf{h})\  \leq B_2$                                                                                           | 9: <b>return</b> $\perp$                                                    |
| 10: <b>if</b> $\ (\mathbf{z}_1, 2^d \mathbf{h})\ _\infty \leq B_\infty$                                                                              |                                                                             |
| 11: <b>return</b> $(\mathbf{w}, \mathbf{z}_1, \mathbf{h})$                                                                                           |                                                                             |
| 12: <b>return</b> $\perp$                                                                                                                            |                                                                             |

Figure 3: NTRU+ident protocol simulator

$[\mathbf{a}\mathbf{z}_1 + \mathbf{z}_2 + \mathbf{c}\hat{q}]_d - [\mathbf{a}\mathbf{z}_1 + \mathbf{c}\hat{q}]_d = [\mathbf{u}]_d - [\mathbf{u} - \mathbf{z}_2 + (1 - b)\mathbf{c}]_d \bmod p$ . This completes the proof, confirming that the distributions of  $\mathbf{h}$  in both algorithms are identical.

### 3.6.2 Commitment Min-Entropy

We claim that the underlying identification scheme has large commitment min-entropy by showing that, for a given public polynomial  $\mathbf{a} \in R_q$ ,

$$\Pr [\mathbf{a}\mathbf{y}_1 + \mathbf{y}_2]_d = \mathbf{w} \mid \mathbf{y}_1, \mathbf{y}_2 \leftarrow D_\sigma^n] \leq \left( \frac{2^d}{\sqrt{2\pi}\sigma - 1} \right)^n \quad \forall \mathbf{w} \in \mathcal{W}.$$

Define  $T$  to be the set containing all the elements  $\mathbf{u}$  such that  $[\mathbf{u}]_d = \mathbf{w}$ . By the definition of the function  $[\cdot]_d$ , the size of  $T$  is at most  $2^{dn}$ . We can rewrite the above probability as

$$\Pr [\mathbf{a}\mathbf{y}_1 + \mathbf{y}_2 \in T \mid \mathbf{y}_1, \mathbf{y}_2 \leftarrow D_\sigma^n] = \Pr [\mathbf{y}_2 \in (T - \mathbf{a}\mathbf{y}_1) \mid \mathbf{y}_2 \leftarrow D_\sigma^n]$$

where the equality follows from the fact that the size of the set  $(T - \mathbf{a}\mathbf{y}_1)$  is the same as the size of  $T$ . Finally, using  $\sum_{x \in \mathbb{Z}} \rho_\sigma(x) \geq \int_{-\infty}^{\infty} \rho_\sigma(x) dx - 1 = \sqrt{2\pi}\sigma - 1$  [Lyu11, Lemma 4.4], we can derive the inequality

$$\Pr [\mathbf{y}_2 = \mathbf{t} \mid \mathbf{y}_2 \leftarrow D_\sigma^n] \leq \left( \frac{1}{\sqrt{2\pi}\sigma - 1} \right)^n \quad \text{for any } \mathbf{t} \in R_q,$$

which in turn implies the claim. For example, the commitment min-entropy  $\alpha$  is at least 989 for NTRU+Sign-1024.

### 3.6.3 UF-NMA Security

We prove that NTRU+Sign is UF-NMA secure if the NTRU assumption and BimodalSelfTargetRSIS assumption hold.

**Theorem 3.3** (UF-NMA Security). For any quantum adversary  $\mathcal{A}$  against the UF-NMA security of NTRU+Sign making at most  $Q_h$  quantum hash queries, there exists an adversary  $\mathcal{B}$  and  $\mathcal{C}$  such that:

$$\text{Adv}_{\text{DS}}^{\text{UF-NMA}}(\mathcal{A}) \leq \text{Adv}_{n,q,\psi_1}^{\text{NTRU}}(\mathcal{B}) + \text{Adv}_{H,n,q,B_2+(2^{d-1}+1)\sqrt{n}}^{\text{BimodalSelfTargetRSIS}}(\mathcal{C}). \quad (2)$$

*Proof.* Given an element  $\mathbf{a} \in R_q$ , the adversary  $\mathcal{C}$  sets  $\mathbf{a}$  as the public key of the signature scheme and sends it to  $\mathcal{A}$ . If the public key  $pk$  generated by KeyGen is indistinguishable from uniform over  $R_q$  (i.e., if the  $\text{NTRU}_{n,q,\psi_1}$  problem is hard), then with probability  $\text{Adv}_{\text{DS}}^{\text{UF-NMA}}(\mathcal{A})$ ,  $\mathcal{A}$  will return a signature  $(\mathbf{z}_1, \mathbf{h}, \mathbf{c})$  of some message  $\mu$  such that  $\|(\mathbf{z}_1 | 2^d \mathbf{h})\| \leq B_2$  satisfying the verification equation

$$\mathbf{c} = H([\mathbf{a}\mathbf{z}_1 + \mathbf{c}\hat{q}]_d + \mathbf{h} \bmod p, \mu).$$

For simplicity, we define the function  $H'$  such that  $H(w_1, \mu) = H'(w_1 \cdot 2^d, \mu)$  to accommodate the compression that does not appear in the BimodalSelfTargetRSIS problem. Notice that the difference between  $H$  and  $H'$  is just a change in the format of the input. The above equality can be rewritten as:

$$\mathbf{c} = H'(\mathbf{a}\mathbf{z}_1 + \mathbf{c}\hat{q} + 2^d \mathbf{h} + \mathbf{e} + \mathbf{k} \bmod q, \mu) \quad (3)$$

where  $\mathbf{e} = \mathbf{a}\mathbf{z}_1 + \mathbf{c}\hat{q} - [\mathbf{a}\mathbf{z}_1 + \mathbf{c}\hat{q}]_d$ ,  $\|\mathbf{e}\|_\infty \leq 2^{d-1}$ , and  $\|\mathbf{k}\|_\infty = 1$ . The term  $\mathbf{k}$  is added for the following reason. Define  $\mathbf{v}_1 = [\mathbf{a}\mathbf{z}_1 + \mathbf{c}\hat{q}]_d + \mathbf{h} \bmod p$ . It can be shown that there exists  $\mathbf{k}$  such that the following equation holds over  $R$ :

$$\mathbf{v}_1 = [\mathbf{a}\mathbf{z}_1 + \mathbf{c}\hat{q}]_d + \mathbf{h} + p\mathbf{k}$$

where all coefficients of  $\mathbf{k}$  are in  $\{0, \pm 1\}$ . Then we obtain  $2^d \mathbf{v}_1 = 2^d([\mathbf{a}\mathbf{z}_1 + \mathbf{c}\hat{q}]_d + \mathbf{h} + p\mathbf{k}) = \mathbf{a}\mathbf{z}_1 + \mathbf{c}\hat{q} + 2^d \mathbf{h} + \mathbf{e} + 2^d \cdot p\mathbf{k}$ . Note that it holds that  $2^d \cdot p = q - 1$ . Applying the modulo  $q$  operation to both sides, we derive equation (3).

Equation (3) provides a polynomial vector  $\mathbf{Y} = [2^d \mathbf{h} + \bar{\mathbf{e}} | \mathbf{z}_1]^T$  such that  $H'([1 | \mathbf{a}] \cdot \mathbf{Y} + \mathbf{c}\hat{q}, \mu) = \mathbf{c}$  where  $\bar{\mathbf{e}} = \mathbf{e} + \mathbf{k}$ . We can show that  $\|\mathbf{Y}\|_\infty \leq (q-2)/4$  since the condition  $\|(\mathbf{z}_1, 2^d \mathbf{h})\|_\infty \leq B_\infty$  is enforced in the Sign algorithm, where  $B_\infty \leq (q-2)/4 - 2^{d-1} - 1$ . Next, we note that  $\|\mathbf{Y}\| = \|(2^d \mathbf{h} + \bar{\mathbf{e}} + \mathbf{k}, \mathbf{z}_1)\| \leq \|(2^d \mathbf{h}, \mathbf{z}_1)\| + \|(\bar{\mathbf{e}}, 0)\| \leq B_2 + (2^{d-1} + 1)\sqrt{n}$ . This completes the proof.  $\square$

### 3.6.4 CUR

To prove the sUF-CMA security of the signature scheme, we must also bound the advantage an adversary may have against the CUR property of the identification scheme. Such a bound is presented in the following lemma.

**Lemma 3.4.** If  $\mathcal{A}$  is an adversary against the CUR property of the identification scheme, then there exist adversaries  $\mathcal{B}$  and  $\mathcal{C}$ , each using  $\mathcal{A}$  once and performing negligible additional work, such that

$$\text{Adv}_{\text{ID}}^{\text{CUR}}(\mathcal{A}) \leq \text{Adv}_{n,q,\psi_1}^{\text{NTRU}}(\mathcal{B}) + \text{Adv}_{H,n,q,2(B_2+(2^{d-1}+1)\sqrt{n})}^{\text{RSIS}}(\mathcal{C}).$$

*Proof.* Given an element  $\mathbf{a} \in R_q$ , the adversary  $\mathcal{C}$  sets  $\mathbf{a}$  as the public key of the identification scheme and sends it to  $\mathcal{A}$ . If the public key  $pk$  generated by KeyGen is indistinguishable from uniform over  $R_q$ , then with probability  $\text{Adv}_{\text{ID}}^{\text{CUR}}(\mathcal{A})$ , An adversary  $\mathcal{A}$  against the CUR property of the identification will return

$\mathbf{w}, \mathbf{c}, \mathbf{z}_1, \mathbf{h}, \mathbf{z}'_1, \mathbf{h}'$  such that  $(\mathbf{z}_1, \mathbf{h}) \neq (\mathbf{z}'_1, \mathbf{h}')$  and both  $\mathbf{w}, \mathbf{c}, \mathbf{z}_1, \mathbf{h}$  and  $\mathbf{w}, \mathbf{c}, \mathbf{z}'_1, \mathbf{h}'$  are accepting transcripts for the identification scheme. Thus,

$$[\mathbf{a}\mathbf{z}_1 + \mathbf{c}\hat{q} \bmod q]_d + \mathbf{h} \bmod p = [\mathbf{a}\mathbf{z}'_1 + \mathbf{c}\hat{q} \bmod q]_d + \mathbf{h}' \bmod p = \mathbf{w}.$$

Therefore, following the same approach used to derive equation (3), we can obtain the following congruence modulo  $q$ :

$$\mathbf{a}(\mathbf{z}_1 - \mathbf{z}'_1) + 2^d(\mathbf{h} - \mathbf{h}') + \mathbf{e} - \mathbf{e}' + \mathbf{k} = 0 \bmod q,$$

where  $\mathbf{e} = \mathbf{a}\mathbf{z}_1 + \mathbf{c}\hat{q} - [\mathbf{a}\mathbf{z}_1 + \mathbf{c}\hat{q}]_d \cdot 2^d \bmod q$ ,  $\mathbf{e}' = \mathbf{a}\mathbf{z}'_1 + \mathbf{c}\hat{q} - [\mathbf{a}\mathbf{z}'_1 + \mathbf{c}\hat{q}]_d \cdot 2^d \bmod q$ ,  $\|\mathbf{e}\|_\infty \leq 2^{d-1}$ ,  $\|\mathbf{e}'\|_\infty \leq 2^{d-1}$  and  $\|\mathbf{k}\|_\infty \leq 2$ . It follows that  $[1 \mid \mathbf{a}] \cdot \mathbf{Y} = 0 \bmod q$  where  $\mathbf{Y} = \begin{bmatrix} 2^d(\mathbf{h} - \mathbf{h}') + \mathbf{e} - \mathbf{e}' + \mathbf{k} \\ \mathbf{z} - \mathbf{z}' \end{bmatrix}$  and  $\|\mathbf{Y}\|_2 \leq 2(B_2 + (2^{d-1} + 1)\sqrt{n})$ . For  $\mathbf{Y}$  to be a solution to the RSIS problem, it must be non-zero. Since  $(\mathbf{z}_1, \mathbf{h}) \neq (\mathbf{z}'_1, \mathbf{h}')$ , if  $\mathbf{z}_1 \neq \mathbf{z}'_1$ , then  $\mathbf{Y}$  is immediately non-zero, and hence constitutes a valid solution. If instead  $\mathbf{z}_1 = \mathbf{z}'_1$ , then by definition  $\mathbf{e} = \mathbf{e}'$  and then  $2^d(\mathbf{h} - \mathbf{h}') + \mathbf{e} - \mathbf{e}' + \mathbf{k} = 2^d(\mathbf{h} - \mathbf{h}') + \mathbf{k}$ . Since  $\mathbf{h} \neq \mathbf{h}'$  and  $\|\mathbf{k}\|_\infty \leq 2$ , this expression is non-zero, implying that  $\mathbf{Y} \neq 0$ . Therefore, in both cases,  $\mathbf{Y}$  is a valid non-zero short vector satisfying the RSIS equation, which completes the proof.  $\square$

## 4 Parameter Settings

### 4.1 Concrete Parameters

Table 2 presents several conditions necessary for choosing the parameters for NTRU+Sign. We provide concrete parameter sets achieving 93 and 211 security bits, with the latter corresponding to NIST level 3, as shown in Table 3. Both parameter sets meet all the requirements outlined in Table 2.

The sizes of the public keys for NTRU+Sign-{512, 1024} are 768 and 1664 bytes, respectively, calculated as  $(n \lceil \log q \rceil) / 8$  bytes. A signature consists of  $(\mathbf{z}_1, \mathbf{h}, \mathbf{c})$ , where the high bits of  $\mathbf{z}_1$  and  $\mathbf{h}$  are encoded using range Asymmetric Numeral System (rANS) encoding [ETWY22]. Then, the resulting signature size is approximately  $(256 + nd + n \log_2(2\pi e(\sigma/2^d)^2)) / 8$  bytes, where 256 represents a hash value related to  $\mathbf{c}$ ,  $nd$  represents low bits of  $\mathbf{z}_1$ , and  $n \log_2(2\pi e(\sigma/2^d)^2)$  represents rANS-encoded values of the high bits of  $\mathbf{z}_1$  and  $\mathbf{h}$ . The sizes of the secret keys are 2336 and 5024 bytes for NTRU+Sign-{512, 1024}, respectively. These are calculated as  $(3n \lceil \log q \rceil + 256) / 8$  bytes, based on the storage of the secret key in NTT format. If the secret key is not stored in NTT format, the sizes would be 1056 and 2208 bytes for NTRU+Sign-{512, 1024}, respectively, computed as  $(n \lceil \log q \rceil + 4n + 256) / 8$  bytes.

In the KeyGen algorithm, the process restarts if the generated secret key  $\mathbf{S}$  satisfies  $\mathcal{N}(\mathbf{S}) > \gamma^2 n$ , where  $\gamma$  is a pre-defined constant. The constant  $\gamma$  is chosen to ensure that  $\mathbf{S}$  is accepted with the  $sk$  acceptance rate in Table 3. Specifically,  $\gamma$  is determined in advance by calculating  $\{\mathcal{N}(\mathbf{S})\}$  for randomly sampled secret keys  $\{\mathbf{S}\}$ , sorting  $\{\mathcal{N}(\mathbf{S})\}$ , and selecting a threshold value  $\mathcal{N}(\mathbf{S}^*)$  such that  $\mathcal{N}(\mathbf{S}^*)$  among the sorted  $\{\mathcal{N}(\mathbf{S})\}$  corresponds to the desired  $sk$  acceptance rate. Once  $\mathcal{N}(\mathbf{S}^*)$  is selected,  $\gamma$  is computed from the equation of  $\mathcal{N}(\mathbf{S}^*) = \gamma^2 n$ . Using Lemma 3.1, we can establish an upper bound for  $\|\mathbf{S}\mathbf{c}\|$  for secret keys satisfying  $\mathcal{N}(\mathbf{S}) \leq \gamma^2 n$ . Specifically, Lemma 3.1 states that for any  $\mathbf{c}$  with Hamming weight  $\tau$ , the bound  $\|\mathbf{S}\mathbf{c}\|^2 \leq (\tau/n) \cdot \mathcal{N}(\mathbf{S}) \leq \tau\gamma^2$  holds. Thus, the upper bound for  $\|\mathbf{S}\mathbf{c}\|$  is computed as  $\gamma\sqrt{\tau}$ .



Table 2: Requirements for Parameter Selection

| Parameter Requirements |                                                     |                             |
|------------------------|-----------------------------------------------------|-----------------------------|
| (1)                    | $n = 2^a$ for $a \in \mathbb{N}$                    | Degree of a polynomial      |
| (2)                    | $q \equiv 1 \pmod{n/2}$                             | Modulus for NTT             |
| (3)                    | $p = (q - 1)/2^d$                                   | Modulus for hint generation |
| (4)                    | $(2^d/(\sqrt{2\pi}\sigma - 1))^n \ll 2^{-2\lambda}$ | Commitment entropy          |
| (5)                    | $\binom{n}{\tau} \geq 2^\lambda$                    | Challenge space             |

## 4.2 Concrete Security Analysis

As shown above, the security of NTRU+Sign is based on the RSIS problem and the NTRU problem, specifically relative to  $\text{Adv}_{H', n, q, B_2 + (2^{d-1} + 1)\sqrt{n}}^{\text{BimodalSelfTargetRSIS}}$  and  $\text{Adv}_{n, q, \psi_1}^{\text{NTRU}}$ . Note that while the Euclidean norm bound,  $B = B_2 + (2^{d-1} + 1)\sqrt{n}$ , for RSIS solutions exceeds the modulus  $q$ , the infinite norm bound remains below  $q$ , specifically at  $(q - 2)/4$ . This ensures that trivial solutions are avoided.

To analyze the concrete security of the above RSIS and NTRU problems, we adapt the concrete security analysis conducted in [CCD<sup>+</sup>24] and [CDH<sup>+</sup>20]. The concrete security strength of the NTRU problem is expected to be similar to the case of Ring Learning with Error (LWE) problems. To analyze the concrete security of the above RSIS and NTRU problems, we employ the BKZ lattice reduction algorithm [CN11], which represents the most effective known lattice attack. Various approaches exist to estimate the running time of BKZ [APS15, ADPS16, CN11]. In general, a Shortest Vector Problem (SVP) solver is the main building block of the BKZ algorithm. To estimate the number of gates required to solve the RSIS and NTRU problems, as in [CCD<sup>+</sup>24], we follow the established Core-SVP methodology, which assumes that an SVP oracle is required only once in a conservative model, regarding the number of SVP oracle calls that the BKZ algorithm makes.

Table 3 presents the concrete security levels for each set of parameters of NTRU+Sign. Our estimates for the RSIS problem are slightly conservative in that those estimates are calculated under the Euclidean norm bound  $B_2 + (2^{d-1} + 1)\sqrt{n}$ , not considering the additional infinite norm bound  $(q - 2)/4$ . As noted in [JRLS24], such cryptanalysis likely underestimates the true complexity of the RSIS problem. Because the modulus  $q$  of NTRU+Sign is relatively small compared to the dimension  $n$  and the RSIS bound  $B_2 + (2^{d-1} + 1)\sqrt{n}$ , we can also use the recent estimator proposed by Ducas *et al.* [DEP23]. The estimated results are 93 and 214 bits of security for the RSIS problem, which slightly reduces the RSIS hardness, but has no impact on the overall target security level of NTRU+Sign- $\{512, 1024\}$ <sup>4</sup>.

Note that the classical reduction from BimodalSelfTargetRSIS to RSIS, as shown in Def. 2.10, increases the  $\ell_2$ -norm of the extracted solution by a factor of 4. In line with prior works such as Dilithium [DKL<sup>+</sup>18], we did not apply this factor in the security estimate when computing the hardness of the SIS problem, as no known attack currently exploits this reduction loss. However, if this conservative perspective is adopted, i.e., analyzing the security under the RSIS bound  $4(B_2 + (2^{d-1} + 1)\sqrt{n}) + 2\sqrt{\tau}$ —our estimator indicates a drop in security to 14 and 12 bits for NTRU+Sign-512 and NTRU+Sign-1024, respectively.

To address this issue, one possible solution is to slightly increase the modulus size, namely  $q = 12289$

<sup>4</sup>While we obtained results for baseline parameter sets at  $n = 512$  and at  $n = 1024$  using this estimator, the runtime was prohibitively high. For conservative bounds such as those arising from the SelfTargetRSIS–RSIS reduction (e.g.,  $4B$  where the value  $B$  is the solution norm bound in the RSIS problem), the computation did not complete even after more than 15 days. A fair comparison would also require re-running this estimator for all schemes in Table 4, which is currently impractical. For these reasons, we did not update the tables with this estimator but instead provide the obtained values as reference numbers in the text.

Table 3: Parameter sets for NTRU+Sign- $\{512, 1024\}$ . Numbers in parentheses for SIS security are for strong unforgeability.

| Parameter sets                         |                                                  | 1         | 2         |
|----------------------------------------|--------------------------------------------------|-----------|-----------|
| $n$                                    | Degree of a polynomial                           | 512       | 1,024     |
| $q$                                    | Modulus                                          | 3,329     | 7,681     |
| $\tau$                                 | Hamming weight of $\mathbf{c}$                   | 20        | 36        |
| $\lfloor \log_2  \mathcal{C}  \rfloor$ | $\lfloor \log_2 \binom{n}{\tau} \rfloor$         | 118       | 221       |
| $d$                                    | # of dropped bits in the commitment              | 7         | 8         |
| $M$                                    | Expected # of repetitions for rejection sampling | 3.25      | 4.28      |
| $M_{Eq}$                               | Expected # of repetitions for equality check     | 1.17      | 1.15      |
| $M_{total}$                            | $= M \times M_{Eq}$                              | 3.80      | 4.92      |
| $B_{Sc}$                               | Upper-bound for Euclidean norm of $\mathbf{Sc}$  | 169       | 341       |
| $\gamma$                               | $sk$ rejection parameter                         | 37.77     | 56.71     |
|                                        | $sk$ acceptance rate                             | 0.25      | 0.25      |
| $\sigma$                               | Standard deviation of $D_\sigma$                 | 110       | 200       |
| $\hat{\alpha}$                         | $= \sigma / B_{Sc}$                              | 0.61      | 0.58      |
| $\alpha$                               | Min-entropy of the commitment                    | 564       | 989       |
| $B_2$                                  | Verification threshold in Euclidean norm         | 4,000     | 10,000    |
| $B_\infty$                             | Verification threshold in infinite norm          | 766       | 1,790     |
| NTRU Hardness (Core-SVP)               |                                                  |           |           |
|                                        | BKZ block-size $b$                               | 339       | 724       |
|                                        | Classical Core-SVP                               | 98        | 211       |
|                                        | Quantum Core-SVP                                 | 88        | 186       |
| SIS Hardness (Core-SVP)                |                                                  |           |           |
|                                        | BKZ block-size $b$                               | 319 (180) | 766 (599) |
|                                        | Classical Core-SVP                               | 93 (52)   | 224 (175) |
|                                        | Quantum Core-SVP                                 | 81 (46)   | 196 (153) |

for the 512-bit setting and  $q = 25601$  for the 1024-bit setting (an increase of 2 bits in each case). Under these parameters, the RSIS security levels increase to 72 and 183 bits, respectively, even when accounting for the four-fold norm increase, while the NTRU hardness slightly decreases to 82 and 186 bits. These changes result in no increase in signature size and only a modest increase in public key size (by 128 and 256 bytes, respectively). Thus, even under this conservative assumption, our parameter sets remain competitive in efficiency compared to prior schemes. However, since no concrete attacks are currently known that exploit the reduction loss from BimodalSelfTargetRSIS to RSIS, we did not reflect these adjusted parameters in Table 3, but rather mention them here as a potential safeguard for future analysis.

### 4.3 Comparison

Table 4 provides a detailed comparison of signature sizes among recent lattice-based signature schemes. For comparison with NTRU+Sign, we include FSWA-based schemes [DKL<sup>+</sup>18, CCD<sup>+</sup>24, DPS23, BBR24, DDLL13], including the re-parametrized BLISS, as well as Falcon [FHK<sup>+</sup>17], which is based on the Hash-and-Sign framework. The results show that NTRU+Sign achieves the smallest signature size among the FSWA-based schemes at a comparable security level. Notably, NTRU+Sign-1024 achieves this compact-

Table 4: Comparison to previous lattice-based signature schemes

|                            | Classical<br>Security | Sig <br>(bytes) | pk <br>(bytes) | Sig + pk <br>(bytes) | Sampling<br>Distribution |
|----------------------------|-----------------------|-----------------|----------------|----------------------|--------------------------|
| Dilithium-2                | 123                   | 2,420           | 1,312          | 3,732                | Hypercube                |
| Dilithium-G-2 <sup>1</sup> | 118                   | 1,921           | 800            | 2,721                | Gaussian                 |
| HAETAE-120                 | 119                   | 1,474           | 992            | 2,466                | Hyperball                |
| G + G-120 <sup>2</sup>     | 121                   | 1,677           | 1,472          | 3,149                | (Convolved) Gaussian     |
| G + G-512 <sup>3</sup>     | 85                    | 1,021           | 992            | 2,013                | (Convolved) Gaussian     |
| Patronus-120               | 120                   | 2,070           | 832            | 2,902                | Polytope                 |
| Falcon-512                 | 120                   | 666             | 897            | 1,563                | -                        |
| BLISS-512 <sup>4</sup>     | 87                    | 831             | 896            | 1,727                | Gaussian                 |
| NTRU+Sign-512              | 93                    | 751             | 768            | 1,519                | Gaussian                 |
| Dilithium-3                | 182                   | 3,293           | 1,952          | 5,245                | Hypercube                |
| Dilithium-G-3 <sup>1</sup> | 183                   | 2,462           | 1,184          | 3,646                | Gaussian                 |
| HAETAE-180                 | 180                   | 2,349           | 1,472          | 3,821                | Hyperball                |
| G + G-180 <sup>2</sup>     | 178                   | 2,143           | 1,952          | 4,095                | (Convolved) Gaussian     |
| G + G-1024 <sup>3</sup>    | 178                   | 1,769           | 2,080          | 3,849                | (Convolved) Gaussian     |
| Patronus-180               | 182                   | 2,575           | 1,152          | 3,727                | Polytope                 |
| Falcon-1024                | 273                   | 1,280           | 1,793          | 3,073                | -                        |
| BLISS-1024 <sup>4</sup>    | 192                   | 1,836           | 1,920          | 3,756                | Gaussian                 |
| NTRU+Sign-1024             | 211                   | 1,551           | 1,664          | 3,215                | Gaussian                 |

<sup>1</sup> Gaussian version of Dilithium optimized in [DFPS22]    <sup>2</sup> Module-LWE version of G + G<sup>3</sup> NTRU version of G + G    <sup>4</sup> BLISS with updated security analysis

ness despite targeting a higher classical security level than most other schemes (excluding Falcon-1024). For example, the signature size of NTRU+Sign-1024 is approximately 55% smaller than Dilithium-3 and 35% smaller than HAETAE-180. Moreover, NTRU+Sign-1024 achieves the smallest combined size of the signature and public key among FSWA-based schemes. Compared to Dilithium-3, the combined size is approximately 40% smaller, and compared to HAETAE-180, it is approximately 15% smaller. A direct comparison with the re-parametrized BLISS highlights the significant impact of the modulus change and the novel key structure in NTRU+Sign.

The performance comparison between Dilithium, HAETAE, and NTRU+Sign is summarized in Table 1. Compared to HAETAE-180, NTRU+Sign-1024 achieves 1.8 times faster key generation and 1.7 times faster signing speeds. However, compared to Dilithium, NTRU+Sign-1024 is approximately 2.9 times slower for key generation and 1.9 times slower for signing. This is primarily due to the additional key rejection process in the KeyGen algorithm of NTRU+Sign, designed to reduce the norm  $\|\mathbf{Sc}\|$  and optimize rejection sampling. The performance difference also stems from the efficiency of Dilithium’s uniform hypercube sampling and its simplified rejection sampling, which relies solely on infinite norm checks.

Notably, the Verify algorithm of NTRU+Sign is 4.6 times faster than Dilithium and 4.9 times faster than HAETAE. While the verification processes in all three schemes share a similar conceptual foundation, several factors contribute to the superior efficiency of NTRU+Sign. Firstly, NTRU+Sign-1024 uses a smaller modulus,  $q = 7681 (\approx 2^{13})$ , compared to  $q = 8380417 (\approx 2^{23})$  in Dilithium and  $q = 64513 (\approx 2^{16})$  in HAETAE. The smaller modulus enables faster polynomial multiplications. Secondly, unlike Dilithium and HAETAE, NTRU+Sign does not require re-generating the public matrix  $\mathbf{A} \in R_q^{k \times \ell}$  from a public seed

during verification, further reducing computational overhead.

Regarding the Sign algorithm, NTRU+Sign achieves computational improvements over HAETAE primarily due to its use of Gaussian sampling instead of hyperball sampling. Hyperball sampling, while simplifying rejection sampling, inherently relies on discrete Gaussian sampling with extremely large standard deviations (e.g.,  $2^{72}$ ), high-precision arithmetic, and additional operations such as inverse square root and  $\ell_2$  norm calculations, all of which introduce significant computational overhead. For example, constant-time Gaussian sampling in NTRU+Sign-1024 achieves a 43% speedup compared to hyperball sampling in HAETAE-180. Also, contrary to what is expected, the rejection sampling process of NTRU+Sign-1024 is about 1.6 times faster than that of HAETAE-180. To perform the rejection sampling process, NTRU+Sign needs to calculate transcendental function approximations based on GALATICS, whereas HAETAE needs to compute Euclidean norm. Fortunately, the coefficients of approximated polynomials are precomputed, so that the rejection sampling process of NTRU+Sign can be done by simple scalar multiplications and shift operations, using the precomputed coefficients.

## 5 Performance Analysis

### 5.1 Reference Implementation

We evaluate the performance of NTRU+Sign- $\{512, 1024\}$ , using our reference code on a 2.4GHz Intel Core i5-6300u running Ubuntu 20.04 LTS. Table 1 shows the performance of the KeyGen, Sign, and Verify algorithms of NTRU+Sign- $\{512, 1024\}$  in comparison to the reference code implementations of Dilithium [DKL<sup>+</sup>18] and HAETAE [CCD<sup>+</sup>24]. The cycle counts presented in Table 1 are averaged over 10,000 executions of each respective algorithm. The reference code for NTRU+Sign is publicly available<sup>5</sup>.

Our implementation partially adapts components from the public reference implementations of Dilithium and HAETAE. Dilithium is distributed under a dual license (CC0 or Apache 2.0), and HAETAE is distributed under the MIT License. We have included the corresponding acknowledgments in our LICENSE file.

### 5.2 Implementation Details

The predecessor of NTRU+Sign, BLISS, became a target of side-channel attacks due to its use of discrete Gaussian distributions [BHL16b] and a rejection sampling [EFGT17]. The main reason of this vulnerability is that the execution of these sampling depends on the secret input values. To address this issue, various constant-time implementation techniques have been proposed [BBE<sup>+</sup>19, PRR19, Por19, ZSS20]. Here, the constant-time implementation means that execution time does not depend on the input.

Assuming that modular arithmetic is a constant-time. NTRU+Sign allows for a straightforward constant-time implementation of the signing algorithm, except for discrete Gaussian sampling and rejection sampling. In section 5.2.1 and 5.2.2, we explain our approach to constant-time implementations for these two components. We note that discrete Gaussian sampling is implemented based on the results of [PRR19], and rejection sampling follows the approach proposed in [BBE<sup>+</sup>19].

On the other hand, in the concrete constant-time implementation, approximations of a certain precision to transcendental functions are adopted, and it makes differences from the ideal distribution, which in turn induce security loss. To measure the security loss, we use the approach proposed in [Pre17, BLRL<sup>+</sup>18, PRR19] which is based on Rényi divergence. A detailed explanation is omitted, and all details can be found in Appendix B.

<sup>5</sup>[https://github.com/KU-Cryptographic-Protocol-Lab/ntruplus\\_sign](https://github.com/KU-Cryptographic-Protocol-Lab/ntruplus_sign)

### 5.2.1 Gaussian Sampling

In line 1 of the Sign algorithm in Figure 1, we produce  $\mathbf{y} = (\mathbf{y}_1, \mathbf{y}_2)$ , where  $\mathbf{y}_1, \mathbf{y}_2$  are polynomials with coefficients sampled according to a centered discrete Gaussian distribution of relatively large standard deviation ( $\sigma = 200$  for NTRU+Sign-1024). In essence, we start by sampling  $x$  from a discrete Gaussian distribution with a smaller standard deviation  $\sigma_1 = \sigma/k$  where  $k = 2^{\lceil \log_2 \sigma \rceil}$  using a Cumulative Distribution Table (CDT), and next we sample a uniform  $y \in \{0, \dots, k-1\}$  and accept the candidate Gaussian sample  $r = xk + y$  with probability  $\exp(-y(y + 2xk)/(2\sigma^2))$ , following the technique of BLISS [DDL13]. Note that, to ensure constant-time execution, all entries of the CDT are accessed when the CDT table is used, and rejection sampling is performed using an approximated polynomial of  $\exp(x/(2\sigma^2))$ , as proposed with the GALACTICS tool. For instance, we use a CDT with  $\sigma_1 = 200/2^7 \approx 1.57$  where  $n = 1024, \sigma = 200$ , and  $k = 2^7$ , which only requires 15 table entries. This CDT implementation stores the cumulative probabilities with 86 bits of precision. Then, under a sufficiently large precision, an approximated polynomial  $P_{\exp}^{I_1}(x)$  of  $\exp(x/(2\sigma^2))$  over the interval  $I_1 = [-\lfloor 2\sigma^2 \rfloor, 0]$  can be used to evaluate a function value at a point  $x$  in constant time.

### 5.2.2 Rejection Sampling

Explicitly in line 6 of the Sign algorithm in Figure 1, we need to implement the rejection sampling in constant time. Specifically, we need to sample a bit  $\tilde{b}$  from the Bernoulli distribution  $\mathcal{B}_\psi$  with parameter  $\psi(\mathbf{z}, \mathbf{v}) = 1/(M \exp(-\|\mathbf{v}\|^2/2\sigma^2) \cosh(\langle \mathbf{z}, \mathbf{v} \rangle / \sigma^2))$  where  $M = \exp(1/2\hat{\sigma}^2)$ . According to the bit  $\tilde{b}$ , the Sign algorithm decides whether or not it proceeds with  $\mathbf{z}$ . To implement this efficiently, we decompose the Bernoulli distribution  $\mathcal{B}_\psi$  into the product of  $\mathcal{B}_{\psi_1}$  and  $\mathcal{B}_{\psi_2}$  where  $\psi_1(\mathbf{v}) = 1/(M \exp(-\|\mathbf{v}\|^2/2\sigma^2))$  and  $\psi_2(\mathbf{z}, \mathbf{v}) = 1/\cosh(\langle \mathbf{z}, \mathbf{v} \rangle / \sigma^2)$ . Following GALACTICS, we approximate the functions  $\exp$  and  $\cosh$  with approximated polynomial of  $\exp(x/(2\sigma^2))$ , and given an input  $(\mathbf{z}, \mathbf{v})$  we calculate those function values in constant time.

### 5.2.3 Signature Packing

In line of 12 in the Sign algorithm in Figure 1, a signature consists of  $(\mathbf{z}_1, \mathbf{h}, \mathbf{c})$ , where  $\mathbf{z}_1$  and  $\mathbf{h}$  are packed to reduce the signature size. Indeed,  $\mathbf{z}_1$  is a polynomial whose coefficients follow a discrete Gaussian distribution with standard deviation  $\sigma$ , and the hint  $\mathbf{h} = [\mathbf{u}]_d - [\mathbf{u} - \mathbf{z}_2 + (1 - b)\mathbf{c}]_d \bmod p$  also nearly follows a discrete Gaussian distribution that is substantially derived from  $\mathbf{z}_2$ . That is,  $\mathbf{z}_1$  and  $\mathbf{h}$  have their coefficients distributed according to probability distributions that are concentrated around elements with small  $\ell_2$  norms. Hence, it is not optimal to simply represent coefficients with their binary representations. Instead, we use the adaptive arithmetic encoding, referred as rANS as described in [ETWY22]. This entropic coding method enables efficient compression by closely matching the signature size to its entropy, while maintaining high performance through finite precision integer arithmetic. Our rANS encoding is based on the implementation from [Gie14], which provides an efficient and practical approach for encoding a signature.

## References

- [ADPS16] Erdem Alkim, Léo Ducas, Thomas Pöppelmann, and Peter Schwabe. Post-quantum key exchange - A new hope. In Thorsten Holz and Stefan Savage, editors, *25th USENIX Security Symposium*, USENIX Security 16, Austin, TX, USA, August 10-12, 2016, pages 327–343, USA, 2016. USENIX Association.

- [APS15] Martin R. Albrecht, Rachel Player, and Sam Scott. On the concrete hardness of learning with errors. J. Math. Cryptol., 9(3):169–203, 2015.
- [BBD<sup>+</sup>23] Manuel Barbosa, Gilles Barthe, Christian Doczkal, Jelle Don, Serge Fehr, Benjamin Grégoire, Yu-Hsuan Huang, Andreas Hülsing, Yi Lee, and Xiaodi Wu. Fixing and mechanizing the security proof of fiat-shamir with aborts and dilithium. In Helena Handschuh and Anna Lysyanskaya, editors, Advances in Cryptology - CRYPTO 2023 - 43rd Annual International Cryptology Conference, CRYPTO 2023, Santa Barbara, CA, USA, August 20-24, 2023, Proceedings, Part V, volume 14085 of Lecture Notes in Computer Science, pages 358–389, Cham, 2023. Springer.
- [BBE<sup>+</sup>19] Gilles Barthe, Sonia Belaïd, Thomas Espitau, Pierre-Alain Fouque, Méliissa Rossi, and Mehdi Tibouchi. GALACTICS: gaussian sampling for lattice-based constant- time implementation of cryptographic signatures, revisited. In Lorenzo Cavallaro, Johannes Kinder, XiaoFeng Wang, and Jonathan Katz, editors, Proceedings of the 2019 ACM SIGSAC Conference on Computer and Communications Security, CCS 2019, London, UK, November 11-15, 2019, pages 2147–2164, New York, NY, USA, 2019. ACM.
- [BBRS24] Henry Bambury, Hugo Beguinet, Thomas Ricosset, and Éric Sageloli. Polytopes in the fiat-shamir with aborts paradigm. In Leonid Reyzin and Douglas Stebila, editors, Advances in Cryptology - CRYPTO 2024 - 44th Annual International Cryptology Conference, Santa Barbara, CA, USA, August 18-22, 2024, Proceedings, Part I, volume 14920 of Lecture Notes in Computer Science, pages 339–372, Cham, 2024. Springer.
- [BDE<sup>+</sup>18] Jonathan Bootle, Claire Delaplace, Thomas Espitau, Pierre-Alain Fouque, and Mehdi Tibouchi. LWE without modular reduction and improved side-channel attacks against BLISS. In Advances in Cryptology - ASIACRYPT 2018 - 24th International Conference on the Theory and Application of Cryptology and Information Security, Brisbane, QLD, Australia, December 2-6, 2018, Proceedings, Part I, volume 11272 of Lecture Notes in Computer Science, pages 494–524, Cham, 2018. Springer.
- [BG14] Shi Bai and Steven D. Galbraith. An improved compression technique for signatures based on learning with errors. In Josh Benaloh, editor, Topics in Cryptology - CT-RSA 2014 - The Cryptographer’s Track at the RSA Conference 2014, San Francisco, CA, USA, February 25-28, 2014. Proceedings, volume 8366 of Lecture Notes in Computer Science, pages 28–47, Cham, 2014. Springer.
- [BHLY16a] Leon Groot Bruinderink, Andreas Hülsing, Tanja Lange, and Yuval Yarom. Flush, gauss, and reload - A cache attack on the BLISS lattice-based signature scheme. In Cryptographic Hardware and Embedded Systems - CHES 2016 - 18th International Conference, Santa Barbara, CA, USA, August 17-19, 2016, Proceedings, volume 9813 of Lecture Notes in Computer Science, pages 323–345, Berlin, Heidelberg, 2016. Springer.
- [BHLY16b] Leon Groot Bruinderink, Andreas Hülsing, Tanja Lange, and Yuval Yarom. Flush, gauss, and reload - A cache attack on the BLISS lattice-based signature scheme. In Benedikt Gierlichs and Axel Y. Poschmann, editors, Cryptographic Hardware and Embedded Systems - CHES 2016 - 18th International Conference, Santa Barbara, CA, USA, August 17-19, 2016, Proceedings, volume 9813 of Lecture Notes in Computer Science, pages 323–345. Springer, 2016.

- [BLRL<sup>+</sup>18] Shi Bai, Tancrede Lepoint, Adeline Roux-Langlois, Amin Sakzad, Damien Stehlé, and Ron Steinfeld. Improved security proofs in lattice-based cryptography: using the rényi divergence rather than the statistical distance. Journal of Cryptology, 31:610–640, 2018.
- [BN06] Mihir Bellare and Gregory Neven. Multi-signatures in the plain public-key model and a general forking lemma. In Ari Juels, Rebecca N. Wright, and Sabrina De Capitani di Vimercati, editors, Proceedings of the 13th ACM Conference on Computer and Communications Security, CCS 2006, Alexandria, VA, USA, October 30 - November 3, 2006, pages 390–399, New York, NY, USA, 2006. ACM.
- [CCD<sup>+</sup>24] Jung Hee Cheon, Hyeongmin Choe, Julien Devevey, Tim Güneysu, Dongyeon Hong, Markus Krausz, Georg Land, Marc Möller, Damien Stehlé, and MinJune Yi. HAETAE: shorter lattice-based fiat-shamir signatures. IACR Trans. Cryptogr. Hardw. Embed. Syst., 2024(3):25–75, 2024.
- [CDH<sup>+</sup>20] Cong Chen, Oussama Danba, Jeffrey Hoffstein, Andreas Hulsing, Joost Rijneveld, John M. Schanck, Peter Schwabe, William Whyte, Zhenfei Zhang, Tsunekazu Saito, Takashi Yamakawa, and Keita Xagawa. Ntru. a submission to the nist post-quantum standardization effort, 2020. Available: <https://ntru.org/>.
- [CN11] Yuanmi Chen and Phong Q. Nguyen. BKZ 2.0: Better lattice security estimates. In Dong Hoon Lee and Xiaoyun Wang, editors, Advances in Cryptology - ASIACRYPT 2011 - 17th International Conference on the Theory and Application of Cryptology and Information Security, Seoul, South Korea, December 4-8, 2011. Proceedings, volume 7073 of Lecture Notes in Computer Science, pages 1–20, Berlin, Heidelberg, 2011. Springer.
- [DDLL13] Léo Ducas, Alain Durmus, Tancrede Lepoint, and Vadim Lyubashevsky. Lattice signatures and bimodal gaussians. In CRYPTO 2013, volume 8042 of Lecture Notes in Computer Science, pages 40–56, Berlin, Heidelberg, 2013. Springer.
- [DEP23] Léo Ducas, Thomas Espitau, and Eamonn W Postlethwaite. Finding short integer solutions when the modulus is small. In Annual International Cryptology Conference, pages 150–176. Springer, 2023.
- [DFPS22] Julien Devevey, Omar Fawzi, Alain Passelègue, and Damien Stehlé. On rejection sampling in lyubashevsky’s signature scheme. In ASIACRYPT 2022, volume 13794 of Lecture Notes in Computer Science, pages 34–64, Cham, 2022. Springer.
- [DFPS23] Julien Devevey, Pouria Fallahpour, Alain Passelègue, and Damien Stehlé. A detailed analysis of fiat-shamir with aborts. In CRYPTO 2023, volume 14085 of Lecture Notes in Computer Science, pages 327–357, Cham, 2023. Springer.
- [DHK<sup>+</sup>23] Julien Duman, Kathrin Hövelmanns, Eike Kiltz, Vadim Lyubashevsky, Gregor Seiler, and Dominique Unruh. A thorough treatment of highly-efficient NTRU instantiations. In Alexandra Boldyreva and Vladimir Kolesnikov, editors, Public-Key Cryptography - PKC 2023 - 26th IACR International Conference on Practice and Theory of Public-Key Cryptography, Atlanta, GA, USA, May 7-10, 2023, Proceedings, Part I, volume 13940 of Lecture Notes in Computer Science, pages 65–94, Cham, 2023. Springer.

- [DKL<sup>+</sup>18] Léo Ducas, Eike Kiltz, Tancrede Lepoint, Vadim Lyubashevsky, Peter Schwabe, Gregor Seiler, and Damien Stehlé. Crystals-dilithium: A lattice-based digital signature scheme. IACR Trans. Cryptogr. Hardw. Embed. Syst., 2018(1):238–268, 2018.
- [dPKPR24] Rafaël del Pino, Shuichi Katsumata, Thomas Prest, and Mélissa Rossi. Raccoon: A masking-friendly signature proven in the probing model. In Leonid Reyzin and Douglas Stebila, editors, Advances in Cryptology - CRYPTO 2024 - 44th Annual International Cryptology Conference, Santa Barbara, CA, USA, August 18-22, 2024, Proceedings, Part I, volume 14920 of Lecture Notes in Computer Science, pages 409–444. Springer, 2024.
- [DPS23] Julien Devevey, Alain Passelègue, and Damien Stehlé. G+G: A fiat-shamir lattice signature based on convolved gaussians. In Jian Guo and Ron Steinfeld, editors, Advances in Cryptology - ASIACRYPT 2023 - 29th International Conference on the Theory and Application of Cryptology and Information Security, Guangzhou, China, December 4-8, 2023, Proceedings, Part VII, volume 14444 of Lecture Notes in Computer Science, pages 37–64, Singapore, 2023. Springer.
- [EFGT17] Thomas Espitau, Pierre-Alain Fouque, Benoît Gérard, and Mehdi Tibouchi. Side-channel attacks on bliss lattice-based signatures: Exploiting branch tracing against strongswan and electromagnetic emanations in microcontrollers. In Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security, CCS '17, page 1857–1874, New York, NY, USA, 2017. Association for Computing Machinery.
- [ETWY22] Thomas Espitau, Mehdi Tibouchi, Alexandre Wallet, and Yang Yu. Shorter hash-and-sign lattice-based signatures. In Yevgeniy Dodis and Thomas Shrimpton, editors, CRYPTO 2022, volume 13508 of Lecture Notes in Computer Science, pages 245–275, Cham, 2022. Springer.
- [FHK<sup>+</sup>17] P.A. Fouque, J. Hoffstein, P. Kirchner, V. Lyubashevsky, T. Pornin, T. Prest, T. Ricosset, G. Seiler, W. Whyte, and Z. Zhang. Falcon: Fast-fourier lattice-based compact signatures over NTRU, 2017. Available: <https://falcon-sign.info/falcon.pdf>.
- [Gär25] Joel Gärtner. Compact lattice signatures via iterative rejection sampling. In Yael Taurman Kalai and Seny F. Kamara, editors, Advances in Cryptology - CRYPTO 2025 - 45th Annual International Cryptology Conference, Santa Barbara, CA, USA, August 17-21, 2025, Proceedings, Part I, volume 16000 of Lecture Notes in Computer Science, pages 514–547. Springer, 2025.
- [Gie14] Fabian Giesen. Interleaved entropy coders. CoRR, abs/1402.3392, 2014.
- [GLP12] Tim Güneysu, Vadim Lyubashevsky, and Thomas Pöppelmann. Practical lattice-based cryptography: A signature scheme for embedded systems. In Emmanuel Prouff and Patrick Schaumont, editors, Cryptographic Hardware and Embedded Systems - CHES 2012 - 14th International Workshop, Leuven, Belgium, September 9-12, 2012. Proceedings, volume 7428 of Lecture Notes in Computer Science, pages 530–547, Berlin, Heidelberg, 2012. Springer.
- [Gä24] Joel Gärtner. Compact lattice signatures via iterative rejection sampling. Cryptology ePrint Archive, Paper 2024/2052, 2024.



- [HHGP<sup>+</sup>03] Jeffrey Hoffstein, Nick Howgrave-Graham, Jill Pipher, Joseph H Silverman, and William Whyte. Ntrusign: Digital signatures using the ntru lattice. In Cryptographers’ track at the RSA conference, pages 122–140. Springer, 2003.
- [HPS98] Jeffrey Hoffstein, Jill Pipher, and Joseph H Silverman. Ntru: A ring-based public key cryptosystem. In Proceedings of the Third International Symposium on Algorithmic Number Theory, pages 267–288, 1998.
- [JRLS24] Corentin Jeudy, Adeline Roux-Langlois, and Olivier Sanders. Phoenix: hash-and-sign with aborts from lattice gadgets. In International Conference on Post-Quantum Cryptography, pages 265–299. Springer, 2024.
- [KLS18] Eike Kiltz, Vadim Lyubashevsky, and Christian Schaffner. A concrete treatment of fiat-shamir signatures in the quantum random-oracle model. In EUROCRYPT 2018, volume 10822 of Lecture Notes in Computer Science, pages 552–586, Cham, 2018. Springer.
- [LPR13] Vadim Lyubashevsky, Chris Peikert, and Oded Regev. A toolkit for ring-lwe cryptography. In Thomas Johansson and Phong Q. Nguyen, editors, Advances in Cryptology - EUROCRYPT 2013, 32nd Annual International Conference on the Theory and Applications of Cryptographic Techniques, Athens, Greece, May 26-30, 2013. Proceedings, volume 7881 of Lecture Notes in Computer Science, pages 35–54, Berlin, Heidelberg, 2013. Springer.
- [Lyu09] Vadim Lyubashevsky. Fiat-shamir with aborts: Applications to lattice and factoring-based signatures. In ASIACRYPT 2009, volume 5912 of Lecture Notes in Computer Science, pages 598–616, Berlin, Heidelberg, 2009. Springer.
- [Lyu11] Vadim Lyubashevsky. Lattice signatures without trapdoors. Cryptology ePrint Archive, Paper 2011/537, 2011.
- [Lyu12] Vadim Lyubashevsky. Lattice signatures without trapdoors. In David Pointcheval and Thomas Johansson, editors, EUROCRYPT 2012, volume 7237 of Lecture Notes in Computer Science, pages 738–755, Berlin, Heidelberg, 2012. Springer.
- [MM11] Daniele Micciancio and Petros Mol. Pseudorandom knapsacks and the sample complexity of LWE search-to-decision reductions. In Phillip Rogaway, editor, Advances in Cryptology - CRYPTO 2011 - 31st Annual Cryptology Conference, Santa Barbara, CA, USA, August 14-18, 2011. Proceedings, volume 6841 of Lecture Notes in Computer Science, pages 465–484. Springer, 2011.
- [PBY17] Peter Pessl, Leon Groot Bruinderink, and Yuval Yarom. To BLISS-B or not to be: Attacking strongswan’s implementation of post-quantum signatures. In Bhavani Thuraisingham, David Evans, Tal Malkin, and Dongyan Xu, editors, Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security, CCS 2017,., pages 1843–1855, New York, NY, USA, 2017. ACM.
- [Por19] Thomas Pornin. New efficient, constant-time implementations of falcon. Cryptology ePrint Archive, Paper 2019/893, 2019.

- [Pre17] Thomas Prest. Sharper bounds in lattice-based cryptography using the rényi divergence. In Tsuyoshi Takagi and Thomas Peyrin, editors, Advances in Cryptology – ASIACRYPT 2017, pages 347–374, Cham, 2017. Springer International Publishing.
- [PRR19] Thomas Prest, Thomas Ricosset, and Mélissa Rossi. Simple, fast and constant-time gaussian sampling over the integers for falcon. Post-Quantum Cryptography Project of NIST, 2019.
- [ZFY23] Jiang Zhang, Dengguo Feng, and Di Yan. NEV: faster and smaller NTRU encryption using vector decoding. In Jian Guo and Ron Steinfeld, editors, Advances in Cryptology - ASIACRYPT 2023 - 29th International Conference on the Theory and Application of Cryptology and Information Security, Guangzhou, China, December 4-8, 2023, Proceedings, Part VII, volume 14444 of Lecture Notes in Computer Science, pages 157–189, Singapore, 2023. Springer.
- [ZSS20] Raymond K. Zhao, Ron Steinfeld, and Amin Sakzad. Facct: Fast, compact, and constant-time discrete gaussian sampler over integers. IEEE Transactions on Computers, 69(1):126–137, 2020.

## A Re-parameterized BLISS Parameters

To re-parametrize BLISS, we adjusted the secret key distribution to a fixed Hamming weight, following the design of NTRU+Sign, and applied the HAETAE method to tighten the bound on  $\|\mathbf{Sc}\|$ . In addition, while keeping the dimension  $n$  unchanged, we considered enlarging the modulus  $q$  to ensure sufficient SIS security levels. The detailed parameter sets are provided in Table 5.

Table 5: Parameter sets for BLISS

| Parameter sets                         |                                                  | 1         | 2         |
|----------------------------------------|--------------------------------------------------|-----------|-----------|
| $n$                                    | Degree of a polynomial                           | 512       | 1,024     |
| $q$                                    | Modulus                                          | 13,313    | 13,313    |
| $\tau$                                 | Hamming weight of $\mathbf{c}$                   | 20        | 36        |
| $\lfloor \log_2  \mathcal{C}  \rfloor$ | $\lfloor \log_2 \binom{n}{\tau} \rfloor$         | 118       | 221       |
| $d$                                    | # of dropped bits in the commitment              | 8         | 9         |
| $M$                                    | Expected # of repetitions for rejection sampling | 5.31      | 4.74      |
| $M_{Eq}$                               | Expected # of repetitions for equality check     | 1.08      | 1.07      |
| $M_{total}$                            | $= M \times M_{Eq}$                              | 5.74      | 5.08      |
| $B_{\mathbf{Sc}}$                      | Upper-bound for Euclidean norm of $\mathbf{Sc}$  | 530       | 1,270     |
| $\gamma$                               | $sk$ rejection parameter                         | 118.55    | 211.53    |
|                                        | $sk$ acceptance rate                             | 0.25      | 0.25      |
| $\sigma$                               | Standard deviation of $D_\sigma$                 | 290       | 720       |
| $\hat{\alpha}$                         | $= \sigma / B_{\mathbf{Sc}}$                     | 0.55      | 0.57      |
| $\alpha$                               | Min-entropy of the commitment                    | 770       | 1,860     |
| $B_2$                                  | Verification threshold in Euclidean norm         | 13,000    | 42,000    |
| $B_\infty$                             | Verification threshold in infinite norm          | 3,198     | 6,142     |
| NTRU Hardness (Core-SVP)               |                                                  |           |           |
|                                        | BKZ block-size $b$                               | 298       | 659       |
|                                        | Classical Core-SVP                               | 87        | 192       |
|                                        | Quantum Core-SVP                                 | 76        | 169       |
| SIS Hardness (Core-SVP)                |                                                  |           |           |
|                                        | BKZ block-size $b$                               | 321 (234) | 675 (542) |
|                                        | Classical Core-SVP                               | 93 (68)   | 197 (158) |
|                                        | Quantum Core-SVP                                 | 82 (60)   | 173 (139) |

## B Discrete Gaussian and Rejection Sampling

We provide a detailed description for the constant-time implementations of discrete Gaussian sampling and rejection sampling, including their algorithm, constraints for related parameters, and concrete parameters. Moreover, the security loss incurred by replacing the ideal sampling with its constant-time implementation is analyzed. To this end, we adopt the methodology presented in [BLRL<sup>+</sup>18, Pre17]. We first recall the definition and useful lemma for Rényi divergence.

**Definition B.1.** Let  $\mathcal{P}, \mathcal{Q}$  be two distributions such that  $\text{Supp}(\mathcal{P}) \subseteq \text{Supp}(\mathcal{Q})$  where  $\text{Supp}()$  means support set of given distribution. For  $a \in (1, +\infty)$ , we define the Rényi divergence of order  $a$  by

$$R_a(\mathcal{P} \mid \mathcal{Q}) = \left( \sum_{x \in \text{Supp}(\mathcal{P})} \frac{\mathcal{P}(x)^a}{\mathcal{Q}(x)^{a-1}} \right)^{\frac{1}{a-1}}.$$

In addition, the Rényi divergence of order  $+\infty$  is defined by  $R_\infty(\mathcal{P} \mid \mathcal{Q}) = \max_{x \in \text{Supp}(\mathcal{P})} \mathcal{P}(x)/\mathcal{Q}(x)$ .

**Lemma B.2** ([BLRL<sup>+</sup>18, Lemma 2.9]). For two distributions  $\mathcal{P}, \mathcal{Q}$  and two families of distributions  $(\mathcal{P}_i)_i, (\mathcal{Q}_i)_i$ , the Rényi divergence verifies the following properties:

- **Data processing inequality.** For any function  $f$ ,  $R_a(f(\mathcal{P}) \mid f(\mathcal{Q})) \leq R_a(\mathcal{P} \mid \mathcal{Q})$ .
- **Multiplicativity.**  $R_a(\prod_i \mathcal{P}_i \mid \prod_i \mathcal{Q}_i) = \prod_i R_a(\mathcal{P}_i \mid \mathcal{Q}_i)$ .
- **Probability preservation.** For any event  $E \subseteq \text{Supp}(\mathcal{Q})$  and  $a \in (1, +\infty)$ ,

$$\mathcal{Q}(E) \geq \mathcal{P}(E)^{a/(a-1)} / R_a(\mathcal{P} \mid \mathcal{Q}), \quad (4)$$

$$\mathcal{Q}(E) \geq \mathcal{P}(E) / R_\infty(\mathcal{P} \mid \mathcal{Q}). \quad (5)$$

- **Weak triangle inequality.** For  $P_1, P_2, P_3$  with  $\text{Supp}(P_i) \subseteq \text{Supp}(P_{i+1})$ :

$$R_a(P_1 \mid P_3) \leq \begin{cases} R_a(P_1 \mid P_2) \cdot R_\infty(P_2 \mid P_3), \\ R_\infty(P_1 \mid P_2)^{a/(a-1)} \cdot R_a(P_2 \mid P_3) & \text{if } a \in (1, +\infty). \end{cases}$$

Security loss between constant-time implementation and ideal one is analyzed with Rényi divergence in the following way. Let  $\Phi$  (resp.  $\Phi'$ ) denote the distribution of signatures in the view of forging adversary when he interact with the real signing oracle (resp. ideal signing oracle), and let  $\epsilon$  denote its success probability of forging a signature against the real signing oracle. Then, the probability preservation property of Rényi divergence directly shows that the adversary succeed in forging a signature with probability  $\epsilon' \geq \epsilon^{a/(a-1)} / R_a(\Phi \mid \Phi')$  against the ideal signing oracle. We now suppose that the real signing oracle differs from the ideal oracle in that the distribution  $\chi'$  is replaced with  $\chi$ , and that the distribution is called at most  $\mathcal{Q}$  times. It is clear that  $R_a(\Phi \mid \Phi')$  is upper bounded by  $R_a(\chi^{\mathcal{Q}} \mid \chi'^{\mathcal{Q}})$ , which is equal to  $R_a(\chi \mid \chi')^{\mathcal{Q}}$  by the multiplicativity property of Rényi divergence. Combining these, we obtain  $\epsilon' \geq \epsilon^{a/(a-1)} / R_a(\chi \mid \chi')^{\mathcal{Q}}$ , which yields a lower bound on the forging probability of the adversary against the ideal signing oracle. Based on this lower bound, we can derive a contradiction under the assumption of an adversary against the real signing oracle, and we further analyze the security loss with respect to the adversary's success probabilities.

To estimate security loss of the constant-time implementation for NTRU+Sign, three signing oracles are introduced as follows.

1. Ideal: The signing algorithm consists of the ideal distribution (including  $D_\sigma$  and  $\mathcal{B}_{\Psi(\mathbf{z}, \mathbf{v})}$ ).
2. Inter: The signing algorithm employs  $\tilde{\mathcal{B}}_{\Psi(\mathbf{z}, \mathbf{v})}$  in place of  $\mathcal{B}_{\Psi(\mathbf{z}, \mathbf{v})}$ , as compared to the Ideal case.
3. Real: The signing algorithm employs  $\tilde{D}_\sigma$  in place of  $D_\sigma$ , as compared to the Inter case.

Here  $\tilde{D}_\sigma$  and  $\tilde{\mathcal{B}}_{\Psi(\mathbf{z}, \mathbf{v})}$  denote the output distributions of the constant-time implementation of  $D_\sigma$  and  $\mathcal{B}_{\Psi(\mathbf{z}, \mathbf{v})}$ , respectively. Let  $\Phi_{\text{case}}$  and  $\epsilon_{\text{case}}$  denote the distribution of signatures in the view of the adversary and the corresponding forging probability for  $\text{case} \in \{\text{Real}, \text{Inter}, \text{Ideal}\}$ . Moreover, given the maximum number of signing queries  $Q_{\text{sign}}$ , the maximum number of call for  $D_\sigma$  is set to  $Q_{\text{gauss}} = M \cdot 2n \cdot Q_{\text{sign}}$  as results of  $M$  restarts during signing and  $2n$  calls to discrete Gaussian sampler for  $y_1$  and  $y_2$ . Similarly, the maximum number of call for  $\mathcal{B}_{\Psi(\mathbf{z}, \mathbf{v})}$  is set to  $Q_{\text{reject}} = M \cdot Q_{\text{sign}}$ , which is resulted from  $M$  restarts during signing.

Let  $\lambda$  be the security parameter of the real cases. The values  $R_{2\lambda+1}(\Phi_{\text{Real}}, \Phi_{\text{Inter}})$  and  $R_{2\lambda+2}(\Phi_{\text{Inter}}, \Phi_{\text{Real}})$  will be used to quantify the differences over each case. In first step, we analyze the security loss with the values of  $R_{2\lambda+1}(\tilde{\mathcal{B}}_{\Psi(\mathbf{z}, \mathbf{v})}, \mathcal{B}_{\Psi(\mathbf{z}, \mathbf{v})})$  and  $R_{2\lambda+2}(\tilde{D}_\sigma, D_\sigma)$  in following theorem. In second step, we define the requirements for constant-time implementation to achieve the desired bounds on the Rényi divergence.

**Theorem B.3.** Suppose that  $R_{2\lambda+1}(\tilde{\mathcal{B}}_{\Psi(\mathbf{z}, \mathbf{v})} | \mathcal{B}_{\Psi(\mathbf{z}, \mathbf{v})}) \leq 1 + \frac{1}{4Q_{\text{reject}}}$  and  $R_{2\lambda+2}(\tilde{D}_\sigma | D_\sigma) \leq 1 + \frac{1}{4Q_{\text{gauss}}}$ . If NTRU+Sign has  $(\lambda + 2)$ -bit secure against any forging adversary in Ideal case, then its security is lower bounded by  $\lambda$ -bit in Real case.

*Proof.* Let  $\epsilon_{\text{Real}}$ ,  $\epsilon_{\text{Inter}}$  and  $\epsilon_{\text{Ideal}}$  denote that forging probability of the any forging adversary against NTRU+Sign in each case, respectively. Suppose that NTRU+Sign in real case does not achieve  $\lambda$ -bit security. Then, we have  $\epsilon_{\text{Real}} \geq 2^{-\lambda}$ , which implies that  $\epsilon_{\text{Real}}^{(2\lambda+1)/2\lambda} \geq \epsilon_{\text{Real}}/\sqrt{2}$ . Moreover, by assumption, we have  $R_{2\lambda+1}(\tilde{\mathcal{B}}_{\Psi(\mathbf{z}, \mathbf{v})} | \mathcal{B}_{\Psi(\mathbf{z}, \mathbf{v})})^{Q_{\text{reject}}} \leq e^{1/4} \leq \sqrt{2}$ . Then, by the probability preservation of Rényi divergence, it follows that

$$\begin{aligned} \epsilon_{\text{Inter}} &\geq \epsilon_{\text{Real}}^{(2\lambda+1)/2\lambda} / R_{2\lambda+1}(\tilde{\mathcal{B}}_{\Psi(\mathbf{z}, \mathbf{v})} | \mathcal{B}_{\Psi(\mathbf{z}, \mathbf{v})})^{Q_{\text{reject}}} \\ &\geq \epsilon_{\text{Real}}/2 \\ &\geq 2^{-\lambda-1}. \end{aligned}$$

Similarly, we can derive  $\epsilon_{\text{Ideal}} \geq 2^{-\lambda-2}$  from  $\epsilon_{\text{Inter}} \geq 2^{-\lambda-1}$  and assumption. In summary,  $\epsilon_{\text{Real}} \geq 2^{-\lambda}$  implies  $\epsilon_{\text{Ideal}} \geq 2^{-\lambda-2}$ . By contraposition, we can derive  $\epsilon_{\text{Real}} < 2^{-\lambda}$ , provided that  $\epsilon_{\text{Ideal}} < 2^{-\lambda-2}$  holds. Therefore, a  $(\lambda + 2)$ -bit secure NTRU+Sign in Ideal case remains at least  $\lambda$ -bit secure when replacing  $\mathcal{B}_{\Psi(\mathbf{z}, \mathbf{v})}$  with  $\tilde{\mathcal{B}}_{\Psi(\mathbf{z}, \mathbf{v})}$  and  $D_\sigma$  with  $\tilde{D}_\sigma$ , assuming the required conditions are satisfied.  $\square$

Given a probability distribution  $\mathcal{D}$ , we denote  $\tilde{\mathcal{D}}$  as a probability distribution similar to  $\mathcal{D}$ . Let  $x \sim \mathcal{D}$  mean that  $x$  follows the probability distribution  $\mathcal{D}$  and  $x \sim \tilde{\mathcal{D}}$  indicate that  $x$  follows a probability distribution similar to  $\mathcal{D}$ . The relative error of  $\mathcal{D}_1$  from  $\mathcal{D}_2$  is defined as  $\left| \frac{\mathcal{D}_1}{\mathcal{D}_2} - 1 \right| = \max_{x \in \text{supp}(\mathcal{D}_1)} \left| \frac{\mathcal{D}_1(x)}{\mathcal{D}_2(x)} - 1 \right|$ . For  $I \subseteq \mathbb{Z}$  and  $x \in I$ , let us define  $D_{I, \sigma}(x) = \frac{\rho_\sigma(x)}{\rho_\sigma(I)}$  as discrete Gaussian distribution over  $I$  with standard deviation  $\sigma$  and center at 0. For  $p \in [0, 1]$ , let  $\mathcal{B}_p$  denote a Bernoulli distribution with probability  $p$ . For  $a, b \in \mathbb{R}$ , we define  $[a, b]_\theta = \{x \in [a, b] : |x| - \lfloor |x| \rfloor = \sum_{i=1}^\theta x_i 2^i, x_i \in \{0, 1\}\}$ . Similarly,  $(a, b)_\theta$ ,  $[a, b)_\theta$  and  $(a, b)_\theta$  are also defined. For a predicate  $P$ ,  $\llbracket P \rrbracket$  returns 1 if  $P$  is true, and 0 otherwise. Lastly, we use the notation  $f(\delta) \underset{\delta \rightarrow 0}{\sim} c \iff f(\delta) \approx c$  only when  $|\delta| \approx 0^6$ .

<sup>6</sup>The notation is derived from [Pre17], and approximate values are used for brevity in this paper.

## B.1 Discrete Gaussian

### B.1.1 Algorithms

To sample from the discrete Gaussian distribution  $\mathcal{D}_\sigma$ , we employ a constant-time discrete Gaussian sampling algorithm whose output distribution closely approximates the target distribution. More specifically, we use a constant-time variant of the approach originally taken in BLISS [DDLL13], which relies on sampling from a positive discrete Gaussian distribution with a small standard deviation followed by rejection sampling.

The constant-time approximated discrete Gaussian sampler is described in Algorithm 1. Briefly explained, the sampler generates a non-negative integer  $y$  from an approximated distribution of the non-negative Gaussian, which is described in Algorithm 2, and then determines its sign. Here are some relevant parameters:

$$k = 2^{\lfloor \log_2 \sigma \rfloor}, \quad \sigma_1 = \sigma/k, \quad w_1 = \lfloor \tau_1 \sigma_1 \rfloor.$$

---

#### Algorithm 1 $\text{ApproxG}_\sigma$

---

**Input** None

**Output**  $y \in [-(w_1+1)k+1, (w_1+1)k-1]$

- 1:  $y \leftarrow \text{ApproxG}_\sigma^+(\cdot)$   $// y \sim D_{\mathbb{Z}_{\geq 0}, \sigma}$
  - 2: **if**  $y = 0$
  - 3:   **restart** with probability  $\frac{1}{2}$
  - 4:  $b \leftarrow \mathcal{U}(\{0, 1\})$
  - 5:  $y \leftarrow (2b - 1)y$
  - 6: **return**  $y$
- 

The approximated sampler  $\text{ApproxG}_\sigma^+$  for the non-negative Gaussian is represented in Algorithm 2. In broad terms,  $\text{ApproxG}_\sigma^+$  samples  $y_1$  from a discrete Gaussian distribution with a smaller standard deviation  $\sigma_1 = \sigma/k$  and  $y_0$  from the uniform distribution on  $[0, k-1]$ , and returns  $y = y_1k + y_0$  conditionally.

---

#### Algorithm 2 $\text{ApproxG}_\sigma^+$

---

**Input** None

**Output**  $y \in [0, (w_1+1)k-1]$

- 1:  $y_1 \leftarrow \text{ApproxBaseG}_{\sigma_1}^+(\cdot)$   $// y_1 \sim D_{\mathbb{Z}_{\geq 0}, \sigma_1}$
  - 2:  $y_0 \leftarrow \mathcal{U}([0, k-1])$
  - 3:  $x \leftarrow -y_0(y_0 + 2y_1k)$
  - 4:  $b \leftarrow \text{SampleBernExp}_\sigma^{I_1}(x)$   $// b \sim \mathcal{B}_{\exp(x/2\sigma^2)}$
  - 5: **if**  $b = 0$
  - 6:   **restart**
  - 7:  $y \leftarrow y_1k + y_0$
  - 8: **return**  $y$
- 

In Algorithm 2, the subroutine  $\text{ApproxBaseG}_{\sigma_1}^+$  outputs an integer  $y_1$  which is distributed according to a

probability distribution  $\tilde{D}_{[0, w_1], \sigma_1}$  that can be considered as a  $\theta_1$ -precision version of  $D_{[0, w_1], \sigma_1}$ :

$$\tilde{D}_{[0, w_1], \sigma_1}(x) = \begin{cases} \lfloor D_{[0, w_1], \sigma_1}(x) \cdot 2^{\theta_1} \rfloor \cdot 2^{-\theta_1} & , (x \in [1, w_1]) \\ 1 - \sum_{i=1}^{w_1} \tilde{D}_{[0, w_1], \sigma_1}(i) & , (x = 0) \end{cases}$$

Note that for sufficiently large  $\tau_1 = w_1/\sigma_1$  the output of Algorithm 3 actually is distributed according to  $D_{\mathbb{Z}_{\geq 0}, \sigma_1}$ . In Algorithm 3,  $\text{cdt}[i] = 2^{\theta_1} \cdot \Pr[z < i : z \leftarrow \tilde{D}_{[0, w_1], \sigma_1}]$  for  $i \in [0, w_1]$ .

---

**Algorithm 3** ApproxBaseG $^+_{\sigma_1}$

---

**Input** None

**Output**  $y_1 \in [0, w_1]$

- 1:  $u \leftarrow \mathcal{U}([0, 2^{\theta_1} - 1])$
  - 2:  $y_1 \leftarrow 0$
  - 3: **for**  $i = 0$  **to**  $w_1 - 1$  **do**
  - 4:    $y_1 \leftarrow y_1 + \llbracket u > \text{cdt}[i] \rrbracket$
  - 5: **return**  $y_1$
- 

Another subroutine  $\text{SampleBernExp}_{\sigma}^{I_1}(x)$  of Algorithm 2 describes an approximated version of the Bernoulli distribution  $\mathcal{B}_{\exp(x/2\sigma^2)}$  for  $x \in I_1$ . To do this, it uses a polynomial approximation  $P_{\exp}^{I_1}(x)$  of  $\exp(x/2\sigma^2)$  over the interval  $I_1$ . Note that the property of  $P_{\exp}^{I_1}(x)$  necessary to ensure security is described in following section. As a result,  $\text{SampleBernExp}_{\sigma}^{I_1}(x)$  returns a bit by comparing a uniform random number  $u$  in  $[0, 1]_{\vartheta_1}$  and the output of  $\text{EvaluatePoly}[P_{\exp}^{I_1}, \vartheta_1](x)$ , which is an approximation of  $P_{\exp}^{I_1}(x)$ .

---

**Algorithm 4** SampleBernExp $_{\sigma}^{I_1}(x)$

---

**Input**  $x \in I_1 \cap \mathbb{Z}$

**Output**  $b \in \{0, 1\}$

- 1:  $\hat{p} \leftarrow \text{EvaluatePoly}[P_{\exp}^{I_1}, \vartheta_1](x)$  //  $\hat{p} \in [0, 1]_{\vartheta_1}$
  - 2:  $u \leftarrow \mathcal{U}([0, 1]_{\vartheta_1})$
  - 3: **if**  $u \leq \hat{p}$
  - 4:   **return** 1
  - 5: **return** 0
- 

In Algorithm 4, the output of  $\text{EvaluatePoly}[P_{\exp}^{I_1}, \vartheta_1](x)$  provides the  $\vartheta_1$ -bit precision value of  $P_{\exp}^{I_1}(x) = \sum_{i=0}^d p_i x^i$  where  $p_0 \in [0, 1]_{\vartheta_1}$  and  $p_i = \tilde{p}_i \cdot 2^{-\ell_i}$ ,  $\tilde{p}_i \in [0, 1]_{\vartheta_1}$ ,  $\ell_i \in \mathbb{N}$  for  $i = 1, \dots, d$ . Basically,  $P_{\exp}^{I_1}(x) = \sum_{i=0}^d p_i x^i$  is computed by adapting Horner's rule as follows:

$$P_{\exp}^{I_1}(x) = (((p_d \cdot x + p_{d-1}) \cdot x) + p_{d-2}) \cdot x + \dots \cdot x + p_0.$$

On the other hand, since  $P_{\exp}^{I_1}$  is an approximation of  $\exp(-x)$ , its coefficients are decreasing in the sense that  $1 = |p_0| \gg |p_1| \gg \dots \gg |p_d|$ , i.e.,  $\ell_0 < \ell_1 < \ell_2 < \dots < \ell_d$ . So the coefficients of  $P_{\exp}^{I_1}$  can be expressed as

$$p_i = \tilde{p}_i \cdot 2^{-\sum_{j=1}^i s_j}, \quad \tilde{p}_i \in [0, 1]_{\vartheta_1}$$

where  $s_1 = \ell_1$  and  $s_j = \ell_j - \ell_{j-1} \in \mathbb{N}$  for  $j = 2, \dots, d$ . Using this representation, the Horner's rule can be written as

$$P_{\text{exp}}^{I_1}(x) = (((\tilde{p}_d \cdot 2^{-s_d} \cdot x + \tilde{p}_{d-1}) \cdot 2^{-s_{d-1}} \cdot x) + \tilde{p}_{d-2}) \cdot 2^{-s_{d-2}} \cdot x + \dots) \cdot x \cdot 2^{-s_1} + p_0,$$

and Algorithm 5 adapts this representation as the following sequential execution:

$$y_d = \tilde{p}_d \cdot x \cdot 2^{-s_d}, y_{d-1} = (y_d + \tilde{p}_{d-1}) \cdot x \cdot 2^{-s_{d-1}}, \dots, y_1 = (y_2 + \tilde{p}_1) \cdot x \cdot 2^{-s_1}, y_0 = y_1 + p_0.$$

Actually, Algorithm 5 maintains  $\vartheta_1$ -bit precision during computation using only the integer arithmetic of a fixed size, which results in computing the approximation of  $y_i$ . For simplicity, when a polynomial  $P = \sum_{i=0}^d p_i x^i$  and  $\vartheta$  are given as the parameters of Algorithm 5, we assume that two sequences  $(\tilde{p}_i)_{0 \leq i \leq d}, (s_i)_{0 \leq i \leq d}$  satisfying  $p_i = \tilde{p}_i \cdot 2^{-\sum_{j=1}^i s_j}, \tilde{p}_i \in [0, 1)_{\vartheta}$  for  $i = 0, \dots, d$  are given.

---

**Algorithm 5** EvaluatePoly $[P_{\text{exp}}^{I_1}, \vartheta_1](x)$

---

**Input**  $x \in I_1$

**Output**  $\tilde{y} \in [0, 1)_{\vartheta_1}$

- 1:  $\tilde{y}_d \leftarrow \lfloor (\tilde{p}_d \cdot x) \cdot 2^{\vartheta_1 - s_d} \rfloor \cdot 2^{-\vartheta_1}$
  - 2: **for**  $i = d - 1$  **to** 1 **do**
  - 3:    $\tilde{y}_i \leftarrow \lfloor ((\tilde{y}_{i+1} + \tilde{p}_i) \cdot x) \cdot 2^{\vartheta_1 - s_i} \rfloor \cdot 2^{-\vartheta_1}$
  - 4:  $\tilde{y} \leftarrow \tilde{y}_1 + p_0$
  - 5: **return**  $\tilde{y}$
-



### B.1.2 Constant-Time Execution

We recall semi-formal definition of constant-time algorithm.

**Definition B.4** ([PRR19, Definition 2]). Let  $\mathcal{A}$  be an (probabilistic or deterministic) algorithm.  $\mathcal{A}$  is said to be constant-time if its execution time is independent of its input and output. In addition,  $\mathcal{A}$  is still considered to be constant-time with the following relaxations:

- If a subset  $\mathcal{I}$  (resp.  $\mathcal{O}$ ) of the input (resp. output) is tagged as sensitive,  $\mathcal{A}$  is still considered as constant-time if its running time is independent of the values taken by  $\mathcal{I}$  (resp.  $\mathcal{O}$ ).
- If its running time is negligibly close (for a clearly identified divergence) to  $\mathcal{D}$  where distribution  $\mathcal{D}$  is independent of the (sensitive) input/output.

In addition, we may extend the notion of being constant-time of algorithmic objects such as conditional loops: if an algorithm contains a conditional loop which loops over a number of time which is independent of sensitive values, we abusively say that this loop is constant-time. Furthermore, it is clear that the composition of constant-time elements is constant-time.

To prove that  $\text{ApproxG}_\sigma$  is constant-time in the sense of Definition B.4, we show that both the element  $\text{ApproxG}_\sigma^+$  (Algorithm 3) and the expected number of iterations in  $\text{ApproxG}_\sigma$  are independent of its output  $y$ . We observe that the expected number of iterations in  $\text{ApproxG}_\sigma$  is  $D_\sigma(0)/2$ , which is a known constant and independent of its output  $y$ . Therefore, it suffices to verify the  $\text{ApproxG}_\sigma^+$  is constant-time. Assume that integer arithmetic is constant-time. If  $\text{ApproxBaseG}_{\sigma_1}^+$  (Algorithm 3),  $\text{SampleBernExp}_\sigma^{I_1}$  (Algorithm 4) and the expected number of iterations are constant-time, then  $\text{ApproxG}_\sigma^+$  is constant-time. In  $\text{ApproxBaseG}_{\sigma_1}^+$ , it operates by reading the entire precomputed table (CDT), thus its execution time is independent of its output value  $y_1$ . In  $\text{SampleBernExp}_\sigma^{I_1}$ ,  $\text{EvaluatePoly}$  (Algorithm 5) is applied to evaluate the approximate polynomial in Horner's form, and a comparing result between the evaluated value and a random value is returned. Since both  $\text{EvaluatePoly}$  and the comparing consist only of integer arithmetics, the execution times are independent of both its input and output. In  $\text{ApproxG}_\sigma^+$ , the average probability of restart is

$$\sum_y \underbrace{\frac{1}{k} \cdot D_{\sigma_1}(y_1)}_{\text{prob. of } y} \cdot \underbrace{\exp\left(-\frac{y_0(y_0 + 2y_1k)}{2\sigma^2}\right)}_{\text{accepting prob. of } y} = \frac{\rho_\sigma(\mathbb{Z}^+)}{k \cdot \rho_{\sigma_1}(\mathbb{Z}^+)} \cdot \sum_y D_\sigma(y) = \frac{\rho_\sigma(\mathbb{Z}^+)}{k \cdot \rho_{\sigma_1}(\mathbb{Z}^+)}$$

where  $y = ky_1 + y_0$ . Accordingly, the expected number of iterations is known constant and independent with output value  $y$ . Therefore,  $\text{ApproxG}_\sigma^+$  is constant-time as a composition of constant-time elements.

In summary,  $\text{ApproxG}_\sigma^+$  and loop in  $\text{ApproxG}_\sigma$  are constant-time. Therefore, we conclude that  $\text{ApproxG}_\sigma$  is constant-time.

### B.1.3 Constraints for Parameters

Let  $Q_{\text{sign}} = 2^{64}$  be the maximum number of signing queries and define  $Q_{\text{gauss}} = M \cdot 2n \cdot Q_{\text{sign}}$ . Recall that  $k = 2^{\lfloor \log_2 \sigma \rfloor}$ ,  $\sigma_1 = \sigma/k$ , and  $w_1 = \lfloor \tau_1 \sigma_1 \rfloor$ .

$\tau_1$  is a value used to restrict the support of a discrete Gaussian distribution with a small standard deviation  $\sigma_1$ , implemented through the cumulative distribution table (CDT). Specifically,  $\tau_1$  is set to satisfy the

following condition:

$$\frac{1}{1 - 2C(\tau_1)} \leq c_g \cdot \left(1 + \frac{1}{16Q_{\text{gauss}}}\right) \quad (6)$$

where  $c_g = \sqrt{1 - 1/(2^5 Q_{\text{gauss}})}$  is a constant used for proper scaling to adjust inequalities and is chosen to be a real number close to  $1 - 2^{-47}$ .

$\theta_1$   $\theta_1$  represents the precision required when approximating  $D_{[0, w_1], \sigma_1}$  using CDT. According to the way CDT is used in Algorithm 4,  $\theta_1$  needs to be set for satisfying  $R_\infty(\tilde{D}_{[0, w_1], \sigma_1} \mid D_{[0, w_1], \sigma_1}) \leq 1 + \frac{1}{16Q_{\text{gauss}}}$ , which holds when

$$2^{\theta_1} \geq w_1 \cdot \rho_{\sigma_1}([0, w_1]) \cdot \left(c_g \cdot \left(1 + \frac{1}{16Q_{\text{gauss}}}\right) - 1\right)^{-1} \quad (7)$$

$P_{\text{exp}}^{I_1}$   $P_{\text{exp}}^{I_1}$  is a polynomial approximation for  $\exp(-\frac{x}{2\sigma^2})$  over  $I_1 = (-\lfloor 2\sigma^2 \rfloor, 0]$ . The required properties for the approximation polynomial  $P_{\text{exp}}^{I_1}$  are as follows:

$$\left| \frac{P_{\text{exp}}^{I_1}(x)}{\exp(x/(2\sigma^2))} - 1 \right| \leq \frac{1}{2^2 \sqrt{(2\lambda+2)Q_{\text{gauss}}}}, \quad (8)$$

$$\left| \frac{\exp(x/(2\sigma^2)) - P_{\text{exp}}^{I_1}(x)}{1 - \exp(x/(2\sigma^2))} \right| \leq \frac{1}{2^2 \sqrt{(2\lambda+2)Q_{\text{gauss}}}} \quad (9)$$

for all  $x \in I_1$ . We find a polynomial satisfying (8) by using the method proposed in GALATICS, and (9) is checked experimentally.

$\vartheta_1$   $\vartheta_1$  represents the precision required for evaluating the polynomial  $P_{\text{exp}}^{I_1}$  at  $x \in I_1$ . It needs to satisfy

$$2^{\vartheta_1} \geq \frac{2}{1 - \exp(-1/2\sigma^2)} \sqrt{(2\lambda+2)Q_{\text{gauss}}}. \quad (10)$$

#### B.1.4 Concrete parameters

The parameters for the constant-time discrete Gaussian sampler are set as follows:

| parameter set                             | 1                 | 2                  |
|-------------------------------------------|-------------------|--------------------|
| $\lambda$                                 | 93                | 211                |
| $n$                                       | 512               | 1024               |
| $B_{\mathbf{Sc}}$                         | 169               | 341                |
| $\sigma$                                  | 110               | 200                |
| $\hat{\alpha} (= \sigma/B_{\mathbf{Sc}})$ | 0.61              | 0.58               |
| $M (= \exp(1/2\hat{\alpha}^2))$           | 3.25              | 4.28               |
| $k (= 2^{\lceil \log_2 \sigma \rceil})$   | $2^6$             | $2^7$              |
| $\sigma_1 (= \sigma/k)$                   | 1.71875           | 1.5625             |
| $\tau_1$                                  | 9                 | 9                  |
| $w_1 (= \lfloor \tau_1 \sigma_1 \rfloor)$ | 15                | 14                 |
| $\theta_1$                                | 86                | 87                 |
| $\vartheta_1$                             | 58                | 61                 |
| $P_{\text{exp}}^{I_1}$                    | $P_{\text{g}512}$ | $P_{\text{g}1024}$ |

Table 6: Parameters for constant-time discrete Gaussian sampler

| $i$ | cdt[ $i$ ]                         |
|-----|------------------------------------|
| 0   | 29,151,226,717,600,037,870,661,691 |
| 1   | 53,763,390,821,665,321,263,532,167 |
| 2   | 68,575,902,413,396,088,331,456,359 |
| 3   | 74,930,580,255,359,372,543,360,029 |
| 4   | 76,873,900,198,393,993,982,160,770 |
| 5   | 77,297,524,641,422,720,945,905,655 |
| 6   | 77,363,351,578,546,129,243,855,570 |
| 7   | 77,370,643,000,298,910,318,530,342 |
| 8   | 77,371,218,714,211,971,522,174,381 |
| 9   | 77,371,251,117,356,994,444,585,424 |
| 10  | 77,371,252,417,387,484,525,835,338 |
| 11  | 77,371,252,454,567,181,491,505,901 |
| 12  | 77,371,252,455,325,137,652,441,583 |
| 13  | 77,371,252,455,336,152,240,596,897 |
| 14  | 77,371,252,455,336,266,338,686,520 |
| 15  | 77,371,252,455,336,267,181,195,263 |

Figure 4: CDT for parameter set 1

| $i$ | cdt[ $i$ ]                          |
|-----|-------------------------------------|
| 0   | 62,946,871,698,369,966,505,011,122  |
| 1   | 114,236,628,729,620,172,213,724,308 |
| 2   | 141,982,611,487,569,176,724,311,357 |
| 3   | 151,947,732,170,118,402,385,414,151 |
| 4   | 154,323,904,823,045,118,016,279,667 |
| 5   | 154,700,076,769,483,578,589,249,467 |
| 6   | 154,739,614,141,168,150,315,796,150 |
| 7   | 154,742,373,080,603,193,508,246,173 |
| 8   | 154,742,500,897,867,616,040,036,121 |
| 9   | 154,742,504,829,291,305,376,616,883 |
| 10  | 154,742,504,909,574,225,795,514,989 |
| 11  | 154,742,504,910,662,678,228,546,535 |
| 12  | 154,742,504,910,672,475,580,890,215 |
| 13  | 154,742,504,910,672,534,130,092,131 |
| 14  | 154,742,504,910,672,534,362,390,527 |

Figure 5: CDT for parameter set 2

$$\begin{aligned}
P_{g512}(x) = & ((((((((((\frac{8004071744321549}{2^{58}} \cdot x \cdot 2^{-19} \\
& + \frac{6144316755720785}{2^{58}}) \cdot x \cdot 2^{-19} \\
& + \frac{3070710202578901}{2^{58}}) \cdot x \cdot 2^{-17} \\
& + \frac{5162723504379443}{2^{58}}) \cdot x \cdot 2^{-18} \\
& + \frac{3817631981177055}{2^{58}}) \cdot x \cdot 2^{-17} \\
& + \frac{4934484841355621}{2^{58}}) \cdot x \cdot 2^{-18} \\
& + \frac{2733198821433297}{2^{58}}) \cdot x \cdot 2^{-19} \\
& + \frac{630792911273043}{2^{58}}) \cdot x \cdot 2^{-13} \\
& + \frac{7453705375596337}{2^{58}}) \cdot x \cdot 2^{-18} \\
& + \frac{2064281503328677}{2^{58}}) \cdot x \cdot 2^{-14} \\
& + \frac{6098097214449223}{2^{58}}) \cdot x \cdot 2^{-9} \\
& + 1
\end{aligned}$$

$$\begin{aligned}
P_{g1024}(x) = & ((((((((((\frac{2226939648802041}{2^{61}} \cdot x \cdot 2^{-11} \\
& + \frac{1576303411248839679}{2^{61}}) \cdot x \cdot 2^{-29} \\
& + \frac{2797046210472055}{2^{61}}) \cdot x \cdot 2^{-18} \\
& + \frac{8636915620785111}{2^{61}}) \cdot x \cdot 2^{-12} \\
& + \frac{1520161196432549375}{2^{61}}) \cdot x \cdot 2^{-28} \\
& + \frac{3624730440691813}{2^{61}}) \cdot x \cdot 2^{-19} \\
& + \frac{3871655815940135}{2^{61}}) \cdot x \cdot 2^{-20} \\
& + \frac{1772303971126881}{2^{61}}) \cdot x \cdot 2^{-20} \\
& + \frac{676080319865065}{2^{61}}) \cdot x \cdot 2^{-15} \\
& + \frac{6602346876136031}{2^{61}}) \cdot x \cdot 2^{-18} \\
& + \frac{6044629098072481}{2^{61}}) \cdot x \cdot 2^{-17} \\
& + \frac{7378697629483815}{2^{61}}) \cdot x \cdot 2^{-8} \\
& + 1
\end{aligned}$$

Figure 6: Polynomial approximation for  $\exp(x/2\sigma^2)$  over  $I_1$

### B.1.5 Security

Let  $\tilde{D}_\sigma$  denote the output distribution of  $\text{ApproxG}_\sigma$ . In NTRU+Sign, when  $D_\sigma$  is replaced to  $\tilde{D}_\sigma$ , it is necessary to ensure that  $R_{2\lambda+2}(\tilde{D}_\sigma | D_\sigma) \leq 1 + \frac{1}{4Q_{\text{gauss}}}$  in order to support the security argument in Theorem B.3. Since the only difference between  $D_\sigma$  and  $\tilde{D}_\sigma$  lies in the difference between  $D_{\mathbb{Z}_{\geq 0}, \sigma}$  and  $\tilde{D}_{\mathbb{Z}_{\geq 0}, \sigma}$ , which is the output distribution of  $\text{ApproxG}_\sigma^+$ , we have  $R_{2\lambda+2}(\tilde{D}_\sigma | D_\sigma) = R_{2\lambda+2}(\tilde{D}_{\mathbb{Z}_{\geq 0}, \sigma} | D_{\mathbb{Z}_{\geq 0}, \sigma})$ . In the following theorem,  $R_{2\lambda+2}(\tilde{D}_{\mathbb{Z}_{\geq 0}, \sigma} | D_{\mathbb{Z}_{\geq 0}, \sigma}) \leq 1 + \frac{1}{4Q_{\text{gauss}}}$  is established under constraints for parameters.

We provide a brief outline for proving Theorem B.5. Since  $D_{\mathbb{Z}_{\geq 0}, \sigma}$  can be realized with discrete half Gaussian sampler with a small standard deviation  $D_{\mathbb{Z}_{\geq 0}, \sigma_1}$  and corresponding rejecting sampling with  $\mathcal{B}_{\exp(x/2\sigma^2)}$ ,  $\tilde{D}_{\mathbb{Z}_{\geq 0}, \sigma}$  can be realized by replacing two components with  $\tilde{D}_{[0, w_1], \sigma_1}$  and  $\tilde{\mathcal{B}}_{\exp(x/2\sigma^2)}$ , respectively. Therefore,  $R_{2\lambda+2}(\tilde{D}_{\mathbb{Z}_{\geq 0}, \sigma} | D_{\mathbb{Z}_{\geq 0}, \sigma})$  can be analyzed by separately evaluating the differences arising from replacing each component. To this end, the difference between  $D_{\mathbb{Z}_{\geq 0}, \sigma_1}$  and  $\tilde{D}_{[0, w_1], \sigma_1}$  is analyzed in Lemma B.7, while the difference between  $\mathcal{B}_{\exp(x/2\sigma^2)}$  and  $\tilde{\mathcal{B}}_{\exp(x/2\sigma^2)}$  is addressed in Lemma B.8.

**Theorem B.5.** Let  $\tilde{D}_{\mathbb{Z}_{\geq 0}, \sigma}$  denote the output distribution of  $\text{ApproxG}_\sigma^+$  in Algorithm 2. If  $\sigma$  satisfies  $\exp(-\frac{1}{2\sigma^2}) \leq 1 - \frac{1}{2\sigma^2+1}$  and  $\tau_1, \theta_1, P_{\text{exp}}^{I_1}, \theta_{\text{gauss}}, \theta_{\text{poly}}^{I_1}$  satisfy (6), (7), (8), (9), (10), then

$$R_{2\lambda+2}(\tilde{D}_{\mathbb{Z}_{\geq 0}, \sigma} | D_{\mathbb{Z}_{\geq 0}, \sigma}) \leq 1 + \frac{1}{4Q_{\text{gauss}}}.$$

*Proof.* The distribution  $\tilde{D}_{\mathbb{Z}_{\geq 0}, \sigma}$  differs from the distribution  $D_{\mathbb{Z}_{\geq 0}, \sigma}$  by two aspects due to the operation of  $\text{ApproxG}_\sigma^+$ . Let  $\hat{D}_{\mathbb{Z}_{\geq 0}, \sigma}$  be the output distribution of Algorithm 2 when replacing  $\text{ApproxG}_{\sigma_1}^+$  with  $D_{\mathbb{Z}_{\geq 0}, \sigma_1}$ , and let  $\tilde{\mathcal{B}}_{\exp(x/2\sigma^2)}$  be the output distribution of Algorithm 4. Then, by Lemma B.6, the Rényi divergence is

bounded as

$$\begin{aligned} R_{2\lambda+2}(\tilde{D}_{\mathbb{Z}_{\geq 0}, \sigma} \mid D_{\mathbb{Z}_{\geq 0}, \sigma}) &\leq R_{2\lambda+2}(\tilde{D}_{\mathbb{Z}_{\geq 0}, \sigma} \mid \hat{D}_{\mathbb{Z}_{\geq 0}, \sigma}) \cdot R_{\infty}(\hat{D}_{\mathbb{Z}_{\geq 0}, \sigma} \mid D_{\mathbb{Z}_{\geq 0}, \sigma}) \\ &= R_{2\lambda+2}(\tilde{\mathcal{B}}_{\exp(x/2\sigma^2)} \mid \mathcal{B}_{\exp(x/2\sigma^2)}) \cdot R_{\infty}(\tilde{D}_{[0, w_1], \sigma_1} \mid D_{\mathbb{Z}_{\geq 0}, \sigma_1}). \end{aligned}$$

Combining this with Lemma B.7 and Lemma B.8, we have the result:

$$R_{2\lambda+2}(\tilde{D}_{\mathbb{Z}_{\geq 0}, \sigma} \mid D_{\mathbb{Z}_{\geq 0}, \sigma}) \leq c_g^2 \cdot \left(1 + \frac{1}{16Q_{\text{gauss}}}\right)^2 \cdot \left(1 + \frac{1}{8Q_{\text{gauss}}}\right) \leq 1 + \frac{1}{4Q_{\text{gauss}}}.$$

□

**Lemma B.6** ([BLRL<sup>+</sup>18, Lemma 2.9]). Let  $a \in [1, \infty)$ . For three distributions  $\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$  with  $\text{Supp}(\mathcal{D}_i) \subset \text{Supp}(\mathcal{D}_{i+1})$ , we have

$$R_a(\mathcal{D}_1 \mid \mathcal{D}_3) \leq \begin{cases} R_a(\mathcal{D}_1 \mid \mathcal{D}_2) \cdot R_{\infty}(\mathcal{D}_2 \mid \mathcal{D}_3) \\ R_{\infty}(\mathcal{D}_1 \mid \mathcal{D}_2)^{\frac{a}{a-1}} \cdot R_a(\mathcal{D}_2 \mid \mathcal{D}_3) & \text{if } a \in (1, \infty) \end{cases}.$$

**Lemma B.7.** Let  $\tilde{D}_{[0, w_1], \sigma_1}$  be the output distribution of  $\text{ApproxBaseG}_{\sigma_1}^+$  in Algorithm 3. If  $\tau_1$  and  $\theta_1$  satisfy

$$\frac{1}{1-2C(\tau_1)} \leq c_g \cdot \left(1 + \frac{1}{16Q_{\text{gauss}}}\right) \quad \text{and} \quad 2^{\theta_1} \geq w_1 \cdot \rho_{\sigma_1}([0, w_1]) \cdot c_g^{-1} \cdot \left(1 + \frac{1}{16Q_{\text{gauss}}}\right)^{-1},$$

then

$$R_{\infty}(\tilde{D}_{[0, w_1], \sigma_1} \mid D_{\mathbb{Z}_{\geq 0}, \sigma_1}) \leq c_g^2 \cdot \left(1 + \frac{1}{16Q_{\text{gauss}}}\right)^2.$$

*Proof.* The result is directly obtained from the following calculations:

$$\begin{aligned} R_{\infty}(\tilde{D}_{[0, w_1], \sigma_1} \mid D_{\mathbb{Z}_{\geq 0}, \sigma_1}) &= \max_{x \in [0, w_1]} \frac{\tilde{D}_{[0, w_1], \sigma_1}(x)}{D_{\mathbb{Z}_{\geq 0}, \sigma_1}(x)} \\ &\leq \max_{x \in [0, w_1]} \frac{\tilde{D}_{[0, w_1], \sigma_1}(x)}{D_{[0, w_1], \sigma_1}(x)} \cdot \max_{x \in [0, w_1]} \frac{D_{[0, w_1], \sigma_1}(x)}{D_{\mathbb{Z}_{\geq 0}, \sigma_1}(x)} \\ &= R_{\infty}(\tilde{D}_{[0, w_1], \sigma_1} \mid D_{[0, w_1], \sigma_1}) \cdot R_{\infty}(D_{[0, w_1], \sigma_1} \mid D_{\mathbb{Z}_{\geq 0}, \sigma_1}) \\ &\leq c_g^2 \cdot \left(1 + \frac{1}{16Q_{\text{gauss}}}\right)^2 \end{aligned}$$

where the last inequality comes from Lemma B.9 and Lemma B.10. □

**Lemma B.8.** Let  $P_{\exp}^{I_1}(x)$  be a approximate polynomial for  $\exp(x/2\sigma^2)$  over  $x \in I_1 = [-\lfloor 2\sigma^2 \rfloor, 0]$  satisfying  $P_{\exp}^{I_1}(0) = 1$ . Assume that  $1 - \exp(-\frac{1}{2\sigma^2}) \leq \frac{1}{e}$ . If  $P_{\exp}^{I_1}(x)$  satisfies

$$\left| \frac{P_{\exp}^{I_1}(x)}{\exp(x/2\sigma^2)} - 1 \right| \leq \frac{1}{2^2 \sqrt{(2\lambda+2)Q_{\text{gauss}}}} \quad \text{and} \quad \left| \frac{\exp(x/2\sigma^2) - P_{\exp}^{I_1}(x)}{1 - \exp(x/2\sigma^2)} \right| \leq \frac{1}{2^2 \sqrt{(2\lambda+2)Q_{\text{gauss}}}}$$

for all  $x \in I_1 \cap \mathbb{Z}$ , then

$$R_{2\lambda+2}(\tilde{\mathcal{B}}_{\exp(x/2\sigma^2)} \mid \mathcal{B}_{\exp(x/2\sigma^2)}) \leq 1 + \frac{1}{8Q_{\text{gauss}}}$$

where  $\tilde{\mathcal{B}}_{\exp(x/2\sigma^2)}$  denote the output distribution of Algorithm 4.

*Proof.* Note that  $1 + \frac{(2\lambda+2)\delta^2}{2} \leq 1 + \frac{1}{8Q_{\text{gauss}}} \Leftrightarrow \delta \leq \frac{1}{2\sqrt{(2\lambda+2)Q_{\text{gauss}}}}$ . From Lemma B.12, it is enough to show that the relative error

$$\delta := \left| \frac{\tilde{\mathcal{B}}_{\text{exp}(x/2\sigma^2)}(0)}{\mathcal{B}_{\text{exp}(x/2\sigma^2)}(0)} - 1 \right| = \max \left( \left| \frac{\tilde{\mathcal{B}}_{\text{exp}(x/2\sigma^2)}(0)}{\mathcal{B}_{\text{exp}(x/2\sigma^2)}(0)} - 1 \right|, \left| \frac{\tilde{\mathcal{B}}_{\text{exp}(x/2\sigma^2)}(1)}{\mathcal{B}_{\text{exp}(x/2\sigma^2)}(1)} - 1 \right| \right)$$

is bounded by  $\frac{1}{2\sqrt{(2\lambda+2)Q_{\text{gauss}}}}$ .

Let  $\tilde{P}_{\text{exp}}^{I_1}(x)$  denote the output of Algorithm 5 with parameters  $P_{\text{exp}}^{I_1}, \vartheta_1$ . Note that  $\tilde{\mathcal{B}}_{\text{exp}(x/2\sigma^2)} = \mathcal{B}_{\text{exp}(x/2\sigma^2)}$  for  $x = 0$  since  $P_{\text{exp}}^{I_1}(0) = 1$ . Now, we only consider the case that  $x \in [-\lfloor 2\sigma^2 \rfloor, -1]$ . Since  $|\tilde{P}_{\text{exp}}^{I_1}(x) - P_{\text{exp}}^{I_1}(x)| < \frac{1 - \exp(-1/2\sigma^2)}{2^2\sqrt{(2\lambda+2)Q_{\text{gauss}}}}$  by Lemma B.13 and (10) of  $\vartheta_1$ , we have

$$\begin{aligned} \left| \frac{\tilde{\mathcal{B}}_{\text{exp}(x/2\sigma^2)}(0)}{\mathcal{B}_{\text{exp}(x/2\sigma^2)}(0)} - 1 \right| &= \left| \frac{1 - \tilde{P}_{\text{exp}}^{I_1}(x)}{1 - \exp(x/2\sigma^2)} - 1 \right| = \left| \frac{\tilde{P}_{\text{exp}}^{I_1}(x) - \exp(x/2\sigma^2)}{1 - \exp(x/2\sigma^2)} \right| \\ &\leq \left| \frac{\tilde{P}_{\text{exp}}^{I_1}(x) - P_{\text{exp}}^{I_1}(x)}{1 - \exp(x/2\sigma^2)} \right| + \left| \frac{P_{\text{exp}}^{I_1}(x) - \exp(x/2\sigma^2)}{1 - \exp(x/2\sigma^2)} \right| \\ &\leq \left| \frac{\tilde{P}_{\text{exp}}^{I_1}(x) - P_{\text{exp}}^{I_1}(x)}{1 - \exp(-1/2\sigma^2)} \right| + \left| \frac{P_{\text{exp}}^{I_1}(x) - \exp(x/2\sigma^2)}{1 - \exp(x/2\sigma^2)} \right| \\ &\leq \frac{1}{2^2\sqrt{(2\lambda+2)Q_{\text{gauss}}}} + \frac{1}{2^2\sqrt{(2\lambda+2)Q_{\text{gauss}}}} \end{aligned}$$

and

$$\begin{aligned} \left| \frac{\tilde{\mathcal{B}}_{\text{exp}(x/(2\sigma^2))}(1)}{\mathcal{B}_{\text{exp}(x/(2\sigma^2))}(1)} - 1 \right| &= \left| \frac{\tilde{P}_{\text{exp}}^{I_1}(x)}{\exp(x/(2\sigma^2))} - 1 \right| \\ &\leq \left| \frac{\tilde{P}_{\text{exp}}^{I_1}(x)}{\exp(x/(2\sigma^2))} - \frac{P_{\text{exp}}^{I_1}(x)}{\exp(x/(2\sigma^2))} \right| + \left| \frac{P_{\text{exp}}^{I_1}(x)}{\exp(x/(2\sigma^2))} - 1 \right| \\ &\leq e \left| \tilde{P}_{\text{exp}}^{I_1}(x) - P_{\text{exp}}^{I_1}(x) \right| + \left| \frac{P_{\text{exp}}^{I_1}(x)}{\exp(x/(2\sigma^2))} - 1 \right| \\ &\leq \frac{e(1 - \exp(-1/2\sigma^2))}{2^2\sqrt{(2\lambda+2)Q_{\text{gauss}}}} + \frac{1}{2^2\sqrt{(2\lambda+2)Q_{\text{gauss}}}} \\ &\leq \frac{1}{2^2\sqrt{(2\lambda+2)Q_{\text{gauss}}}} + \frac{1}{2^2\sqrt{(2\lambda+2)Q_{\text{gauss}}}}. \end{aligned}$$

Then, these two inequalities show that the relative error is bounded by  $\frac{1}{2\sqrt{(2\lambda+2)Q_{\text{gauss}}}}$ .  $\square$

**Lemma B.9.** If  $2^{\theta_1} \geq w_1 \cdot \rho_{\sigma_1}([0, w_1]) \cdot c_g^{-1} \cdot \left(1 + \frac{1}{16Q_{\text{gauss}}}\right)^{-1}$ , then

$$R_{\infty}(\tilde{D}_{[0, w_1], \sigma_1} \mid D_{[0, w_1], \sigma_1}) \leq c_g \cdot \left(1 + \frac{1}{16Q_{\text{gauss}}}\right).$$

*Proof.* Recall that  $R_{\infty}(\tilde{D}_{[0, w_1], \sigma_1} \mid D_{[0, w_1], \sigma_1}) = \max_{x \in [0, w_1]} \frac{\tilde{D}_{[0, w_1], \sigma_1}(x)}{D_{[0, w_1], \sigma_1}(x)}$ . Then, by the definition of  $\tilde{D}_{[0, w_1], \sigma_1}$ ,

it is trivial that  $\frac{\tilde{D}_{[0,w_1],\sigma_1}(x)}{D_{[0,w_1],\sigma_1}(x)} < 1$  for any  $x \neq 0$ . In addition, one can easily see that

$$\begin{aligned}
\frac{\tilde{D}_{[0,w_1],\sigma_1}(0)}{D_{[0,w_1],\sigma_1}(0)} &= \left(1 - \sum_{i=1}^{w_1} \lfloor D_{[0,w_1],\sigma_1}(i) \cdot 2^{\theta_1} \rfloor \cdot 2^{-\theta_1} \right) \cdot \frac{\rho_{\sigma_1}([0, w_1])}{\rho_{\sigma_1}(0)} \\
&< \left(1 - \sum_{i=1}^{w_1} (D_{[0,w_1],\sigma_1}(i) \cdot 2^{\theta_1} - 1) \cdot 2^{-\theta_1} \right) \cdot \rho_{\sigma_1}([0, w_1]) \\
&= \left(1 - \sum_{i=1}^{w_1} (D_{[0,w_1],\sigma_1}(i) - 2^{-\theta_1}) \right) \cdot \rho_{\sigma_1}([0, w_1]) \\
&= \rho_{\sigma_1}([0, w_1]) - \sum_{i=1}^{w_1} \rho_{\sigma_1}(i) + w_1 \cdot 2^{-\theta_1} \cdot \rho_{\sigma_1}([0, w_1]) \\
&= 1 + w_1 \cdot 2^{-\theta_1} \cdot \rho_{\sigma_1}([0, w_1]) \\
&\leq c_g \cdot \left(1 + \frac{1}{16Q_{\text{gauss}}}\right).
\end{aligned}$$

□

**Lemma B.10.** If  $\tau_1$  satisfies  $\frac{1}{1-2C(\tau_1)} \leq c_g \cdot \left(1 + \frac{1}{16Q_{\text{gauss}}}\right)$ , then

$$R_{\infty}(D_{[0,w_1],\sigma_1} \mid D_{\mathbb{Z}_{\geq 0},\sigma_1}) \leq c_g \cdot \left(1 + \frac{1}{16Q_{\text{gauss}}}\right).$$

*Proof.* Using the inequality that  $\frac{a}{b} < \frac{a-1/2}{b-1/2}$  for  $a > b > 1$ , we have

$$\begin{aligned}
R_{\infty}(D_{[0,w_1],\sigma_1} \mid D_{\mathbb{Z}_{\geq 0},\sigma_1}) &= \max_{x \in [0,w_1]} \frac{D_{[0,w_1],\sigma_1}(x)}{D_{\mathbb{Z}_{\geq 0},\sigma_1}(x)} = \frac{\rho_{\sigma_1}(\mathbb{Z}_{\geq 0})}{\rho_{\sigma_1}([0, w_1])} \\
&\leq \frac{\rho_{\sigma_1}(\mathbb{Z}_{\geq 0}) - 1/2}{\rho_{\sigma_1}([0, w_1]) - 1/2} = \frac{\rho_{\sigma_1}(\mathbb{Z})}{\rho_{\sigma_1}([-w_1, w_1])} \\
&\leq \frac{1}{1 - 2C(\tau_1)} \\
&\leq c_g \cdot \left(1 + \frac{1}{16Q_{\text{gauss}}}\right),
\end{aligned}$$

where the second inequality comes from the last holds by Lemma B.11. □

**Lemma B.11** ([MM11, Section 2.3]). For random  $n \in \mathbb{Z}_{>0}$ ,  $\sigma \in \mathbb{R}_{>0}$ ,  $\tau > 1 \in \mathbb{R}$ ,

$$\rho_{\sigma}(\mathbb{Z}^n \setminus \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| \leq \tau\sigma\sqrt{n}\}) < 2C(\tau)^n \cdot \rho_{\sigma}(\mathbb{Z})^n$$

with  $C(\tau) = \tau \exp\left(\frac{1-\tau^2}{2}\right) < 1$ .

**Lemma B.12** ([Pre17, Lemma 3]). Assume that two distributions,  $\mathcal{D}_1$  and  $\mathcal{D}_2$ , satisfy the following conditions:

$$- \text{supp}(\mathcal{D}_1) = \text{supp}(\mathcal{D}_2)$$

–  $\exists \delta > 0$  such that  $\left| \frac{\mathcal{D}_1}{\mathcal{D}_2} - 1 \right| \leq \delta$  over  $\text{supp}(\mathcal{D}_1)$

Given that  $a \in (1, \infty)$ ,

$$R_a(\mathcal{D}_1 | \mathcal{D}_2) \leq \left( 1 + \frac{a(a-1)\delta^2}{2(1-\delta)^{a+1}} \right)^{\frac{1}{a-1}} \underset{\delta \rightarrow 0}{\sim} 1 + \frac{a\delta^2}{2}.$$

**Lemma B.13.** Let  $\tilde{P}(x)$  denote the output of Algorithm 5 with parameters  $P$  and  $\vartheta$ . If  $\tilde{p}_1 \in [\frac{1}{2}, 2^{s_1 - \log_2 \lfloor 2\sigma^2 \rfloor})$ ,  $\log_2 \lfloor 2\sigma^2 \rfloor - s_1 \leq 1$ , and  $\lceil \log_2 \lfloor 2\sigma^2 \rfloor \rceil - s_i \leq -1$ ,  $\tilde{p}_i \in [\frac{1}{2}, 1)_{\vartheta}$  for  $i \in \{2, \dots, d\}$ , then

$$\left| \tilde{P}(x) - P(x) \right| \leq 2^{-(\vartheta+1)} \quad \text{and} \quad \tilde{P}(x) \in [0, 1]_{\vartheta}$$

for all  $x \in I \subseteq [-\lfloor 2\sigma^2 \rfloor, 0]$ .

*Proof.* Let  $\eta = \lceil \log_2 \lfloor 2\sigma^2 \rfloor \rceil$ . For  $x \in I_1$ , we have

$$\begin{aligned} |y_d - \tilde{y}_d| &= \left| (\tilde{p}_d \cdot x) \cdot 2^{-s_d} - \lfloor (\tilde{p}_d \cdot x) \cdot 2^{\vartheta-s_d} \rfloor \cdot 2^{-\vartheta} \right| \\ &= 2^{-\vartheta} \cdot \left| (\tilde{p}_d \cdot x) \cdot 2^{\vartheta-s_d} - \lfloor (\tilde{p}_d \cdot x) \cdot 2^{\vartheta-s_d} \rfloor \right| \\ &\leq 2^{-(\vartheta+1)}. \end{aligned}$$

Moreover, since  $\tilde{p}_d \in [\frac{1}{2}, 1)_{\vartheta}$  and  $x \in (-2^{\eta}, 0]$ , we have  $\lfloor (\tilde{p}_d \cdot x) \cdot 2^{\vartheta-s_d} \rfloor \in [-2^{\eta+\vartheta-s_d}, 0]$ , which in turn implies  $\tilde{y}_d = \lfloor (\tilde{p}_d \cdot x) \cdot 2^{\vartheta-s_d} \rfloor \cdot 2^{-\vartheta} \in [-2^{\eta-s_d}, 0] \subseteq [-\frac{1}{2}, 0]_{\vartheta}$  by  $\eta - s_d \leq -1$ .

Now, let us consider the case that  $i = d-1$ . Given that  $\tilde{p}_{d-1} \in [\frac{1}{2}, 1)_{\vartheta}$  and  $\tilde{y}_d \in [-\frac{1}{2}, 0]_{\vartheta}$ , we have  $\tilde{y}_d + \tilde{p}_{d-1} \in [0, 1)_{\vartheta}$ . As before, letting  $\tilde{z}_{d-1} = \tilde{y}_d + \tilde{p}_{d-1}$ , we then have

$$\begin{aligned} |y_{d-1} - \tilde{y}_{d-1}| &= \left| (\tilde{z}_{d-1} \cdot x) \cdot 2^{-s_{d-1}} - \lfloor (\tilde{z}_{d-1} \cdot x) \cdot 2^{\vartheta-s_{d-1}} \rfloor \cdot 2^{-\vartheta} \right| \\ &= 2^{-\vartheta} \cdot \left| (\tilde{z}_{d-1} \cdot x) \cdot 2^{\vartheta-s_{d-1}} - \lfloor (\tilde{z}_{d-1} \cdot x) \cdot 2^{\vartheta-s_{d-1}} \rfloor \right| \\ &\leq 2^{-(\vartheta+1)} \end{aligned}$$

and  $\tilde{y}_{d-1} \in [-\frac{1}{2}, 0]_{\vartheta}$  by  $\eta - s_{d-1} \leq -1$ .

Since  $\tilde{y}_i$  and  $y_i$  are computed in the same manner under the identical conditions on  $\tilde{p}_i$  and  $\tilde{y}_{i+1}$  from  $i = d-1$  down to 1, we have  $|y_1 - \tilde{y}_1| \leq 2^{-\vartheta+1}$ , and thus

$$|\tilde{P}(x) - P(x)| = |y_0 - \tilde{y}_0| = |(\tilde{p}_0 + y_1) - (\tilde{p}_0 + \tilde{y}_1)| = |y_1 - \tilde{y}_1| \leq 2^{-(\vartheta+1)}.$$

Since  $\tilde{y}_2 \in [-\frac{1}{2}, 0]_{\vartheta}$  and  $\tilde{p}_1 \in [\frac{1}{2}, 2^{s_1 - \log_2 \lfloor 2\sigma^2 \rfloor})$ , we have  $((\tilde{y}_2 + \tilde{p}_1) \cdot x) \cdot 2^{\vartheta-s_1} \in (-2^{\vartheta}, 0]$ , which implies  $\tilde{y}_1 = \lfloor ((\tilde{y}_2 + \tilde{p}_1) \cdot x) \cdot 2^{\vartheta-s_1} \rfloor \cdot 2^{-\vartheta} \in [-1, 0]_{\vartheta}$ . Thus,  $\tilde{P}(x) = \tilde{y}_0 = 1 + \tilde{y}_1 \in [0, 1]_{\vartheta}$ .  $\square$

## B.2 Rejection Sampling

### B.2.1 Algorithm

Essentially, Algorithm 6 describes a way of sampling bits from the Bernoulli distribution with parameter  $\Psi(\mathbf{z}, \mathbf{v}) = 1/(M \exp(-\|\mathbf{v}\|^2/(2\sigma^2)) \cosh(\langle \mathbf{z}, \mathbf{v} \rangle / \sigma^2))$  where  $M = \exp(1/(2\hat{\alpha}^2))$ . To do this, Algorithm 6 combines two Bernoulli distributions by observing that the Bernoulli distribution  $\mathcal{B}_{\Psi}$  can be



decomposed into the product of  $\mathcal{B}_{\Psi_1}$  and  $\mathcal{B}_{\Psi_2}$  where  $\Psi_1(\mathbf{v}) = 1/(M \exp(-\|\mathbf{v}\|^2/(2\sigma^2)))$ ,  $\Psi_2(\mathbf{z}, \mathbf{v}) = 1/\cosh(\langle \mathbf{z}, \mathbf{v} \rangle/\sigma^2)$ .

---

**Algorithm 6** RejectSample $_{\sigma, \hat{\alpha}}(\mathbf{z}, \mathbf{v})$

---

**Input**  $\mathbf{z} \in R^2, \mathbf{v} \in R$  satisfying  $\|\mathbf{v}\| \leq \sigma/\hat{\alpha}$  and  $|\langle \mathbf{z}, \mathbf{v} \rangle| \leq B_2\sigma/\hat{\alpha}$   
**Output**  $b \in \{0, 1\}$

- 1:  $x \leftarrow \|\mathbf{v}\|^2 - \frac{\sigma^2}{\hat{\alpha}^2}$   $// x \in I_2 = [-\frac{\sigma^2}{\hat{\alpha}^2}, 0]$
- 2:  $y \leftarrow 2\langle \mathbf{z}, \mathbf{v} \rangle$   $// y \in [-2B_2\sigma/\hat{\alpha}, 2B_2\sigma/\hat{\alpha}]$
- 3:  $b_1 \leftarrow \text{SampleBernExp}_{\sigma}^{I_2}(x)$   $// b_1 \sim \mathcal{B}_{\exp(x/(2\sigma^2))}$
- 4:  $b_2 \leftarrow \text{SampleBernCosh}_{\sigma}(y)$   $// b_2 \sim \mathcal{B}_{1/\cosh(y/(2\sigma^2))}$
- 5: **return**  $b_1 b_2$

---

$\text{SampleBernExp}_{\sigma}^{I_2}(x)$  implements an approximated version of the Bernoulli distribution  $\mathcal{B}_{\exp(x/(2\sigma^2))}$  for  $x \in I_2$ , based on the following approximations: for  $x = x_1\hat{c} + x_0, x_0 \in I'_2 = (-\hat{c}, 0]$ ,

$$\begin{aligned} \exp\left(\frac{x}{2\sigma^2}\right) &= \exp\left(\frac{x_1\hat{c} + x_1(\hat{c}-c) + x_0}{2\sigma^2}\right) = 2^{x_1} \exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^{x_1} \exp\left(\frac{x_0}{2\sigma^2}\right) \\ &\approx 2^{x_1}(1 + \omega x_1) \cdot P_{\exp}'^{I'_2}(x_0) \end{aligned}$$

where

$$\hat{c} \approx c = 2\sigma^2 \ln(2), \quad \exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^{x_1} \approx (1 + \omega x_1), \quad \exp\left(\frac{x_0}{2\sigma^2}\right) \approx P_{\exp}'^{I'_2}(x_0).$$

In particular, the output of  $\text{SampleBernExp}_{\sigma}^{I_2}(x)$  is distributed according to the Bernoulli distribution with parameter similar to  $2^{x_1}(1 + \omega x_1) \cdot P_{\exp}'^{I'_2}(x_0)$ . Note that  $\text{EvaluatePoly}[P_{\exp}'^{I'_2}, \vartheta_2](x_0)$  is described in the Algorithm 5 with parameter  $P_{\exp}'^{I'_2}$  and  $\vartheta_2$ .

---

**Algorithm 7** SampleBernExp $_{\sigma}^{I_2}(x)$

---

**Input**  $x \in I_2 = [-\frac{\sigma^2}{\hat{\alpha}^2}, 0]$   
**Output**  $b \in \{0, 1\}$

- 1:  $x_1 \leftarrow \lceil x/\hat{c} \rceil$   $// \hat{c} \approx 2\sigma^2 \ln(2)$
- 2:  $x_0 \leftarrow x - x_1\hat{c}$   $// x_0 \in I'_2 = (-\hat{c}, 0]$
- 3:  $\hat{p}_0 \leftarrow \text{EvaluatePoly}[P_{\exp}'^{I'_2}, \vartheta_2](x_0)$
- 4:  $\hat{p}_1 \leftarrow 2^{x_1}(1 + \omega x_1)$
- 5:  $\hat{p} \leftarrow \text{Mul}[\vartheta_2](\hat{p}_0, \hat{p}_1)$
- 6:  $u \leftarrow \mathcal{U}([0, 1]_{\vartheta_2})$
- 7: **if**  $u \leq \hat{p}$  **then return** 1 **else return** 0

---

$\text{SampleBernCosh}_{\sigma}(y)$  implements an approximated version of the Bernoulli distribution  $\mathcal{B}_{1/\cosh(y/(2\sigma^2))}$ , based on the following approximations:

$$\begin{aligned} 2 \cosh\left(\frac{y}{2\sigma^2}\right) &= \exp\left(\frac{y}{2\sigma^2}\right) + \exp\left(\frac{-y}{2\sigma^2}\right) \\ &= 2^{y_1} \exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^{y_1} \exp\left(\frac{y_0}{2\sigma^2}\right) + 2^{-y_1+1} \exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^{-y_1+1} \exp\left(\frac{-y_0-\hat{c}}{2\sigma^2}\right) \\ &\approx 2^{y_1}(1 + y_1\omega)P_{\exp}'^{I'_2}(y_0) + 2^{-y_1+1}(1 + (1 - y_1)\omega)P_{\exp}'^{I'_2}(-y_0 - \hat{c}). \end{aligned}$$

where  $y = y_1\hat{c} + y_0$ ,  $y_0 \in I'_2$ . In particular, the output of  $\text{SampleBernCosh}_\sigma(y)$  is distributed according to the Bernoulli distribution with parameter similar to

$$\frac{2}{2^{y_1}(1 + y_1\omega)P_{\text{exp}}^{I'_2}(y_0) + 2^{-y_1+1}(1 + (1 - y_1)\omega)P_{\text{exp}}^{I'_2}(-y_0 - \hat{c})}.$$

Note that  $u\hat{p}$  is multiplied with exactly and then compared with 1.

---

**Algorithm 8**  $\text{SampleBernCosh}_\sigma(y)$

---

**Input**  $y \in [-2B_2\sigma/\hat{\alpha}, 2B_2\sigma/\hat{\alpha}]$

**Output**  $b \in \{0, 1\}$

- 1:  $y_1 \leftarrow \lceil y/\hat{c} \rceil$
  - 2:  $y_0 \leftarrow y - y_1\hat{c}$
  - 3:  $\hat{p}_0 \leftarrow \text{EvaluatePoly}[P_{\text{exp}}^{I'_2}, \vartheta_2](-y_0 - \hat{c})$
  - 4:  $\hat{p}_1 \leftarrow 2^{-y_1+1}(1 + \omega(1 - y_1))$
  - 5:  $\hat{q}_0 \leftarrow \text{EvaluatePoly}[P_{\text{exp}}^{I'_2}, \vartheta_2](y_0)$
  - 6:  $\hat{q}_1 \leftarrow 2^{y_1}(1 + \omega y_1)$
  - 7:  $\hat{p} \leftarrow 2^{-1}(\text{Mul}[\vartheta_2](\hat{p}_0, \hat{p}_1) + \text{Mul}[\vartheta_2](\hat{q}_0, \hat{q}_1))$
  - 8:  $u \leftarrow \mathcal{U}([0, 1]_{\vartheta_2})$
  - 9: **if**  $u\hat{p} \leq 1$  **then return** 1 **else return** 0
- 

$\text{Mul}[\vartheta](a, b)$  returns an approximate value of  $a \cdot b$  based on the following approximations:

$$a \cdot b \approx \lfloor a' \cdot b' \cdot 2^\vartheta \rfloor \cdot 2^{-\vartheta + \lceil \log_2 |a| \rceil + \lceil \log_2 |b| \rceil}$$

where  $a' = a \cdot 2^{-\lfloor \log_2 a \rfloor} \in (\frac{1}{2}, 1]$  and  $b' = b \cdot 2^{-\lfloor \log_2 b \rfloor} \in (\frac{1}{2}, 1]$ . For  $a, b \in (0, 1)$ , the difference between  $\text{Mul}[\vartheta](a, b)$  and  $a \cdot b$  is bounded by the following inequality:

$$\begin{aligned} |\text{Mul}[\vartheta](a, b) - a \cdot b| &= \left| \lfloor a' \cdot b' \cdot 2^\vartheta \rfloor \cdot 2^{-\vartheta + \lceil \log_2 a \rceil + \lceil \log_2 b \rceil} - a \cdot b \right| \\ &= 2^{-\vartheta + \lceil \log_2 a \rceil + \lceil \log_2 b \rceil} \cdot \left| \lfloor a' \cdot b' \cdot 2^\vartheta \rfloor - a' \cdot b' \cdot 2^\vartheta \right| \\ &\leq 2^{-(\vartheta+1) + \lceil \log_2 a \rceil + \lceil \log_2 b \rceil} \leq 2^{-(\vartheta+1)}. \end{aligned}$$

---

**Algorithm 9**  $\text{Mul}[\vartheta](a, b)$

---

**Input**  $a, b > 0$

**Output**  $c$

- 1:  $a' \leftarrow a \cdot 2^{-\lfloor \log_2 a \rfloor}$
  - 2:  $b' \leftarrow b \cdot 2^{-\lfloor \log_2 b \rfloor}$
  - 3:  $c \leftarrow \lfloor a' \cdot b' \cdot 2^\vartheta \rfloor \cdot 2^{-\vartheta + \lceil \log_2 a \rceil + \lceil \log_2 b \rceil}$
  - 4: **return**  $c$
-

### B.2.2 Constant-Time Execution

To prove that  $\text{RejectSample}_{\sigma, \hat{\alpha}}$  is constant-time in the sense of Definition B.4, we show that both the elements of  $\text{SampleBernExp}_{\sigma}^{I_2}$  (Algorithm 7) and  $\text{SampleBernCosh}_{\sigma}$  (Algorithm 8) are independent of both its input and output.

$\text{SampleBernExp}_{\sigma}^{I_2}$  and  $\text{SampleBernCosh}_{\sigma}$  consist of  $\text{EvaluatePoly}$  (Algorithm 5),  $\text{Mul}$  (Algorithm 9) and integer arithmetics, and evaluate a transcendental function using an approximate polynomial, followed by a comparing with a random value. Since the operations used in the algorithms involve only integer arithmetics, the execution times of each component are independent of both its input and output. Therefore, both algorithms are constant time, which implies that  $\text{RejectSample}_{\sigma, \hat{\alpha}}$  is also constant-time, as it is composition of constant-time elements.

Moreover, we note that the expected number of iterations for generating signature is constant-time. The average probability of a restart in generating signature is given by

$$\sum_{\mathbf{z} \in R^2} \underbrace{\frac{D_{\mathbf{v}, \sigma}^{2n}(\mathbf{z}) + D_{-\mathbf{v}, \sigma}^{2n}(\mathbf{z})}{2}}_{\text{prob. of } \mathbf{z}} \cdot \underbrace{\Psi(\mathbf{z}, \mathbf{v})}_{\text{accepting prob. of } \mathbf{z}} = \sum_{\mathbf{z} \in R^2} \frac{D_{\sigma}^{2n}(\mathbf{z})}{M} = \frac{1}{M},$$

by supposing  $\text{RejectSample}_{\sigma, \hat{\alpha}}$  has an ideal distribution  $\mathcal{B}_{\Psi}$ . Therefore, the expected number of iterations is a known constant and independent of its input and output.

### B.2.3 Constraints for parameters

$\delta_{\text{reject}}, c_{r,1}, c_{r,2}$   $\delta_{\text{reject}} = \frac{1}{2^5 \sigma^2 \sqrt{2(2\lambda+1)Q_{\text{reject}}}} < 1$ , and

$$c_{r,1} = \frac{1}{2\delta_{\text{reject}}} \left( 1 - \sqrt{\frac{1}{4\delta_{\text{reject}}(c_{r,2}-5/8)+1}} \right) < 1, \quad (11)$$

$$c_{r,2} = \frac{1}{8\delta_{\text{reject}}} \left( \sqrt{1 + 16\delta_{\text{reject}}} - 1 \right) < 1. \quad (12)$$

are small positive constants used for proper scaling to adjust some inequalities. Note that  $c_{r,1}, c_{r,2}$  satisfy some inequalities such as  $c_{r,1} < c_{r,2} - \frac{5}{8}$ ,

$$2(1 + 2^{\lfloor \log_2(1+\omega\zeta) \rfloor}) \left( 1 + \frac{2 \cdot c_{r,1} \cdot \delta_{\text{reject}}}{1 - 2 \cdot c_{r,1} \cdot \delta_{\text{reject}}} \right)^2 < 5, \quad (13)$$

$$\frac{c_{r,1} \cdot \delta_{\text{reject}}}{1 - 2 \cdot c_{r,1} \cdot \delta_{\text{reject}}} + \left( \frac{c_{r,1} \cdot \delta_{\text{reject}}}{1 - 2 \cdot c_{r,1} \cdot \delta_{\text{reject}}} \right)^2 \leq (c_{r,2} - \frac{5}{8}) \cdot \delta_{\text{reject}} \quad (14)$$

where  $\zeta = \lceil \frac{B_2}{\sigma \hat{\alpha} \ln 2} \rceil$ .

$\hat{c}$   $\hat{c}$  is an approximate value of  $c = 2\sigma^2 \ln 2$ , and it is set to satisfy

$$(\hat{c} - c)^2 \leq \frac{8\sigma^4}{e\zeta^2} \cdot c_{r,1} \cdot \delta_{\text{reject}}. \quad (15)$$

$\omega$   $\omega$  is a approximate value of  $\frac{\hat{c}-c}{2\sigma^2}$  and satisfies

$$\frac{\hat{c} - c}{2\sigma^2} - \omega \leq \frac{1}{e(\zeta + 1)} \cdot c_{r,1} \cdot \delta_{\text{reject}}. \quad (16)$$

$P_{\text{exp}}^{I'_2}$  is a polynomial approximation for  $\exp(-\frac{x}{2\sigma^2})$  over  $I'_2 = (-\hat{c}, 0]$ , which satisfies

$$\left| \frac{P_{\text{exp}}^{I'_2}(x)}{\exp(x/2\sigma^2)} - 1 \right| \leq c_{r,1} \cdot \delta_{\text{reject}} \quad (17)$$

for all  $x \in I'_2$ .

$\vartheta_2$  represents the precision required for evaluating the polynomial  $P_{\text{exp}}^{I'_2}$  at  $x \in I'_2$ . It needs to satisfy

$$2^{\vartheta_2} \geq (2 \cdot c_{r,1} \cdot \delta_{\text{reject}})^{-1}. \quad (18)$$

#### B.2.4 Concrete parameters

The parameters for the constant time rejection sampling are set as follows:

( $Q_{\text{sign}} = 2^{64}$ ,  $Q_{\text{reject}} = M \cdot Q_{\text{sign}}$ )

| parameter set           | 1                   | 2                     |
|-------------------------|---------------------|-----------------------|
| $n$                     | 512                 | 1024                  |
| $\sigma$                | 110                 | 200                   |
| $\hat{c}$               | $4397245863/2^{18}$ | $116290799681/2^{21}$ |
| $\omega$                | $58857/2^{52}$      | $29419/2^{53}$        |
| $\vartheta_2$           | 58                  | 60                    |
| $P_{\text{exp}}^{I'_2}$ | $P_{r512}$          | $P_{r1024}$           |

Table 7: Parameters for the constant-time Bernoulli sampler

$$\begin{aligned}
P_{r512}(x) = & ((((((((((\frac{270450876660152039}{2^{56}} \cdot x \cdot 2^{-37} \\
& + \frac{202500528913723915}{2^{58}}) \cdot x \cdot 2^{-36} \\
& + \frac{214649841049349961}{2^{58}}) \cdot x \cdot 2^{-37} \\
& + \frac{99522673477595099}{2^{58}}) \cdot x \cdot 2^{-35} \\
& + \frac{165431048337028741}{2^{58}}) \cdot x \cdot 2^{-35} \\
& + \frac{244354746788411695}{2^{58}}) \cdot x \cdot 2^{-39} \\
& + \frac{19738067975512813}{2^{58}}) \cdot x \cdot 2^{-35} \\
& + \frac{21865597162968399}{2^{58}}) \cdot x \cdot 2^{-32} \\
& + \frac{161482986980653483}{2^{58}}) \cdot x \cdot 2^{-40} \\
& + \frac{3726852688779707}{2^{58}}) \cdot x \cdot 2^{-32} \\
& + \frac{16514252026700727}{2^{58}}) \cdot x \cdot 2^{-30} \\
& + \frac{195139110862382031}{2^{58}}) \cdot x \cdot 2^{-32} \\
& + \frac{288230376151711743}{2^{58}} \\
P_{r1024}(x) = & ((((((((((\frac{671599188646772819}{2^{60}} \cdot x \cdot 2^{-42} \\
& + \frac{450080750041433613}{2^{60}}) \cdot x \cdot 2^{-42} \\
& + \frac{215076359977559233}{2^{60}}) \cdot x \cdot 2^{-39} \\
& + \frac{725265309224830985}{2^{60}}) \cdot x \cdot 2^{-43} \\
& + \frac{138381184076022813}{2^{60}}) \cdot x \cdot 2^{-38} \\
& + \frac{760165731345387857}{2^{60}}) \cdot x \cdot 2^{-41} \\
& + \frac{463968758545868589}{2^{60}}) \cdot x \cdot 2^{-41} \\
& + \frac{247786057534681841}{2^{60}}) \cdot x \cdot 2^{-38} \\
& + \frac{907419645104301689}{2^{60}}) \cdot x \cdot 2^{-40} \\
& + \frac{692306247804410657}{2^{60}}) \cdot x \cdot 2^{-39} \\
& + \frac{845100400152149459}{2^{60}}) \cdot x \cdot 2^{-40} \\
& + \frac{386856262276681323}{2^{60}}) \cdot x \cdot 2^{-38} \\
& + \frac{472236648286964521}{2^{60}}) \cdot x \cdot 2^{-36} \\
& + \frac{1152921504606846975}{2^{60}}
\end{aligned}$$

Figure 7: Polynomial approximation for  $\exp(x/2\sigma^2)$  over  $I'_2 = (-\hat{c}, 0]$

### B.2.5 Security

Let  $\tilde{\mathcal{B}}_{\Psi(\mathbf{z}, \mathbf{v})}$  denote the output distribution of  $\text{RejectSample}_{\sigma, \hat{\alpha}}(\mathbf{z}, \mathbf{v})$ . In NTRU+Sign, when  $\mathcal{B}_{\Psi(\mathbf{z}, \mathbf{v})}$  is replaced to  $\tilde{\mathcal{B}}_{\Psi(\mathbf{z}, \mathbf{v})}$ , it is necessary to ensure that  $R_{2\lambda+1}(\tilde{\mathcal{B}}_{\Psi(\mathbf{z}, \mathbf{v})} \mid \mathcal{B}_{\Psi(\mathbf{z}, \mathbf{v})}) < 1 + \frac{1}{4Q_{\text{reject}}}$  in order to support the security argument in Theorem B.3. In the following theorem, the inequality is established with constraints for parameters.

We provide a brief outline for proving Theorem B.14. Since  $\mathcal{B}_{\Psi(\mathbf{z}, \mathbf{v})}$  is decomposed with  $\mathcal{B}_{\exp(x/2\sigma^2)}$  and  $\mathcal{B}_{\cosh(y/2\sigma^2)}$ ,  $\tilde{\mathcal{B}}_{\Psi(\mathbf{z}, \mathbf{v})}$  can be realized by replacing with  $\tilde{\mathcal{B}}_{\exp(x/2\sigma^2)}$  and  $\tilde{\mathcal{B}}_{\cosh(y/2\sigma^2)}$ , respectively. Therefore,  $R_{2\lambda+1}(\tilde{\mathcal{B}}_{\Psi(\mathbf{z}, \mathbf{v})} \mid \mathcal{B}_{\Psi(\mathbf{z}, \mathbf{v})})$  is analyzed by relative errors arising from two components. To this end, the relative error associated with probability  $\exp(x/2\sigma^2)$  is analyzed in Lemma B.15, while relative error associated with probability  $\cosh(y/2\sigma^2)$  is addressed in Lemma B.16.

**Theorem B.14.** Let  $\tilde{\mathcal{B}}_{\Psi(\mathbf{z}, \mathbf{v})}$  be a probability distribution of  $\text{RejectSample}_{\sigma, \hat{\alpha}}(\mathbf{z}, \mathbf{v})$  in Algorithm 6. If  $P_{\exp}^{I'_2}(x)$ ,  $\hat{c}$  and  $\omega$  satisfy (15), (16), (17), then

$$R_{2\lambda+1}(\tilde{\mathcal{B}}_{\Psi(\mathbf{z}, \mathbf{v})} \mid \mathcal{B}_{\Psi(\mathbf{z}, \mathbf{v})}) \leq 1 + \frac{1}{4Q_{\text{reject}}}$$

for all  $\mathbf{z}, \mathbf{v}$  satisfying  $\|\mathbf{v}\| \leq \frac{\sigma}{\hat{\alpha}}$ ,  $|\langle \mathbf{z}, \mathbf{v} \rangle| \leq \frac{B_2 \sigma}{\hat{\alpha}}$  where  $\Psi(\mathbf{z}, \mathbf{v}) = 1/(M \exp(-\|\mathbf{v}\|^2/(2\sigma^2)) \cosh(\langle \mathbf{z}, \mathbf{v} \rangle/\sigma^2))$ ,  $M = \exp(\frac{1}{2\hat{\alpha}^2})$ .

*Proof.* Let us denote the output distributions of  $\text{SampleBernExp}_{\sigma}(x)$  in Algorithm 7 and  $\text{SampleBernCosh}_{\sigma}(y)$  in Algorithm 8 as  $\tilde{\mathcal{B}}_{\exp(x/2\sigma^2)}$  and  $\tilde{\mathcal{B}}_{1/\cosh(y/2\sigma^2)}$ , respectively. Then, by Lemma B.12, the Rényi divergence is bounded by its relative error.

Letting  $A = \frac{\tilde{\mathcal{B}}_{\exp(x/2\sigma^2)}(1)}{\mathcal{B}_{\exp(x/2\sigma^2)}(1)}$  and  $B = \frac{\tilde{\mathcal{B}}_{1/\cosh(y/2\sigma^2)}(1)}{\mathcal{B}_{1/\cosh(y/2\sigma^2)}(1)}$ , we have, for  $x \in [-\sigma^2/\hat{\alpha}^2, 0]$ ,

$$\begin{aligned} \left| \frac{\tilde{\mathcal{B}}_{\Psi(\mathbf{z}, \mathbf{v})}(1)}{\mathcal{B}_{\Psi(\mathbf{z}, \mathbf{v})}(1)} - 1 \right| &= \left| \frac{\tilde{\mathcal{B}}_{\exp(x/2\sigma^2)}(1) \cdot \tilde{\mathcal{B}}_{1/\cosh(y/2\sigma^2)}(1)}{\mathcal{B}_{\exp(x/2\sigma^2)}(1) \cdot \mathcal{B}_{1/\cosh(y/2\sigma^2)}(1)} - 1 \right| = |AB - 1| \\ &\leq |A - 1| + |B - 1| + |(A - 1)(B - 1)| \\ &\leq 2^4 \cdot c_{r,2} \cdot \delta_{\text{reject}} + 2^6 \cdot c_{r,2}^2 \cdot \delta_{\text{reject}}^2 \\ &\leq 2^4 \cdot \delta_{\text{reject}} \end{aligned}$$

where the second inequality from the last comes from Lemma B.15, Lemma B.16, and the last equality holds by (12). Similarly, we have, for  $x \in [-\sigma^2/\hat{\alpha}^2, 0]$ ,

$$\begin{aligned} \left| \frac{\tilde{\mathcal{B}}_{\Psi(\mathbf{z}, \mathbf{v})}(0)}{\mathcal{B}_{\Psi(\mathbf{z}, \mathbf{v})}(0)} - 1 \right| &= \left| \frac{1 - \tilde{\mathcal{B}}_{\exp(x/2\sigma^2)}(1) \cdot \tilde{\mathcal{B}}_{1/\cosh(y/2\sigma^2)}(1)}{1 - \mathcal{B}_{\exp(x/2\sigma^2)}(1) \cdot \mathcal{B}_{1/\cosh(y/2\sigma^2)}(1)} - 1 \right| \\ &= \left| \frac{\mathcal{B}_{\exp(x/2\sigma^2)}(1) \cdot \mathcal{B}_{1/\cosh(y/2\sigma^2)}(1)}{1 - \mathcal{B}_{\exp(x/2\sigma^2)}(1) \cdot \mathcal{B}_{1/\cosh(y/2\sigma^2)}(1)} \right| \cdot \left| \frac{\tilde{\mathcal{B}}_{\exp(x/2\sigma^2)}(1) \cdot \tilde{\mathcal{B}}_{1/\cosh(y/2\sigma^2)}(1)}{\mathcal{B}_{\exp(x/2\sigma^2)}(1) \cdot \mathcal{B}_{1/\cosh(y/2\sigma^2)}(1)} - 1 \right| \\ &\leq \left| \frac{\mathcal{B}_{\exp(x/2\sigma^2)}(1)}{1 - \mathcal{B}_{\exp(x/2\sigma^2)}(1)} \right| \cdot |AB - 1| \\ &\leq \left| \frac{\exp(-1/2\sigma^2)}{1 - \exp(-1/2\sigma^2)} \right| \cdot |AB - 1| \\ &\leq \left| \frac{\exp(-1/2\sigma^2)}{1 - \exp(-1/2\sigma^2)} \right| \cdot 2^4 \cdot \delta_{\text{reject}} \\ &\leq 2^5 \cdot \sigma^2 \cdot \delta_{\text{reject}} \end{aligned}$$

where the last inequality holds since  $\exp(\frac{1}{2\sigma^2}) > 1 + \frac{1}{2\sigma^2}$  and  $\left| \frac{\exp(-1/2\sigma^2)}{1 - \exp(-1/2\sigma^2)} \right| < 2\sigma^2$ .

Thus, the relative error between the two distributions are bounded by  $2^5 \cdot \sigma^2 \cdot \delta_{\text{reject}}$ , which proves the result by Lemma B.12.  $\square$

**Lemma B.15.** Let  $\tilde{\mathcal{B}}_{\exp(x/2\sigma^2)}$  be the output distribution of  $\text{SampleBernExp}_\sigma^{I_2}(x)$  in Algorithm 7. If  $P_{\exp}^{I_2'}(x)$ ,  $\hat{c}$ ,  $\vartheta_2$  and  $\omega$  satisfy (15), (16), (17), (18), then

$$\left| \frac{\tilde{\mathcal{B}}_{\exp(x/2\sigma^2)}(1)}{\mathcal{B}_{\exp(x/2\sigma^2)}(1)} - 1 \right| \leq 2^3 \cdot c_{r,2} \cdot \delta_{\text{reject}}$$

for all  $x \in I_2 = [-\frac{\sigma^2}{\hat{\alpha}^2}, 0]$ .

*Proof.* Note that

$$\left| \frac{\tilde{\mathcal{B}}_{\exp(x/2\sigma^2)}(1)}{\mathcal{B}_{\exp(x/2\sigma^2)}(1)} - 1 \right| \leq \underbrace{\left| \frac{\tilde{\mathcal{B}}_{\exp(x/2\sigma^2)}(1)}{\mathcal{B}_{\exp(x/2\sigma^2)}(1)} - \frac{\mathcal{B}_{\widetilde{\exp}(x/2\sigma^2)}(1)}{\mathcal{B}_{\exp(x/2\sigma^2)}(1)} \right|}_{(*)} + \underbrace{\left| \frac{\mathcal{B}_{\widetilde{\exp}(x/2\sigma^2)}(1)}{\mathcal{B}_{\exp(x/2\sigma^2)}(1)} - 1 \right|}_{(*)},$$

where  $\widetilde{\exp}(x/2\sigma^2) = 2^{x_1}(1 + \omega x_1)\tilde{P}_{\exp}^{I_2'}(x_0)$  and  $x = x_1\hat{c} + x_0$ . We will prove the result by finding an upper bound of  $(*)$  and an upper bound of  $(*)$  respectively, and then by combining them.

Let us define  $A = \frac{1+\omega x_1}{\exp(\frac{\hat{c}-c}{2\sigma^2})^{x_1}}$  and  $B = \frac{\tilde{P}'_{\exp}(x_0)}{\exp(\frac{x_0}{2\sigma^2})}$ . Then, for  $x \in (-\hat{c}, 0]$ , we have  $x_1 = 0$  and so

$$(\star) = \left| \frac{2^{x_1}(1+\omega x_1)\tilde{P}'_{\exp}(x_0)}{\exp(x/2\sigma^2)} - 1 \right| = \left| \frac{\tilde{P}'_{\exp}(x_0)}{\exp(x_0/2\sigma^2)} - 1 \right| \leq 2 \cdot c_{r,1} \cdot \delta_{\text{reject}} < 2^3 \cdot (c_{r,2} - \frac{5}{8}) \cdot \delta_{\text{reject}},$$

where the inequality holds by Lemma B.17 and  $c_{r,1} < c_{r,2} - \frac{5}{8} < 1$ . Moreover, for  $x_0 \in (-\hat{c}, 0]$ , we have

$$\begin{aligned} (\star) &\leq \left| \frac{2^{x_1}(1+\omega x_1)\tilde{P}'_{\exp}(x_0)}{2^{x_1} \exp(\frac{\hat{c}-c}{2\sigma^2})^{x_1} \exp(\frac{x_0}{2\sigma^2})} - 1 \right| \\ &= |AB - 1| \\ &\leq |A - 1| + |B - 1| + |A - 1||B - 1| \\ &\leq 4 \cdot c_{r,1} \cdot \delta_{\text{reject}} + 4 \cdot c_{r,1}^2 \cdot \delta_{\text{reject}}^2 \\ &< 2^3 \cdot (c_{r,2} - \frac{5}{8}) \cdot \delta_{\text{reject}} \end{aligned}$$

where two inequalities from the last come from Lemma B.17, Lemma B.18, and  $c_{r,1} < c_{r,2} - \frac{5}{8} < 1$ .

Since  $\text{SampleBernExp}_{\sigma}^{I_2}(x)$  computes its value with  $\text{Mul}[\vartheta_2](\cdot)$ , we have

$$\begin{aligned} (*) &= \left| \frac{2^{x_1} \text{Mul}[\vartheta_2](1+\omega x_1, \tilde{P}'_{\exp}(x_0)) - 2^{x_1}(1+\omega x_1)\tilde{P}'_{\exp}(x_0)}{2^{x_1} \exp(\frac{\hat{c}-c}{2\sigma^2})^{x_1} \exp(\frac{x_0}{2\sigma^2})} \right| \\ &\leq e \cdot \left| \text{Mul}[\vartheta_2](1+\omega x_1, \tilde{P}'_{\exp}(x_0)) - (1+\omega x_1)\tilde{P}'_{\exp}(x_0) \right| \\ &\leq e \cdot 2^{-(\vartheta_2+1)} \end{aligned}$$

where the last inequality comes from  $(1+\omega x_1)\tilde{P}'_{\exp}(x_0) < 1$ .

Finally, combining these two upper bounds, we have

$$\begin{aligned} \left| \frac{\tilde{\mathcal{B}}_{\exp(x/2\sigma^2)}(1)}{\mathcal{B}_{\exp(x/2\sigma^2)}(1)} - 1 \right| &\leq e \cdot 2^{-(\vartheta_2+1)} + 2^3 \cdot (c_{r,2} - \frac{5}{8}) \cdot \delta_{\text{reject}} \\ &\leq e \cdot c_{r,1} \cdot \delta_{\text{reject}} + 2^3 \cdot (c_{r,2} - \frac{5}{8}) \cdot \delta_{\text{reject}} < 2^3 \cdot c_{r,2} \cdot \delta_{\text{reject}} \end{aligned}$$

where the last inequality comes from  $c_{r,1} < 1$ . □

**Lemma B.16.** Let  $\tilde{\mathcal{B}}_{1/\cosh(y/2\sigma^2)}$  be the output distribution of  $\text{SampleBernCosh}_{\sigma}(y)$  in Algorithm 8. If  $P_{\exp}^{I_2}(x)$ ,  $\hat{c}$ ,  $\omega$ ,  $\vartheta_2$  satisfy (15), (16), (17), (18), then

$$\left| \frac{\tilde{\mathcal{B}}_{1/\cosh(y/2\sigma^2)}(1)}{\mathcal{B}_{1/\cosh(y/2\sigma^2)}(1)} - 1 \right| \leq 2^3 \cdot c_{r,2} \cdot \delta_{\text{reject}}$$

for all  $y \in [-\frac{2B_2\sigma}{\hat{\alpha}}, \frac{2B_2\sigma}{\hat{\alpha}}]$ .

*Proof.* Note that

$$\left| \frac{\tilde{\mathcal{B}}_{1/\cosh(y/2\sigma^2)}(1)}{\mathcal{B}_{1/\cosh(y/2\sigma^2)}(1)} - 1 \right| \leq \underbrace{\left| \frac{\tilde{\mathcal{B}}_{1/\cosh(y/2\sigma^2)}(1)}{\mathcal{B}_{1/\cosh(y/2\sigma^2)}(1)} - \frac{\mathcal{B}_{1/\widetilde{\cosh}(y/2\sigma^2)}(1)}{\mathcal{B}_{1/\cosh(y/2\sigma^2)}(1)} \right|}_{(*)} + \underbrace{\left| \frac{\mathcal{B}_{1/\widetilde{\cosh}(y/2\sigma^2)}(1)}{\mathcal{B}_{1/\cosh(y/2\sigma^2)}(1)} - 1 \right|}_{(*)}.$$

We will prove the result by finding an upper bound of  $(\star)$  and an upper bound of  $(*)$  respectively, and then by combining them.

Let us define  $\widetilde{\cosh}(y/2\sigma^2) = \frac{2^{y_1(1+\omega y_1)} \tilde{P}_{\exp}'^{I_2'}(y_0) + 2^{1-y_1(1+\omega(1-y_1))} \tilde{P}_{\exp}'^{I_2'}(-y_0-\hat{c})}{2}$  and

$$A = \frac{\exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^{y_1}}{1+\omega y_1}, \quad B = \frac{\exp\left(\frac{y_0}{2\sigma^2}\right)}{\tilde{P}_{\exp}'^{I_2'}(y_0)}, \quad C = \frac{\exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^{1-y_1}}{1+\omega(1-y_1)}, \quad D = \frac{\exp\left(-\frac{y_0+\hat{c}}{2\sigma^2}\right)}{\tilde{P}_{\exp}'^{I_2'}(-y_0-\hat{c})}.$$

Then, for  $y = y_1\hat{c} + y_0$ ,  $y_0 \in (-\hat{c}, 0]$ , we have

$$\begin{aligned} (\star) &= \left| \frac{\exp\left(\frac{y}{2\sigma^2}\right) + \exp\left(-\frac{y}{2\sigma^2}\right)}{2^{y_1(1+\omega y_1)} \tilde{P}_{\exp}'^{I_2'}(y_0) + 2^{1-y_1(1+\omega(1-y_1))} \tilde{P}_{\exp}'^{I_2'}(-y_0-\hat{c})} - 1 \right| \\ &\leq \left| \frac{\exp\left(\frac{y}{2\sigma^2}\right)}{2^{y_1(1+\omega y_1)} \tilde{P}_{\exp}'^{I_2'}(y_0)} - 1 \right| + \left| \frac{\exp\left(-\frac{y}{2\sigma^2}\right)}{2^{1-y_1(1+\omega(1-y_1))} \tilde{P}_{\exp}'^{I_2'}(-y_0-\hat{c})} - 1 \right| \\ &= \left| \frac{\exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^{y_1} \exp\left(\frac{y_0}{2\sigma^2}\right)}{(1+\omega y_1) \tilde{P}_{\exp}'^{I_2'}(y_0)} - 1 \right| + \left| \frac{\exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^{1-y_1} \exp\left(-\frac{y_0+\hat{c}}{2\sigma^2}\right)}{(1+\omega(1-y_1)) \tilde{P}_{\exp}'^{I_2'}(-y_0-\hat{c})} - 1 \right| \\ &\leq |A - 1| + |B - 1| + |(A - 1)(B - 1)| + |C - 1| + |D - 1| + |(C - 1)(D - 1)|. \end{aligned}$$

Since  $\left|\frac{1}{A} - 1\right|, \left|\frac{1}{B} - 1\right|, \left|\frac{1}{C} - 1\right|, \left|\frac{1}{D} - 1\right|$  are less than or equal to  $2c_{r,1}\delta_{\text{reject}}$  by Lemma B.18, Lemma B.17, we have

$$(\star) \leq 4 \frac{2 \cdot c_{r,1} \cdot \delta_{\text{reject}}}{1 - 2 \cdot c_{r,1} \cdot \delta_{\text{reject}}} + 2 \left( \frac{2 \cdot c_{r,1} \cdot \delta_{\text{reject}}}{1 - 2 \cdot c_{r,1} \cdot \delta_{\text{reject}}} \right)^2 \leq 2^3 \cdot (c_{r,2} - \frac{5}{8}) \cdot \delta_{\text{reject}}$$

where two inequalities come from Lemma B.19 and (14).

Let us define

$$\begin{aligned} E &= \exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^{y_1} \exp\left(\frac{y_0}{2\sigma^2}\right), \quad F = \exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^{1-y_1} \exp\left(-\frac{y_0+\hat{c}}{2\sigma^2}\right), \\ G &= \text{Mul}[\vartheta_2](1 + \omega y_1, \tilde{P}_{\exp}'^{I_2'}(y_0)), \quad H = \text{Mul}[\vartheta_2](1 + \omega(1 - y_1), \tilde{P}_{\exp}'^{I_2'}(-y_0 - \hat{c})), \\ I &= (1 + \omega y_1) \tilde{P}_{\exp}'^{I_2'}(y_0), \quad J = (1 + \omega(1 - y_1)) \tilde{P}_{\exp}'^{I_2'}(-y_0 - \hat{c}). \end{aligned}$$

Since  $\text{SampleBernCosh}_\sigma(y)$  computes its value with  $\text{Mul}[\vartheta_2](\cdot)$ , we have the following inequalities. Without loss of generality, we assume  $y \geq 0$ . Then, for  $y \in [0, \hat{c})$ , we have  $y_1 = 0$  and so

$$\begin{aligned} (*) &= \left| \frac{(\exp\left(\frac{y_0}{2\sigma^2}\right) + 2F)(\tilde{P}_{\exp}'^{I_2'}(y_0) + 2H - \tilde{P}_{\exp}'^{I_2'}(y_0) - 2J)}{(\tilde{P}_{\exp}'^{I_2'}(y_0) + 2H) \cdot (\tilde{P}_{\exp}'^{I_2'}(y_0) + 2J)} \right| \\ &\leq \left| \frac{\exp\left(\frac{y_0}{2\sigma^2}\right) + 2F}{\tilde{P}_{\exp}'^{I_2'}(y_0) + 2J} \right| \cdot 2 |H - J| \\ &\leq \left( \left| \frac{\exp\left(\frac{y_0}{2\sigma^2}\right)}{\tilde{P}_{\exp}'^{I_2'}(y_0)} \right| + \left| \frac{F}{J} \right| \right) \cdot 2 |H - J| = (|B| + |C \cdot D|) \cdot 2 |H - J| \\ &\leq 2 \left( \left(1 + \frac{c_{r,1} \cdot \delta_{\text{reject}}}{1 - c_{r,1} \cdot \delta_{\text{reject}}}\right) + \left(1 + \frac{2 \cdot c_{r,1} \cdot \delta_{\text{reject}}}{1 - 2 \cdot c_{r,1} \cdot \delta_{\text{reject}}}\right)^2 \right) 2^{-(\vartheta_2+1)} \\ &\leq 5 \cdot 2^{-(\vartheta_2+1)} \end{aligned}$$



where last two inequalities come from  $J < 1$ , Lemma B.17, Lemma B.18 and (13). Moreover, for  $y \in [\hat{c}, 2B_2\sigma/\hat{\alpha}]$ , we have

$$\begin{aligned}
(*) &= \left| \frac{(2^{y_1} E + 2^{1-y_1} F)(2^{y_1} G + 2^{1-y_1} H - 2^{y_1} I - 2^{1-y_1} J)}{(2^{y_1} G + 2^{1-y_1} H) \cdot (2^{y_1} I + 2^{1-y_1} J)} \right| \\
&= \left| \frac{(E + 2^{1-2y_1} F)(G + 2^{1-2y_1} H - I - 2^{1-2y_1} J)}{(G + 2^{1-2y_1} H) \cdot (I + 2^{1-2y_1} J)} \right| \\
&\leq \left| \frac{(E + 2^{1-2y_1} F)(G + 2^{1-2y_1} H - I - 2^{1-2y_1} J)}{I + 2^{1-2y_1} J} \right| \\
&\leq \left( \left| \frac{E}{I + 2^{1-2y_1} J} \right| + \left| \frac{2^{1-2y_1} F}{I + 2^{1-2y_1} J} \right| \right) (|G - I| + 2^{1-2y_1} |H - J|) \\
&\leq (| \frac{E}{I} | + | \frac{F}{J} |) (|G - I| + 2^{1-2y_1} |H - J|) = (|A \cdot B| + |C \cdot D|) (|G - I| + 2^{1-2y_1} |H - J|) \\
&\leq 2 \left( 1 + \frac{2 \cdot c_{r,1} \cdot \delta_{\text{reject}}}{1 - 2 \cdot c_{r,1} \cdot \delta_{\text{reject}}} \right)^2 (2^{\lfloor \log_2(1+\omega y_1) \rfloor} + 2^{1-2y_1}) 2^{-(\vartheta_2+1)} \\
&\leq 2 \left( 1 + \frac{2 \cdot c_{r,1} \cdot \delta_{\text{reject}}}{1 - 2 \cdot c_{r,1} \cdot \delta_{\text{reject}}} \right)^2 (1 + 2^{\lfloor \log_2(1+\omega \zeta) \rfloor}) 2^{-(\vartheta_2+1)} \\
&\leq 5 \cdot 2^{-(\vartheta_2+1)}
\end{aligned}$$

where  $\zeta = \lceil \frac{B_2}{\sigma \hat{\alpha} \ln 2} \rceil$  and last four inequalities come from  $J < 1$ , Lemma B.17, Lemma B.18, and (13).

Finally, combining these two upper bounds, we have

$$\begin{aligned}
\left| \frac{\tilde{\mathcal{B}}_{1/\cosh(y/2\sigma^2)}(1)}{\mathcal{B}_{1/\cosh(y/2\sigma^2)}(1)} - 1 \right| &\leq 5 \cdot 2^{-(\vartheta_2+1)} + 2^3 \cdot (c_{r,2} - \frac{5}{8}) \cdot \delta_{\text{reject}} \\
&\leq 5 \cdot c_{r,1} \cdot \delta_{\text{reject}} + 2^3 \cdot (c_{r,2} - \frac{5}{8}) \cdot \delta_{\text{reject}} < 2^3 \cdot c_{r,2} \cdot \delta_{\text{reject}}
\end{aligned}$$

where the last inequality comes from  $c_{r,1} < 1$ . □

**Lemma B.17.** Let  $\tilde{P}_{\text{exp}}^{I'_2}(x)$  be the output value of Algorithm 5 with parameters  $P_{\text{exp}}^{I'_2}$  and  $\vartheta_2$ . If  $P_{\text{exp}}^{I'_2}(x)$ ,  $\vartheta_2$  satisfies (17), (18), then

$$\left| \frac{\tilde{P}_{\text{exp}}^{I'_2}(x)}{\exp(x/(2\sigma^2))} - 1 \right| \leq 2 \cdot c_{r,1} \cdot \delta_{\text{reject}}$$

for  $x \in I'_2$ .

*Proof.* The result is derived from following inequality:

$$\begin{aligned}
\left| \frac{\tilde{P}_{\text{exp}}^{I'_2}(x)}{\exp(x/(2\sigma^2))} - 1 \right| &\leq \left| \frac{\tilde{P}_{\text{exp}}^{I'_2}(x)}{\exp(x/(2\sigma^2))} - \frac{P_{\text{exp}}^{I'_2}(x)}{\exp(x/(2\sigma^2))} \right| + \left| \frac{P_{\text{exp}}^{I'_2}(x)}{\exp(x/(2\sigma^2))} - 1 \right| \\
&\leq \left| \tilde{P}_{\text{exp}}^{I'_2}(x) - P_{\text{exp}}^{I'_2}(x) \right| + \left| \frac{P_{\text{exp}}^{I'_2}(x)}{\exp(x/(2\sigma^2))} - 1 \right| \\
&\leq 2 \cdot c_{r,1} \cdot \delta_{\text{reject}},
\end{aligned}$$

where the last inequality comes from the (17), (18) and Lemma B.13. □

**Lemma B.18.** If  $\hat{c}, \omega$  satisfy (15) and (16), then

$$\left| \frac{1 + \omega x}{\exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^x} - 1 \right| \leq 2 \cdot c_{r,1} \cdot \delta_{\text{reject}}, \quad (19)$$

$$\left| \frac{1 + \omega(1-x)}{\exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^{1-x}} - 1 \right| \leq 2 \cdot c_{r,1} \cdot \delta_{\text{reject}} \quad (20)$$

for all  $x \in (-\zeta, 0]$  where  $\zeta = \lceil \frac{B_2}{\sigma \hat{\alpha} \ln 2} \rceil$ .

*Proof.* Note that  $|e^t - (1+t)| \leq \frac{t^2}{2}$  for  $t \in (-1, 0]$  and  $|e^t - (1+t)| \leq \frac{t^2}{2}e$  for  $t \in (0, 1)$ . Since  $x \in (-\zeta, 0]$  and  $0 < \frac{\zeta(\hat{c}-c)}{2\sigma^2} < 1$ , we have

$$\begin{aligned} \left| \frac{1+\omega x}{\exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^x} - 1 \right| &= \left| \exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^x - (1+\omega x) \right| \cdot \exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^{-x} \\ &\leq \underbrace{\left| \exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^x - (1+\omega x) \right|}_{(\star)} \cdot e. \end{aligned}$$

Then

$$\begin{aligned} (\star) &\leq \left( \left| \exp\left(\frac{\hat{c}-c}{2\sigma^2}x\right) - \left(1 + \frac{\hat{c}-c}{2\sigma^2}x\right) \right| + \left| \left(1 + \frac{\hat{c}-c}{2\sigma^2}x\right) - (1+\omega x) \right| \right) \cdot e \\ &\leq \left( \frac{1}{2} \left( \frac{\zeta(\hat{c}-c)}{2\sigma^2} \right)^2 + \left| \left(1 + \frac{\hat{c}-c}{2\sigma^2}x\right) - (1+\omega x) \right| \right) \cdot e \\ &\leq \frac{e}{2} \left( \frac{\zeta(\hat{c}-c)}{2\sigma^2} \right)^2 + e\zeta \left| \frac{\hat{c}-c}{2\sigma^2} - \omega \right| \\ &\leq 2 \cdot c_{r,1} \cdot \delta_{\text{reject}} \end{aligned}$$

where the last inequality comes from the (15), (16).

On the other hand, since  $-1 < \frac{(x-1)(\hat{c}-c)}{2\sigma^2} \leq 0$ , we have

$$\begin{aligned} \left| \frac{1+\omega(1-x)}{\exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^{1-x}} - 1 \right| &= \left| \exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^{1-x} - (1+\omega(1-x)) \right| \cdot \exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^{x-1} \\ &\leq \underbrace{\left| \exp\left(\frac{\hat{c}-c}{2\sigma^2}\right)^{1-x} - (1+\omega(1-x)) \right|}_{(\star)}. \end{aligned}$$

Then

$$\begin{aligned} (\star) &\leq \left| \exp\left(\frac{\hat{c}-c}{2\sigma^2}(1-x)\right) - \left(1 + \frac{\hat{c}-c}{2\sigma^2}(1-x)\right) \right| + \left| \left(1 + \frac{\hat{c}-c}{2\sigma^2}(1-x)\right) - (1+\omega(1-x)) \right| \\ &\leq \frac{e}{2} \left( \frac{\zeta(\hat{c}-c)}{2\sigma^2} \right)^2 + \left| \left(1 + \frac{\hat{c}-c}{2\sigma^2}(1-x)\right) - (1+\omega(1-x)) \right| \\ &\leq \frac{e}{2} \left( \frac{\zeta(\hat{c}-c)}{2\sigma^2} \right)^2 + (\zeta+1) \left| \frac{\hat{c}-c}{2\sigma^2} - \omega \right| \\ &\leq 2 \cdot c_{r,1} \cdot \delta_{\text{reject}} \end{aligned}$$

where the last inequality also comes from the (15), (16). □

**Lemma B.19.** If  $|a - 1| \leq \kappa$ , then  $\left|\frac{1}{a} - 1\right| \leq \frac{\kappa}{1-\kappa}$ .

*Proof.* From the hypothesis,  $\left|\frac{1}{a}\right| \leq \frac{1}{1-\kappa}$ . Thus, we have  $\left|\frac{1}{a} - 1\right| = |a - 1| \cdot \left|\frac{1}{a}\right| \leq \frac{\kappa}{1-\kappa}$ . □